

Abstract Algebra

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Contents

1	Prerequisite	80
1.1	Sets and Functions	80
1.2	Arithmetics of Integers	80
2	Groups	81
2.1	Definition and Examples	81
2.2	The most basic properties	83
2.3	Subgroups	84
2.4	Lagrange's Theorem	86
2.4.1	Elementary Number Theory	89
2.5	Homomorphisms and normal subgroups	90
2.6	Quotient Groups (or Quotient Groups)	94
2.7	The homomorphism theorems	96
3	Permutation Groups	99
3.1	The symmetric group	99
3.2	Cycle Decomposition	100
3.3	Odd and Even permutations	101
3.4	The group A_n is simple for $n \geq 5$	102
4	Finite Groups	105
4.1	Cauchy's Theorem	105
4.2	Direct Products (internal and external)	108
4.3	Finite Abelian Groups	110
4.4	Conjugacy and Sylow Theorems	113
4.5	Small Groups	115
4.6	Ideals, Homomorphisms, and Quotient Ring	117
4.7	Polynomial Rings	122
4.8	Polynomials over Rationals	128

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Homework 1: due August 31st

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Herstein, section 1.3, page 13 : Questions 6,12,16,23,30

1. 6. If $f : S \rightarrow T$ is onto and $g : T \rightarrow U$ and $h : T \rightarrow U$ are such that $g \circ f = h \circ f$, prove that $g = h$.

Proof. To prove $g = h$, we need to show that for all $t \in T$, $g(t) = h(t)$. Now fix some t . Because f is onto, there exists some $s \in S$ such that $f(s) = t$. By assumption, $g \circ f(s) = h \circ f(s)$. Simplify and we get $g(t) = h(t)$, as desired. $\square$

2. 12. Let S be the set of nonnegative rational numbers, that is, $S = \{m/n \mid m, n \text{ nonnegative integers, } n \neq 0\}$, and let T be the set of all integers.
(a) Does $f : S \rightarrow T$ defined by $f(m/n) = 2^m 3^n$ define a legitimate function from S to T ?

Solution. No. There is no unique fractional representation of any rational number. For example, $\frac{2}{3} = \frac{4}{6}$ refer to the same rational number, say q . Hence $f(q)$ is not well-defined since we do not know whether it should be $2^2 3^3$ or $4^2 6^3$, or something else. $\square$

(b) If not, how could you modify the definition of f so as to get a legitimate function?

Solution. We can modify the definition of f as follows: Let $f(0) = 0$, and for all $q > 0$, let $f(q) := 2^m 3^n$ for the unique integers m, n such that $m > 0$, $n > 0$, $q = m/n$, and m, n are relatively prime. $\square$

3. 16. If f is a 1-1 mapping of S onto itself, show that $(f^{-1})^{-1} = f$.

Proof. $(f^{-1})^{-1} \circ f^{-1} = \iota_{f(S)}$ by Lemma 1.3.4. Similarly, $f \circ f^{-1} = \iota_{f(S)}$. Therefore $(f^{-1})^{-1} \circ f^{-1} = f \circ f^{-1}$. Since f is a bijection, so is f^{-1} , and in particular f^{-1} is onto. Hence by Exercise 6, we have the result desired. $\square$

4. 23. Let S be the set of all integers of the form $2^m 3^n$, $m \geq 0, n \geq 0$, and let T be the set of all positive integers. Show that there is a 1-1 correspondence of S onto T .

Proof. Because 2 and 3 are prime numbers, for any integer x of the form $2^m 3^n$, $n \geq 0, m \geq 0$, there exists unique m_x and $n_x \in \mathbb{N}$ such that $x = 2^{m_x} 3^{n_x}$ (by unique factorization of integers). Conversely, it is obvious that for each $m \in T$ and $n \in T$ there exists a unique number $2^m 3^n \in S$ in the form $2^m 3^n$. We have shown that there is a bijection μ between S and T^2 .

Now we show that there is a bijection between T^2 and T . We use the standard bijection $\langle \cdot, \cdot \rangle : T^2 \rightarrow T$ defined by $\langle x, y \rangle := \frac{(x+y-2) \cdot (x+y-1)}{2} + x$. Combine $\langle \cdot, \cdot \rangle$ and μ and we have a bijection between S and T . $\square$

5. 30. If S is a finite set and f is a 1-1 mapping of S , show that for some integer $n > 0$,

$$\underbrace{f \circ f \circ f \circ \cdots \circ f}_{n \text{ times}} = i_S.$$

Proof. We call a n -tuple $u_1, u_2, \dots, u_n$ of elements in S a *closed orbit wrt* f , if $f(u_1) = u_2, f(u_2) = u_3, \dots$ and $f(u_n) = u_1$.

Fix some $x \in S$. Define $R(x) := \{f^{(n)}(x) : n \in \mathbb{N}\} \subset S$. Since S is finite, so is $R(x)$. Hence, there exists some integers $i < j$ such that $f^{(i)}(x) = f^{(j)}(x)$ (if not, we can conclude that $R(x)$ is infinite, a contradiction). In other words, $(f^i(x), \dots, f^{j-1}(x))$ forms a closed orbit wrt f . We denote the underlying set of this orbit as $\bar{R}(x)$. Without loss of generality, we may assume that there is no duplication of elements in the orbit.

Our next step is to prove that $R(x) \setminus \bar{R}(x) = \emptyset$. Suppose there is some element $y \in R(x) \setminus \bar{R}(x)$. Then there exists some integer k such that $f^{(k)}(y) \notin \bar{R}(x)$ and $f^{(k+1)}(y) \in \bar{R}(x)$. By property of orbit, there exists some $z \in \bar{R}(x)$ such that $f(z) = f^{(k+1)}(y)$. Hence $f^{(k)}(y)$ and z are both mapped by f to $f^{(k+1)}(y)$, but they are clearly distinct, since one is outside the orbit and the other is inside. This contradicts the assumption that f is 1-1.

Now let $n_x := |R(x)|$ be the orbit length of the point $x \in S$ wrt f . Note that $R(x)$ is a closed orbit. Therefore we can conclude that $f^{(n_x)}(x) = x$. Now define $N(S) := \prod_{x \in S} n_x$. Since S is finite and n_x is always finite, $N \in \mathbb{N}$ is well-defined. N is the desired integer. $\square$

After meeting with Prof. Wooley, I came up with a simpler proof, which I give below.

Proof. (Simpler) Since f is 1-1 and S is finite, f is a bijection and hence $f \in A(S)$. Take $x \in S$. We claim that there exists some integer $n_x \leq |S|$ such that $f^{n_x}(x) = x$. If the claim is true, we take $N := \text{factorial}|S|$. Note that N is finite (since S is finite), and is divided by each n_x . Hence f^N will send each $x \in S$ to itself.

Now we verify the claim. Compute $f^1(x)$. If $f^1(x) = x$ then $n_x = 1$ and we are done. If $n_x \neq x$, we move on and compute $f^2(x)$. If $f^2(x) = x$ we are also done. If not, then we must have $f^2(x) \neq f^1(x)$, otherwise we have two distinct elements both mapping to $f^1(x)$, a contradiction to the 1-1 property. Continue in this fashion. If we are still not done at step $|S|$, note that $x, f^1(x), f^2(x), \dots, f^{|S|-1}(x)$ are all distinct, and $f^{|S|}$ cannot map x to any of $f^1(x), f^2(x), \dots, f^{|S|-1}(x)$, by the same argument applied $|S| - 1$ times (if $f^{|S|}x = fx$ then 1-1 property is violated since x also gets mapped to fx , etc). Hence $f^{|S|}x = x$, $n_x = |S|$ and we are done. $\square$

Herstein, section 1.4, page 19: Questions 4,16,23,27

6. 4. If $f, g, h \in A(S)$, show that $(f^{-1}gf)(f^{-1}hf) = f^{-1}(gh)f$. What can you say about $(f^{-1}gf)^n$?

Proof. We assume the notation for product in $A(S)$ is right-associative.

$$\begin{aligned}
 & (f^{-1}gf)(f^{-1}hf) \\
 &= (f^{-1}(gf))(f^{-1}(hf)) && \text{Notation} \\
 &= ((f^{-1}(gf))f^{-1})(hf) && \text{Associativity} \\
 &= (f^{-1}((gf)f^{-1}))(hf) && \text{Associativity} \\
 &= (f^{-1}(g(ff^{-1})))(hf) && \text{Associativity} \\
 &= (f^{-1}(g\iota_S))(hf) && \text{Inverse} \\
 &= (f^{-1}g)(hf) && \text{Identity} \\
 &= ((f^{-1}g)h)f && \text{Associativity} \\
 &= (f^{-1}(gh))f && \text{Associativity} \\
 &= f^{-1}((gh)f) && \text{Associativity} \\
 &= f^{-1}(gh)f && \text{Notation.}
 \end{aligned}$$

By induction we can show that $(f^{-1}gf)^n = f^{-1}g^n f$. $\square$

7. 16. Let S be an infinite set and let $M \subset A(S)$ be the set of all elements $f \in A(S)$ such that $f(s) \neq s$ for at most a finite number of $s \in S$. Prove that:

(a) $f, g \in M$ implies that $fg \in M$.

Proof of (a), simpler. Since symmetry group is closed under element multiplication, $fg \in A(S)$. It remains to show that there is at most finitely many points in S where fg is nontrivial ($fg(s) \neq s$).

Define $D_f := \{s \in S : f(s) \neq s\}$ and D_g similarly. Take any $s \in S \setminus (D_f \cup D_g)$. By definition, $f(s) = g(s) = s$, hence $fg(s) = s$. By assumption, D_f and D_g are both finite, so their union is finite. Therefore, $fg(s) = s$ on all but a finite subset of S . Hence, $fg(s) \neq s$ on an at most finite set. By definition, $fg \in M$. $\square$

(b) $f \in M$ implies that $f^{-1} \in M$.

Proof of (b), simpler. Since $f \in A(S)$, f^{-1} exists and $\in A(S)$. It remains to show that there is at most finitely many points in S where f^{-1} is nontrivial ($f^{-1}(s) \neq s$). Define D_f as in the previous proof. Take any $s \in S \setminus D_f$. Since $f(s) = s$ and f is a bijection, we know that $f^{-1}(s) = s$. We have proved that $f^{-1}(s) = s$ on all but a finite subset of S , so $f^{-1}(s) \neq s$ on an at most finite subset of S . By definition, $f^{-1} \in M$. $\square$

8. 23. Call an element in S_n a transposition if it interchanges two elements, leaving the others fixed. Show that any element in S_n is a product of transpositions. (This sharpens the result of Problem 22.)

Definition 1 (Union of functions with disjoint domain). If $f : X \rightarrow A$ and $g : Y \rightarrow B$ where $X \cap Y = \emptyset$, we define $f \cup g$ to be the function

$$f \cup g : X \cup Y \longrightarrow A \cup B$$

$$x \longmapsto f \cup g(x) = \begin{cases} f(x) & x \in X \\ g(x) & x \in Y \end{cases}.$$

Lemma 1. For any element f in S_n , we can partition S into k non-empty disjoint closed orbits, where $k \leq n$, which we denote by $R_1, R_2, \dots, R_k$, such that

$$f = f|_{R_1} \cup f|_{R_2} \cup \dots \cup f|_{R_k}.$$

Proof of lemma. We pick an element $x \in S$ and find a closed orbit

$$R_f(x) := \{f^k x : k \in \mathbb{Z}, k \geq 0\} \subset S.$$

$R_f(x)$ is nonempty since $x \in R_f(x)$. We may repeat the process with $S \setminus R_f(x)$ instead of S . The process must terminate because S is finite and we take away at least one element in each repetition. $\square$

Lemma 2. For any element f in $A(R)$ where $|R| = n$, if R is a closed orbit wrt f , f can be decomposed into product of precisely $n - 1$ transpositions.

Proof of lemma. Fix some $x \in R$, and make the list

$$(x, fx, f^2x, \dots, f^{n-1}x).$$

By properties of closed orbit, this is an exhaustive list of elements in R . Now we perform these $n - 1$ transpositions sequentially:

$$(1 \leftrightarrow 2), (1 \leftrightarrow 3), \dots, (1 \leftrightarrow n),$$

where we use integer to represent location on the list. After the first transposition, x moves to the second place, undisturbed thereafter. After $n - 1$ transpositions, $n - 1$ elements have moved to the right spot, and the last item automatically stays at where it belongs. $\square$

Proof. Apply the two lemmas above. $\square$

9. 27. If $f \in A(S)$, call, for $s \in S$, the orbit of s (relative to f) the set $O(s) = \{f^j(s) \mid j \in \mathbb{Z}\}$. Show that if $s, t \in S$, then either $O(s) \cap O(t) = \emptyset$ or $O(s) = O(t)$.

Proof. Assume $O(s) \cap O(t)$ is nonempty, and it suffices to show that $O(s) = O(t)$. Take $x \in O(s) \cap O(t)$. Now $x = f^j(s) = f^k(t)$ for $s, t \in \mathbb{Z}$. Now

$$s = f^{-j} f^j(s) = f^{-j} f^k(t) = f^{k-j}(t).$$

$$\begin{aligned} y \in O(s) &\Leftrightarrow y = f^l(s), l \in \mathbb{Z} \\ &\Leftrightarrow y = f^l(f^{k-j}(t)), l \in \mathbb{Z} \\ &\Leftrightarrow y = f^{l+k-j}(t), l+k-j \in \mathbb{Z} \\ &\Leftrightarrow y \in O(t). \end{aligned}$$

□

Herstein, section 1.5, page 28: Questions 10, 14

10. 10. To check that a given integer $n > 1$ is a prime, prove that it is enough to show that n is not divisible by any prime p with $p \leq \sqrt{n}$.

Proof. We assume $p < n$ throughout the proof. Assume some $p > \sqrt{n}$ divides n . Now we must have $n/p \leq \sqrt{n}$. The contrapositive of this statement is: If all $p \leq \sqrt{n}$ do not divide n , then there does not exist $p > \sqrt{n}$ that divide n . The claim is already proved.

Correction

We assume $p, q < n$ throughout the proof. Assume some $p > \sqrt{n}$ divides n . Now we must have $n/p \leq \sqrt{n}$. Let $q = \frac{n}{p}$, we have $q \mid n$ since $n = pq$. In other words,

$$(\exists p) (p > \sqrt{n} \wedge p \mid n) \Rightarrow (\exists q) (q \leq \sqrt{n} \wedge q \mid n).$$

The contrapositive of this statement is:

$$(\neg \exists q) (q \leq \sqrt{n} \wedge q \mid n) \Rightarrow (\neg \exists p) (p > \sqrt{n} \wedge p \mid n).$$

By trichotomy of order we have $(\forall p)(p \leq \sqrt{n} \vee p > \sqrt{n})$, therefore

$$\begin{aligned} (\neg \exists q) (q \leq \sqrt{n} \wedge q \mid n) &\Rightarrow (\neg \exists q)(q \mid n) \\ &\Rightarrow n \text{ is prime.} \end{aligned}$$

□

11. 14. Adapt the proof of Theorem 1.5.9 to prove:

(a) There is an infinite number of primes of the form $4n + 3$. [\[\[review \]\]](#)

Proof. Suppose there are only a finite number of primes of the form $4n + 3$. Let $p_1 < p_2 < \dots < p_n$ be all the primes of the form $4n + 3$. Now we have all prime numbers less than or equal to p_n , $q_1 < q_2 < \dots < q_m = p_n$. Consider the number $-1 + 2q_1q_2 \dots q_m$. Clearly it is of the form $4n + 3$ (note that $q_1 = 2$), and it is larger than p_n .

For any q_j , this number is not divisible by q_j since this number plus 1 is divisible, i.e. it has a remainder of $q_j - 1$ when divided by q_j . Hence this number is not divisible by **any prime** $\rightarrow$ **any prime smaller than p_n** . (This is not sufficient to show that this number of prime). By Theorem 1.5.7, it is a prime. □

(b) There is an infinite number of primes of the form $6n + 5$.

Proof. Replace ‘ $4n + 3$ ’ with ‘ $6n + 5$ ’ in the previous proof, and note that $q_1 = 2$ and $q_2 = 3$. $\square$

Problem. If we only consider $N = 4p_1p_2 \dots p_n - 1$, how can we be sure that no prime that is not of the form $4n + 3$ divides N ?

Correction

If $N := 4p_1p_2 \dots p_m - 1$ is divisible by some prime factor of the shape $q = 4n + 3$, then the claim is proved since $q \neq p_1, p_2, \dots, p_m$.

Now suppose that N is not divisible by any prime factor of shape $4n + 3$. This means that all its prime factors are of the shape $4n + 1$. Therefore,

$$N \equiv \prod_{\text{factors}} (4k + 1) \equiv 1 \pmod{4}.$$

But clearly we have

$$N \equiv -1 \equiv 3 \pmod{4}$$

which is a contradiction. Hence N has some prime factor of the shape $q = 4n + 3$.

Herstein, section 1.6, page 31: Questions 3,7,12,15.

12. 3. Prove that a set having $n \geq 2$ elements has $\frac{1}{2}n(n-1)$ subsets having exactly two elements.

Proof. Prove by induction. First consider $n = 2$, then the only subset it can have with 2 elements is the set itself. Therefore there’s exactly one such subset, and $\frac{1}{2}n(n-1) = \frac{1}{2}(2)(2-1) = 1$, as desired.

Now let S be a set with n elements, where $n \in \mathbb{N}$, and assume inductively that S has exactly $\frac{1}{2}n(n-1)$ subsets with exactly two elements. Now let $x \notin S$ and consider the set $S \cup \{x\}$ with $n+1$ elements.

For any subset $\{a, b\} \subset S \cup \{x\}$ such that $a \neq b$, we have $\{a, b\} \subset S \Leftrightarrow a \neq x \wedge b \neq x$. Therefore, any subset of $S \cup \{x\}$ that does not contain x is not a "new" subset, i.e. it is also a subset of S . Therefore the new subsets are exactly those containing x and one other element from S .

Hence the number of new subsets is exactly $|S| = n$. Apply the induction hypothesis, we have the number of subsets of $S \cup \{x\}$ with exactly two elements $= \frac{1}{2}n(n-1) + n = \frac{1}{2}n(n+1) = \frac{1}{2}(n+1)((n+1)-1)$, as desired. $\square$

13. 7. If $a \neq 1$, prove that $1 + a + a^2 + \dots + a^n = (a^{n+1} - 1) / (a - 1)$ by induction.

Proof. Use induction. The base case is $n = 0$. In this case, the left hand side equals 1, and the right hand side is

$$\frac{a^{0+1} - 1}{a - 1} = \frac{a - 1}{a - 1} = 1.$$

Now assume inductively that for some fixed n , $1 + a + a^2 + \cdots + a^n = (a^{n+1} - 1) / (a - 1)$ for all $a \neq 0$. Now

$$\begin{aligned} 1 + a + a^2 + \cdots + a^n + a^{n+1} &= (1 + a + a^2 + \cdots + a^n) + a^{n+1} \\ &= (a^{n+1} - 1) / (a - 1) + a^{n+1} \\ &= \frac{a^{n+1} - 1 + a^{n+1}(a - 1)}{a - 1} \\ &= \frac{a^{n+1}a - 1}{a - 1} \\ &= \frac{a^{(n+1)+1} - 1}{a - 1}. \end{aligned}$$

□

14. 12. Prove by induction that a set having n elements has exactly 2^n subsets.

Proof. Base case: a set having 0 element has exactly 1 subset, which is itself (the empty set). Now $2^0 = 1$ as desired.

Let S be some set with n elements, and assume inductively that S has 2^n subsets. Let $x \notin S$ and consider the set $S \cup \{x\}$ with $n + 1$ elements.

We examine the subsets of $S \cup \{x\}$. Now we construct a 1-1 correspondence between subsets containing x and subsets not containing x . For each subset $A \subset S$, the set $A \cup \{x\} \subset S \cup \{x\}$. Conversely, for each subset $B \subset S \cup \{x\}$ such that $x \in B$, we have $B \setminus \{x\} \subset S$.

Since there is a correspondence, there are same number of subsets containing and not containing x . Now that the subsets of S are exactly those subsets of $S \cup \{x\}$ not containing x . Now the number of subsets of $S \cup \{x\}$ is:

$$2^n + 2^n = 2^{n+1}.$$

□

15. 15. Prove by induction that for a set having n elements the number of 1-1 mappings of this set onto itself is $n!$.

Proof. We skip the degenerate case $n = 0$ and start proving at base case $n = 1$. A 1-1 mapping of a singleton set must map the only element to itself, hence there is only one such mapping. Note that $1! = 1$, as desired.

Now consider a set S with n elements, and $x \notin S$. Assume inductively that there are exactly $n!$ 1-1 mappings $S \rightarrow S$. Now for each such mapping $\varphi : S \rightarrow S$ and each element $s \in S$, define

$$\begin{aligned} \varphi_s : S \cup \{x\} &\longrightarrow S \cup \{x\} \\ a &\longmapsto \varphi_s(a) = \begin{cases} x & a = s \\ \varphi(s) & a = x \\ \varphi(a) & \text{otherwise} \end{cases} \end{aligned}$$

and

$$\begin{aligned}\varphi' : S \cup \{x\} &\longrightarrow S \cup \{x\} \\ a &\longmapsto \varphi'(a) = \begin{cases} x & a = x \\ \varphi(a) & \text{otherwise} \end{cases}.\end{aligned}$$

Claim. φ_s is 1-1. Proof: (injectivity) Let a, b be distinct elements from $S \cup \{x\}$. If one of them equals s , say $a = s$, then $\varphi_s(a) = x$. If $b = x$, then $\varphi_s(b) = \varphi(s) \in S$. If $b \neq x$, then $\varphi_s(b) = \varphi(b) \in S$. In both cases $\varphi_s(a) \neq \varphi_s(b)$. If $a, b \in S$, then their images are not equal since φ is 1-1. Surjectivity follows since S is assumed to be finite.

Claim. $((\varphi, s) \mapsto \varphi_s \text{ is injective})$ If φ, s and ψ, t are distinct (either $\varphi \neq \psi$ or $s \neq t$), then $\varphi_s \neq \psi_t$. Proof:

- (a) in the first case, there exists $a \in S$ such that $\varphi(a) \neq \psi(a)$. Now if $a = s$, φ_s and ψ_s map x to different elements; if $a \neq s$, then φ_s and ψ_s map a to different elements.
- (b) in the second case, we assume without loss of generality that $\varphi = \psi$. Now since $s \neq t$, note that $\varphi_s(x) = s \neq t = \psi_t(x)$.

Claim. $((\varphi, s) \mapsto \varphi_s \text{ is surjective})$ For any 1-1 map $f : S \cup \{x\} \rightarrow S \cup \{x\}$ such that $f(x) \neq x$, there exists some $\varphi : S \rightarrow S$ and $s \in S$ such that $f = \varphi_s$. Proof: Define φ by

$$\begin{aligned}\varphi : S &\longrightarrow S \\ x &\longmapsto \varphi(x) = \begin{cases} f(x) & f(s) = x \\ f(s) & \text{otherwise} \end{cases}\end{aligned}$$

and then verify $\varphi_s = f$.

Claim. φ' is 1-1. Proof: (injectivity) Consider $a, b \in S \cup \{x\}$. If both do not equal x then the claim follows from the fact that φ is 1-1. If one of them equals x , say $a = x$, then $\varphi'(a) = \varphi'(x) = x$ and $\varphi'(b) = \varphi(b) \in S$. Hence their images are not equal. Surjectivity follows since S is finite.

Claim. $(\varphi \mapsto \varphi' \text{ is injective})$ For any $\varphi \neq \psi$, $\varphi' \neq \psi'$. Proof: There exists $s \in S$ such that $\varphi(s) \neq \psi(s)$. Now $\varphi'(s) = \varphi(s) \neq \psi(s) = \psi'(s)$.

Claim. $(\varphi \mapsto \varphi' \text{ is surjective})$ For any 1-1 map $f : S \cup \{x\} \rightarrow S \cup \{x\}$ such that $f(x) = x$, there exists some $\varphi : S \rightarrow S$ such that $f = \varphi'$. Proof: define φ as

$$\begin{aligned}\varphi : S &\longrightarrow S \\ x &\longmapsto \varphi(x) = f(x)\end{aligned}$$

and verify that $\varphi' = f$.

By the above claims, the set of all 1-1 mappings on $S \cup \{x\}$ is isomorphic to the set of all the φ_s and φ' . The number of φ_s is $n! \times |S| = n!n$. The number of φ' is $n!$. Summing them up we have $n!n + n! = n!(n + 1) = (n + 1)!$, as desired.

□

Remark. The problems below are not originally assigned. They are actually from section 1.7.

16. 3. Show that $(\bar{z})^{-1} = \overline{(z^{-1})}$.

Proof. For any complex number $z = a + bi$, $z^{-1} = \frac{\bar{z}}{|z|^2}$. Verification:
 $zz^{-1} = \frac{z\bar{z}}{|z|^2} = \frac{|z|^2}{|z|^2} = 1$.

$$\begin{aligned}
 (\bar{z})^{-1} &= \frac{\bar{\bar{z}}}{|\bar{z}|^2} && \text{use the claim above} \\
 &= \bar{\bar{z}} \frac{1}{|\bar{z}|^2} && \text{rewriting} \\
 &= \bar{\bar{z}} \frac{1}{|z|^2} && \text{conjugate of real number is trivial} \\
 &= \bar{z} \frac{1}{|z|^2} && \text{Lemma 1.7.1 (c)} \\
 &= \overline{\left(\frac{\bar{z}}{|z|^2} \right)} && \text{rewriting} \\
 &= \overline{z^{-1}} && \text{use the claim above.}
 \end{aligned}$$

□

17. 7. Verify the commutative law of multiplication $zw = wz$ in $\mathbb{C}$.

Proof. Let $z = a + bi$ and $w = c + di$.

$$\begin{aligned}
 zw &= (a + bi)(c + di) && \text{definition} \\
 &= (a + bi)c + (a + bi)di && \text{distributivity} \\
 &= (ac + bic) + (adi + bidi) && \text{distributivity} \\
 &= ac + bci + adi - bd && \text{associativity, property of } i \\
 &= ca + dai + cbi - db && \text{reals form a commutative ring} \\
 &= (c + di)a + (c + di)bi && \text{distributivity} \\
 &= (c + di)(a + bi) && \text{distributivity} \\
 &= wz.
 \end{aligned}$$

□

18. 12. Prove that $\left(\cos\left(\frac{1}{2}\theta\right) + i \sin\left(\frac{1}{2}\theta\right) \right)^2 = \cos\theta + i \sin\theta$.

Proof.

$$\begin{aligned} \left(\cos\left(\frac{1}{2}\theta\right) + i \sin\left(\frac{1}{2}\theta\right) \right)^2 &= \cos^2 \frac{1}{2}\theta + i^2 \sin^2 \frac{1}{2}\theta + 2i \cos \frac{1}{2}\theta \sin \frac{1}{2}\theta \\ &= \left(\cos^2 \frac{1}{2}\theta - \sin^2 \frac{1}{2}\theta \right) + i \left(2 \cos \frac{1}{2}\theta \sin \frac{1}{2}\theta \right) \\ &= \cos \theta + i \sin \theta. \end{aligned}$$

The last equality follows from the trigonometric identities

$$\cos 2\theta = \cos^2 \theta - \sin^2 \theta, \sin 2\theta = 2 \sin \theta \cos \theta.$$

□

19. 15. Show that $(\cos \theta + i \sin \theta)^r = \cos(r\theta) + i \sin(r\theta)$ for all rational numbers r .

Proof. Let $r = \frac{p}{q}$ where $p, q \in \mathbb{Z}$ and $q \neq 0$.

$$\begin{aligned} (\cos(\theta) + i \sin(\theta))^r &= (\cos \theta + i \sin \theta)^{\frac{p}{q}} \\ &= (\cos p\theta + i \sin p\theta)^{\frac{1}{q}} \quad \text{Integer De Moivre Theorem} \end{aligned}$$

On the other hand,

$$\begin{aligned} (\cos p\theta + i \sin p\theta) &= \cos qr\theta + i \sin qr\theta \\ &= (\cos r\theta + i \sin r\theta)^q \quad \text{Integer De Moivre Theorem} \end{aligned}$$

Hence

$$\begin{aligned} (\cos r\theta + i \sin r\theta) &= (\cos p\theta + i \sin p\theta)^{\frac{1}{q}} \\ &= (\cos(\theta) + i \sin(\theta))^r. \end{aligned}$$

□

Homework 2: Week 2 due Sep 7

Mon 29 Aug 2022 21:32

Herstein, section 2.1, page 47 : Questions 8, 9, 18, 26, 27, 28

1. 8. If G is an abelian group, prove that $(a * b)^n = a^n * b^n$ for all integers n .

Proof. Use induction. Clearly $(a * b)^1 = (a * b) = a^1 * b^1$. Now assume the statement is true for n .

$$\begin{aligned} (a * b)^{n+1} &= (a * b)^n * (a * b) && \text{Notation} \\ &= (a^n * b^n) * (a * b) && \text{Induction Hypothesis} \\ &= (a^n * (b^n * a)) * b && \text{Associativity, multiple times} \\ &= (a^n * (a * b^n)) * b && \text{Commutativity} \\ &= (a^n * a) * (b^n * b) && \text{Associativity, multiple times} \\ &= (a^{n+1}) * (b^{n+1}) && \text{Notation.} \end{aligned}$$

Correction

Need to also show for negative n . The induction only covered nonnegative n .

□

2. 9. If G is a group in which $a^2 = e$ for all $a \in G$, show that G is abelian.

Proof.

$$\begin{aligned}(ab)(ba) &= a(bb)a \\ &= aea \\ &= aa \\ &= e \\ &= (ab)(ab).\end{aligned}$$

By left cancellation, $ab = ba$.

□

3. 18. If G is a finite group of even order, show that there must be an element $a \neq e$ such that $a = a^{-1}$. (Hint: Try to use the result of Problem 17.)

Proof. Suppose G has elements $e, a_1, a_2, \dots, a_n$ where e is the identity element, and all elements listed are distinct. Suppose for all element $a \neq e$, $a \neq a^{-1}$. a_1^{-1} cannot be e , otherwise $a_1 = e$ which contradicts the assumption that elements are distinct. By our supposition, $a_1^{-1} \neq a_1$. Hence it must be some element in $a_2, \dots, a_n$. Without loss of generality, we may call this element a_2 .

Now $a_2^{-1} = (a_1^{-1})^{-1} = a_1$ by Exercise 17. Continue in this fashion, we see that a_3 must have an inverse in $a_4, \dots, a_n$, which we without loss of generality call a_4 . In each step we consume two elements of $a_1, \dots, a_n$. For the process to terminate without contradiction, n must be even. Therefore $e, a_1, \dots, a_n$ has an odd number of elements. Since it is assumed that G has even order, we have a contradiction. □

4. 26. If G is a finite group, prove that, given $a \in G$, there is a positive integer n , depending on a , such that $a^n = e$.

Proof. Consider the orbit $O(a) = \{a^n : n \in \mathbb{Z}\}$. Since $O(a) \subset G$, it is finite. By pigeonhole principle, there exists distinct $n, m \in \mathbb{Z}$ such that $a^n = a^m$. Hence $a^n = a^m a^{n-m} = a^m$, so $a^{n-m} = e$. Also, $a^{m-n} = (a^{n-m})^{-1} = e^{-1} = e$. It is assumed that n, m are distinct, so $n - m \neq 0$ and $m - n \neq 0$. By trichotomy of order, one of them must be positive. □

5. 27. In Problem 26, show that there is an integer $m > 0$ such that $a^m = e$ for all $a \in G$.

Proof. For each element $a \in G$ we have $n_a \in \mathbb{N}$ where $a^{n_a} = e$ by the last exercise. For any $N \in \mathbb{N}$ such that $n_a | N$, we have $a^N = a^{cn_a} = (a^{n_a})^c = e^c = e$, where $c \in \mathbb{N}$ and $N = cn_a$.

Now define $N = \prod_{a \in G} n_a$. Since G is finite, N is finite. Clearly for each $a \in G$, $N = n_a \prod_{b \in G, b \neq a} n_b$, hence $n_a | N$. Therefore $a^N = e$. □

6. 28. Let G be a set with an operation $*$ such that: 1. G is closed under $*$. 2. $*$ is associative. 3. There exists an element $e \in G$ such that $e * x = x$ for all $x \in G$. 4. Given $x \in G$, there exists a $y \in G$ such that $y * x = e$. Prove that G is a group. (Thus you must show that $x * e = x$ and $x * y = e$ for e, y as above.) [\[\[review \]\]](#)

Proof. We use property 4 to pick a left inverse $f(x)$ for each element $x \in G$, where $f(x) * x = e$. We use a right-associative convention to avoid cumbersome notation, so $x * y * z$ always means $x * (y * z)$ rather than $(x * y) * z$.

First we show that e is also a right identity.

$$\begin{aligned}
 x * e &= x * f(x) * x && \text{left inverse} \\
 &= e * x * f(x) * x && \text{left identity} \\
 &= (f(f(x)) * f(x)) * x * f(x) * x && \text{left inverse} \\
 &= f(f(x)) * (f(x) * x) * f(x) * x && \text{associativity} \\
 &= f(f(x)) * e * f(x) * x && \text{left inverse} \\
 &= f(f(x)) * (e * f(x)) * x && \text{associativity} \\
 &= f(f(x)) * f(x) * x && \text{left identity} \\
 &= (f(f(x)) * f(x)) * x && \text{associativity} \\
 &= e * x && \text{left inverse} \\
 &= x && \text{left identity.}
 \end{aligned}$$

Then we prove the left inverse is also a right inverse.

$$\begin{aligned}
 x * f(x) &= e * (x * f(x)) && \text{left identity} \\
 &= (f(f(x)) * f(x)) * (x * f(x)) && \text{left inverse} \\
 &= f(f(x)) * ((f(x) * x) * f(x)) && \text{associativity} \\
 &= f(f(x)) * (e * f(x)) && \text{left inverse} \\
 &= f(f(x)) * f(x) && \text{left identity} \\
 &= e && \text{left inverse.}
 \end{aligned}$$

□

Herstein, section 2.2, page 50: Questions 1, 3, 5

7. 1. Suppose that G is a set closed under an associative operation such that
 1) given $a, y \in G$, there is an $x \in G$ such that $ax = y$, and
 2) given $a, w \in G$, there is a $u \in G$ such that $ua = w$.
 Show that G is a group. [\[\[review \]\]](#)

Proof. Pick any element $a \in G$. Use property 2 to obtain an element $e \in G$ such that $ea = a$. We verify that e is a left identity. Pick any element $b \in G$, by property 1, there exists $c \in G$ such that $b = ac$. Now

$$eb = e(ac) = (ea)c = ac = b.$$

Next we verify the existence of left inverse. For any element b , by property 2, there exists an element b^{-1} such that $b^{-1}b = e$. Now we are in the same situation as the last exercise, so this exercise can be considered solved. $\square$

8. 3. If G is a group in which $(ab)^i = a^i b^i$ for three consecutive integers i , prove that G is abelian.

Proof. Take a natural number n and it is assumed that $(ab)^i = a^i b^i$ holds for $i = n, n+1, n+2$.

$$\begin{aligned}
 (ab)^{n+1} &= ab(ab)^n && \text{associativity} \\
 (ab)^{n+1} &= ab(a^n b^n) && \text{assumption} \\
 (ab)^{n+1} &= a^{n+1} b^{n+1} && \text{assumption} \\
 aba^n b^n &= a^{n+1} b^{n+1} && \text{equivalence transitivity} \\
 ba^n b^n &= a^n b^{n+1} && \text{left cancellation} \\
 ba^n &= a^n b && \text{right cancellation.}
 \end{aligned}$$

Hence a^n commutes with b . Make the same argument with $(ab)^{n+2}$, we conclude that a^{n+1} commutes with b . Now

$$ba^{n+1} = ba^n a = a^n ba = a^{n+1} b.$$

Take the last equality and apply left cancellation, we have $ba = ab$. $\square$

9. 5. Let G be a group in which $(ab)^3 = a^3 b^3$ and $(ab)^5 = a^5 b^5$ for all $a, b \in G$. Show that G is abelian.

Proof.

$$\begin{aligned}
 (ab)^3 &= a^3 b^3 && \text{assumption} \\
 &= a(a^2 b^2)b && \text{associativity, line 1} \\
 &= ababab && \text{notation} \\
 &= a(baba)b && \text{associativity, line 3} \\
 &= a((ba)(ba))b && \text{associativity, line 4} \\
 &= a((ba)^2)b && \text{notation, line 5.}
 \end{aligned}$$

Hence, $a(a^2 b^2)b = a((ba)^2)b$. By left and right cancellation,

$$a^2 b^2 = (ba)^2. \tag{1}$$

Apply a similar argument to $(ab)^5$:

$$\begin{aligned}
 (ab)^5 &= a^5 b^5 && \text{assumption} \\
 &= a(a^4 b^4)b && \text{associativity} \\
 &= ababababab && \text{notation} \\
 &= a(babababa)b && \text{associativity} \\
 &= a((ba)(ba)(ba)(ba))b && \text{associativity} \\
 &= a((ba)^4)b && \text{notation.}
 \end{aligned}$$

Hence, $a(a^4b^4)b = a((ba)^4)b$. By left and right cancellation,

$$a^4b^4 = (ba)^4 \quad (2)$$

Now

$$\begin{aligned} a^2(b^2a^2)b^2 &= (a^2b^2)(a^2b^2) && \text{associativity} \\ &= (ba)^2(ba)^2 && \text{eqn. (1)} \\ &= (ba)^4 && \text{notation} \\ &= a^4b^4 && \text{eqn. (2)} \\ &= a^2(a^2b^2)b^2 && \text{associativity.} \end{aligned}$$

Hence, $a^2(b^2a^2)b^2 = a^2(a^2b^2)b^2$. By left and right cancellation,

$$b^2a^2 = a^2b^2. \quad (3)$$

Now

$$\begin{aligned} (ab)^4 &= (ab)^2(ab)^2 && \text{associativity} \\ &= (b^2a^2)(b^2a^2) && \text{eqn. (1)} \\ &= (a^2b^2)(a^2b^2) && \text{eqn. (3)} \\ &= a^2(b^2a^2)b^2 && \text{associativity} \\ &= a^2(a^2b^2)b^2 && \text{eqn. (3)} \\ &= (a^2a^2)(b^2b^2) && \text{associativity} \\ &= a^4b^4 && \text{associativity and notation.} \end{aligned}$$

Now we see that $(ab)^i = a^ib^i$ is true for $i = 3, 4, 5$, which are three consecutive integers. By using the last exercise we complete the proof. $\square$

Herstein, section 2.3, page 54: Questions 3, 12, 14, 24, 26, 29

10. 3. Let S_3 be the symmetric group of degree 3. Find all the subgroups of S_3 .

Solution. There are 6 elements in S_3 , namely

- (a) $(123) = e$, the identity mapping
- (b) (132) , that maps x_1 to x_1 , x_2 to x_3 , and x_3 to x_2
- (c) (231)
- (d) (213)
- (e) (312)
- (f) (321)

We first consider subgroup of order 1. Clearly the identity element forms a trivial subgroup. Any other element cannot form a subgroup because it

is not the identity of S_3 (since the identity of a group must be the identity of its subgroup).

Next we consider subgroup of order 2. This can only be the identity element with an inversion (an element a such that $a^2 = e$). There are three elements besides e that satisfy this property:

$$(a) \quad (132)(132) = e$$

$$(b) \quad (321)(321) = e$$

$$(c) \quad (213)(213) = e$$

Therefore there are three subgroups of order 2: $\{e, (132)\}$, $\{e, (321)\}$, and $\{e, (213)\}$

Finally, subgroups of order 3. We write such a group as $\{e, a, b\}$ where e is the identity. Note that $ab \neq a$, otherwise we have $ab = ae$ and $b = e$ by cancellation, which contradicts the assumption that a, b, e are distinct. By the same reasoning $ab \neq b$, therefore $ab = e$. By the same reasoning we have $ba = e$. Therefore $b = a^{-1}$. Hence we are looking at two distinct elements from S_3 that are each other's inverse. There is only one such pair in S_3 :

$$(a) \quad \{(231), (312)\}$$

Therefore $\{e, (231), (312)\}$ is a subgroup of S_3 with order 3.

By Lagrange's Theorem, there's no other proper subgroup with higher order, since 4 and 5 do not divide 6. The last subgroup would be order 6, which is S_3 itself. $\square$

11. 12. Prove that a cyclic group is abelian.

Proof. Take some cyclic group G . Because G is cyclic, there exists some element $x \in G$ such that $G = \langle x \rangle$. This means for any element $a, b \in G$ there exists $i, j \in \mathbb{Z}$ such that $a = x^i, b = x^j$. Now

$$\begin{aligned} ab &= x^i x^j \\ &= x^{i+j} \\ &= x^{j+i} \\ &= x^j x^i \\ &= ba. \end{aligned}$$

$\square$

12. 14. If G has no proper subgroups, prove that G is cyclic.

Proof. We prove the contrapositive. Assume G is not cyclic. Then for any $a \in G$, we have $\langle a \rangle \neq G$, otherwise G is by definition cyclic. But $\langle a \rangle \leq G$ (as proved on p.53 of the textbook), so $\langle a \rangle < G$. This is true for arbitrary a , so we can pick a non-identity a so that $a \in \langle a \rangle$ and $e \in \langle a \rangle$, making $\langle a \rangle$ nontrivial. Now $\langle a \rangle$ is a proper subgroup of G . $\square$

13. 24. If H is a subgroup of G , let $N = \cap_{x \in G} x^{-1} H x$. Prove that N is a subgroup of G such that $y^{-1} N y = N$ for every $y \in G$.

Proof. First we prove that N is a subgroup of G . Take any $n, m \in N$. Let $x \in G$ be arbitrary. Then there exists h_n and h_m such that $n = x^{-1}h_nx$ and $m = x^{-1}h_mx$. Now

$$\begin{aligned} nm &= (x^{-1}h_nx)(x^{-1}h_mx) \\ &= x^{-1}h_n(xx^{-1})h_mx \\ &= x^{-1}h_nh_mx \\ &= x^{-1}(h_nh_m)x \\ &\in x^{-1}Hx. \end{aligned}$$

Because H is a subgroup of G , $h_nh_m \in H$, so the last line holds. Because x is arbitrary, $nm \in \bigcap_{x \in G} x^{-1}Hx = N$. Hence N is closed under multiplication.

Also, consider

$$\begin{aligned} (x^{-1}h_n^{-1}x)n &= (x^{-1}h_n^{-1}x)(x^{-1}h_nx) \\ &= x^{-1}h_n^{-1}(xx^{-1})h_nx \\ &= x^{-1}h_n^{-1}h_nx \\ &= x^{-1}(h_n^{-1}h_n)x \\ &= x^{-1}x \\ &= e \\ &= n^{-1}n. \end{aligned}$$

Because H is a group, $h_n^{-1} \in H$ exists. By cancellation we have $n^{-1} = x^{-1}h_n^{-1}x \in x^{-1}Hx$. Because x is arbitrary, $n^{-1} \in N$. We have verified that any element in N has a inverse. By Lemma 2.3.1, N is a subgroup of G .

Fix some $y \in G$. Now we verify $y^{-1}Ny \subset N$. Take any $n \in y^{-1}Ny$ and any $x \in G$. Now there exists some $h_n \in H$ such that

$$(xy^{-1})^{-1}h_n(xy^{-1}) = n.$$

Now

$$\begin{aligned} y^{-1}ny &= y^{-1}((xy^{-1})^{-1}h_n(xy^{-1}))y \\ &= y^{-1}((yx^{-1})h_n(xy^{-1}))y \\ &= (y^{-1}y)x^{-1}h_nx(y^{-1}y) \\ &= ex^{-1}h_nxe \\ &= x^{-1}h_nx \\ &\in x^{-1}Hx. \end{aligned}$$

Since x is arbitrary, $y^{-1}ny \in N$.

Fix some $w \in G$. Finally we verify $N \subset w^{-1}Nw$. Take any $n \in N$, then

$$\begin{aligned} n &= (w^{-1}w)n(w^{-1}w) \\ &= w^{-1}(wnw^{-1})w \\ &= w^{-1}\left((w^{-1})^{-1}n(w^{-1})\right)w. \end{aligned}$$

Now $w^{-1} \in G$ by existence of group inverse. Because $y^{-1}Ny \subset N$ for all $y \in G$,

$$\left((w^{-1})^{-1}n(w^{-1})\right) \in N.$$

Therefore

$$n \in w^{-1}Nw.$$

□

14. 26. Let G be a group, H a subgroup of G . Let $Hx = \{hx \mid h \in H\}$. Show that, given $a, b \in G$, then $Ha = Hb$ or $Ha \cap Hb = \emptyset$.

Proof. Fix $a, b \in G$ and assume $Ha \cap Hb$ is nonempty. It suffices to show that $Ha = Hb$. Take an element $x \in Ha \cap Hb$. Now there $\exists h_1, h_2 \in H$ such that $h_1a = h_2b = x$.

Now we show that $Ha \subset Hb$, and $Hb \subset Ha$ will follow by symmetry. Take $y \in Ha$. By definition of Ha , there exists $h_y \in H$ such that $h_ya = y$. Because H is a group, the $h_yh_1^{-1}h_1$ is an element of H (by closure and existence of inverse) and equals h_y (by associativity, identity and existence of inverse). Now

$$\begin{aligned} h_ya &= (h_yh_1^{-1}h_1)a \\ &= h_yh_1^{-1}(h_1a) \\ &= h_yh_1^{-1}(h_2b) \\ &= (h_yh_1^{-1}h_2)b. \end{aligned}$$

By group closure of H , the element $(h_yh_1^{-1}h_2) \in H$. Hence $y = h_ya \in Hb$. □

15. 29. If M is a subgroup of G such that $x^{-1}Mx \subset M$ for all $x \in G$, prove that actually $x^{-1}Mx = M$

Proof. For all $x \in G$, define

$$\begin{aligned} f_x : M &\longrightarrow M \\ y &\longmapsto f_x(y) = x^{-1}yx. \end{aligned}$$

In the definition above we used the assumption that $f_x(M) = x^{-1}Mx \subset M$.

We prove that f_x is injective. For $y, z \in M$, whenever $f_x(y) = f_x(z)$, we

have

$$\begin{aligned}
 f_x(y) &= f_x(z) \\
 x^{-1}yx &= x^{-1}zx \\
 x(x^{-1}yx) &= x(x^{-1}zx) \\
 (xx^{-1})yx &= (xx^{-1})zx \\
 eyx &= ezx \\
 yx &= zx \\
 yxx^{-1} &= zxx^{-1} \\
 y(xx^{-1}) &= z(xx^{-1}) \\
 ye &= ze \\
 y &= z.
 \end{aligned}$$

Now we prove that f_x is surjective. Fix any $y \in M$. Using the inverse axiom, we have $x^{-1} \in M$. Using the assumption, we have $(x^{-1})^{-1} M x^{-1} \subset M$, hence $(x^{-1})^{-1} y x^{-1} \in M$. Now

$$\begin{aligned}
 &f_x\left((x^{-1})^{-1} y x^{-1}\right) \\
 &= x^{-1} \left((x^{-1})^{-1} y x^{-1} \right) x \\
 &= \left(x^{-1} (x^{-1})^{-1} \right) y (x^{-1} x) \\
 &= eye \\
 &= y.
 \end{aligned}$$

Since f_x is both injective and surjective, it is bijective. Therefore it is a 1-1 mapping from M to itself. In other words, $f_x(M) = x^{-1} M x = M$.

Correction

1-1 means injective. onto means surjective. 1-1 correspondence means bijective.

□

Homework 3: Week 3 due Sep 14

Tue 06 Sep 2022 14:42

Herstein, section 2.4, page 64 : Questions 8,13,16,18,27,31,37

1. 8. If every right coset of H in G is a left coset of H in G , prove that $aHa^{-1} = H$ for all $a \in G$

Proof. We interpret the assumption as: For any right coset Ha in G , $Ha = aH$. Note that this also implies every left coset of H in G is a right coset. Now

$$\begin{aligned}
 x &\in aHa^{-1} \\
 &\Leftrightarrow x \in a(Ha^{-1}) \\
 &\Leftrightarrow x \in a(a^{-1}H) \\
 &\Leftrightarrow x \in (aa^{-1})H \\
 &\Leftrightarrow x \in H.
 \end{aligned}$$

□

Correction

The assumption is: For any right coset Ha in G , there exists $b \in G$ such that $Ha = bH$.

Since $Ha = bH$, we conclude that $b \in Ha$, so $ba^{-1} \in H$, so $ab^{-1} \in H$.

Now $aH = a(b^{-1}b)H = (ab^{-1})bH = (ab^{-1})Ha = Ha$.

2. 13. Find the orders of all the elements of U_{18} . Is U_{18} cyclic?

Proof. $U_{18} = \{[1], [5], [7], [11], [13], [17]\}$. We find order of each element:

$$\begin{aligned}
 \langle [1] \rangle &= \{[1]\} \\
 \langle [5] \rangle &= \{[5], [7], [17], [13], [11], [1]\} \\
 \langle [7] \rangle &= \{[7], [13], [1]\} \\
 \langle [11] \rangle &= \{[11], [13], [17], [7], [5], [1]\} \\
 \langle [13] \rangle &= \{[13], [7], [1]\} \\
 \langle [17] \rangle &= \{[17], [1]\}.
 \end{aligned}$$

Hence, they have order 1, 6, 3, 6, 3, 2 respectively. The group U_{18} is hence cyclic since it is generated by $[5]$ and $[11]$. □

3. 16. If G is a finite abelian group and $a_1, \dots, a_n$ are all its elements, show that $x = a_1a_2 \cdots a_n$ must satisfy $x^2 = e$.

Proof. Without loss of generality we assume $a_1, \dots, a_n$ are distinct. Let $a'_1a'_2 \dots a'_n$ be a permutation of $a_1a_2 \dots a_n$. Because G is finite and abelian, permutation does not change the product. Hence $x^2 = (a_1a_2 \dots a_n)(a_1a_2 \dots a_n) = (a_1a_2 \dots a_n)(a'_1a'_2 \dots a'_n)$.

Now by the inverse axiom, there must exist some $a \in G$ such that $a_1a = e$. Without loss of generality we can call it a'_1 . Hence

$$(a_1a_2 \dots a_n)(a'_1a'_2 \dots a'_n) = (a_1a'_1)(a_2 \dots a_n)(a'_2 \dots a'_n) = (a_2 \dots a_n)(a'_2 \dots a'_n).$$

Then we consider the inverse of a_2 . It cannot be a'_1 , otherwise $a_2 = (a'_1)^{-1} = (a_1^{-1})^{-1} = a_1$, a contradiction. Without loss of generality we may call its inverse $a'_2 = a_2^{-1}$. Now

$$(a_2 \dots a_n)(a'_2 \dots a'_n) = (a_2a'_2)(a_3 \dots a_n)(a'_3 \dots a'_n) = (a_3 \dots a_n)(a'_3 \dots a'_n).$$

Continue in this fashion we can cancel all terms and conclude that $x^2 = e$. □

Proof. (simpler)

The inverse

$$\begin{aligned} i : G &\longrightarrow G \\ a &\longmapsto i(a) = a^{-1} \end{aligned}$$

is a bijection. Since G is abelian,

$$(a_1 a_2 \dots a_n) = (a_1^{-1} a_2^{-1} \dots a_n^{-1}).$$

Now

$$\begin{aligned} x^2 &= (a_1 a_2 \dots a_n)(a_1 a_2 \dots a_n) \\ &= (a_1 a_2 \dots a_n)(a_1^{-1} a_2^{-1} \dots a_n^{-1}) \\ &= (a_1 a_1^{-1}) \dots (a_n a_n^{-1}) \\ &= e. \end{aligned}$$

□

4. 18. Using the results of Problems 15 and 16, prove that if p is an odd prime number, then $(p-1)! \equiv -1 \pmod{p}$. (This is known as Wilson's Theorem.) It is, of course, also true if $p = 2$. [\[\[review \]\]](#)

Proof. Consider the group U_p . Since p is prime, the elements of the group are exactly the natural numbers from 1 up to $p-1$. Now

$$(p-1)! = 1 \cdot 2 \cdot \dots \cdot (p-1) = a_1 a_2 \dots a_n = x.$$

By Exercise 16, $x^2 = 1$. By Exercise 15, any member y of U_p that is not 1 or $p-1$ does not satisfy $y^2 \equiv 1 \pmod{p}$. Note that:

$$y^2 \equiv 1 \pmod{p} \Leftrightarrow y^2 = e \Leftrightarrow y = y^{-1}.$$

Therefore all members except 1 or $p-1$ are not their own inverse. Hence they must be pairs that are inverse of each other. Because U_p has even order, and there are even number of elements excluding 1 and $p-1$, the remaining elements can all pair up correctly. Because U_p is abelian, we may group the pairs together and cancel them out. Now

$$(p-1)! = 1 \cdot (p-1) \equiv -1 \pmod{p}.$$

□

5. 27. If $aH = bH$ forces $Ha = Hb$ in G , show that $aHa^{-1} = H$ for every $a \in G$. [\[\[review \]\]](#)

Proof. Note that

$$\begin{aligned} aH &= bH \\ \Leftrightarrow a^{-1}b &\in H && \text{by definition of left coset} \\ \Leftrightarrow b^{-1}a &\in H. \end{aligned}$$

Using the assumption we also have

$$Ha = Hb \Leftrightarrow ab^{-1} \in H.$$

We have proved that for all $a, b \in G$,

$$b^{-1}a \in H \Rightarrow ab^{-1} \in H.$$

Therefore, for all $a, b \in G$,

$$ab \in H \Leftrightarrow ba \in H.$$

Fix any $a \in G$. We pick arbitrary $h \in H$ and let $b = ah$. Now $a^{-1}b = a^{-1}ah = h \in H$. Hence $ba^{-1} \in H$. Since $a(a^{-1}b) \in aH$, we have

$$\begin{aligned} a(a^{-1}b) &= (aa^{-1})b \\ &= b \\ &= b(a^{-1}a) \\ &= (ba^{-1})a \\ &\in Ha. \end{aligned}$$

Since h is picked arbitrarily, b can take all elements from aH . Since $b \in Ha$, we have shown that $aH \subset Ha$. Apply the same argument to the other side symmetrically, we obtain

$$aH = Ha$$

which means $aHa^{-1} = H$. □

6. 31. If $o(a) = m$ and $a^s = e$, prove that $m \mid s$.

Proof. We first introduce

Lemma 3. For a cyclic group with order n and generating element a , we have for all integers q and r ,

$$a^q = a^r \Leftrightarrow q \equiv r \pmod{n}.$$

Proof of lemma.

$$\begin{aligned} a^q = a^r &\Leftrightarrow (a^q)(a^r)^{-1} = e \\ &\Leftrightarrow a^{q-r} = e. \end{aligned}$$

Now we apply euclidean's algorithm to $q-r$, and obtain x and y such that $0 \leq y < n$, and

$$q - r = nx + y.$$

Now

$$a^{q-r} = a^{nx+y} = (a^n)^x a^y = e^x a^y = a^y = e.$$

Since $0 \leq y < n$, by properties of cyclic group, $y = 0$ (All elements $e, a^1, \dots, a^n$ must be distinct, so $a^y = e$ implies $y = 0$). Hence

$$q - r = nx \equiv 0 \pmod{n}.$$

Now we prove the other direction, and assume $q \equiv r \pmod{n}$. Apply euclidean algorithm to $q - r$. We obtain integers x, y such that $0 \leq y < n$ and

$$q - r = nx + y.$$

We have $q - r \equiv y \pmod{n}$. Since $q - r \equiv 0 \pmod{n}$, $y \equiv 0 \pmod{n}$. Because it is assumed that $0 \leq y < n$, we have $y = 0$. Hence $q - r = nx$. Now

$$\begin{aligned} a^{q-r} &= a^{nx} \\ &= (a^n)^x \\ &= e^x \\ &= e. \end{aligned}$$

Hence $a^q = a^r$. $\square$

By assumption we have $o(a) = |\langle a \rangle| = m$. By Theorem 2.4.5, $a^m = e$. Apply the last lemma. Since $a^s = e = a^m$,

$$s \equiv m \pmod{m}.$$

Hence

$$s \equiv 0 \pmod{m}.$$

Which implies $m|s$. $\square$

7. 37. In a cyclic group of order n , show that for each positive integer m that divides n (including $m = 1$ and $m = n$) there are $\varphi(m)$ elements of order m . [\[\[review \]\]](#)

Proof. An element $a^k \in \langle a \rangle = G$ has order m iff

- (a) $(a^k)^m = a^{km} = e$. In other words, $n|km$, and
- (b) m is the smallest positive integer such that km is a multiple of n . In other words, for any $p < m$, we have $n \nmid kp$.

Note that $1 \leq m \leq n$ (since m divides n), and $0 \leq k \leq n-1$ (since a^k are distinct elements of G).

We introduce a claim that is central to the proof, which we prove later:

Claim. The two conditions above are met, if and only if

$$(k, n) = \frac{n}{m}.$$

Note that

$$(k, n) = \frac{n}{m} \Leftrightarrow \left(\frac{k}{n/m}, \frac{n}{n/m} \right) = \left(\frac{k}{n/m}, m \right) = 1.$$

Hence $o(a^k) = m \Leftrightarrow \left(\frac{k}{n/m}, m \right) = 1$. Since $k < n$, $\frac{k}{n/m} < m$. There are precisely $\varphi(m)$ choices of $\frac{k}{n/m}$ such that $\left(\frac{k}{n/m}, m \right) = 1$ and $\frac{k}{n/m} < m$. Each choice of $\frac{k}{n/m}$ then uniquely corresponds to a choice of k . $\square$

Notation. For integer m and prime x , write $d_x(m)$ for the *multiplicity* of x in the unique prime factorization of m . For simplicity of notation, we may write $d_x(m, n)$ for $d_x((m, n))$.

Lemma 4. (Properties of Multiplicity) The following statements hold:

- (a) $d_x(n) \geq 0$.
- (b) $d_x(n) \geq a \Leftrightarrow x^a | n$.
- (c) $d_x(nm) = d_x(n) + d_x(m)$.
- (d) $d_x\left(\frac{n}{m}\right) = d_x(n) - d_x(m)$ if $m|n$.
- (e) $d_x(m, n) = \min(d_x(m), d_x(n))$.
- (f) If $m|n$, then for any prime x one has $d_x(m) \leq d_x(n)$. Equivalently, if $d_x(m) > d_x(n)$ then $m \nmid n$.
- (g) $m|n$ if and only if for all prime x such that $x|m$, $d_x(m) \leq d_x(n)$.

Proof. (a) Explicitly stated in the definition of multiplicity.

- (b) Apply unique factorization to n . Then there exists primes $p_1, p_2, \dots, p_k$ (all distinct), where k is a natural number, and positive integers $a_1, \dots, a_k$. We have

$$n = p_1^{a_1} p_2^{a_2} \dots p_k^{a_k}.$$

Without loss of generality, let $x = a_1$. If $d_x(n) \geq a$, then $a_1 \geq a$. Hence there exists a nonnegative integer c such that $a_1 = a + c$. Now

$$n = x^{a_1} p_2^{a_2} \dots p_k^{a_k} = x^{a+c} p_2^{a_2} \dots p_k^{a_k} = x^a (x^c p_2^{a_2} \dots p_k^{a_k})$$

Therefore $x^a | n$.

Now assume $x^a | n$ and we want to show $d_x(n) \geq a$. Since $p_2^{a_2}, \dots, p_k^{a_k}$ are all prime factors other than x , we have

$$x^a | p_1^{a_1}.$$

Hence $x^{a_1} \geq x^a$. Since $x > 0$, we have $d_x(n) = a_1 \geq a$, as desired.

- (c) We apply unique factorization to both n and m . Then there exists primes $p_1, p_2, \dots, p_k$ (all distinct), primes $q_1, q_2, \dots, q_l$ (all distinct), where k, l are natural numbers, and positive integers $a_1, \dots, a_k, b_1, \dots, b_l$. We have

$$\begin{aligned} n &= p_1^{a_1} p_2^{a_2} \dots p_k^{a_k} \\ m &= q_1^{b_1} q_2^{b_2} \dots q_l^{b_l}. \end{aligned}$$

Now choose some x out of the p s and q s. Without loss of generality we let $x = p_1$. We have two cases.

Case 1: $q_1, \dots, q_l \neq x$. Now

$$nm = p_1^{a_1} p_2^{a_2} \dots p_k^{a_k} q_1^{b_1} q_2^{b_2} \dots q_l^{b_l}.$$

$p_2, \dots, p_k, q_1, \dots, q_l$ are not equal to p_1 . Hence $d_x(nm) = a_1$. Since $d_x(m) = 0$, we have $d_x(nm) = d_x(n) + d_x(m)$.

Case 2: There exists some positive integer $i \leq l$ such that $q_i = x$. Without loss of generality, we assume $i = 1$. Now we use commutativity to obtain

$$\begin{aligned} nm &= p_1^{a_1} p_2^{a_2} \dots p_k^{a_k} q_1^{b_1} q_2^{b_2} \dots q_l^{b_l} \\ &= p_1^{a_1} q_1^{b_1} p_2^{a_2} \dots p_k^{a_k} q_2^{b_2} \dots q_l^{b_l} \\ &= x^{a_1} x^{b_1} p_2^{a_2} \dots p_k^{a_k} q_2^{b_2} \dots q_l^{b_l} \\ &= x^{a_1+b_1} p_2^{a_2} \dots p_k^{a_k} q_2^{b_2} \dots q_l^{b_l}. \end{aligned}$$

Now $d_x(nm) = a_1 + b_1 = d_x(n) + d_x(m)$, as desired.

- (d) Apply $\frac{n}{m}$ and m to the last property. The assumption $m|n$ ensures that $\frac{n}{m}$ is an integer.

- (e) We have $(m, n) \mid m$ and $(m, n) \mid n$. Hence $d_x(m, n) \leq d_x(m)$ and $d_x(m, n) \leq d_x(n)$. Hence $d_x(m, n) \leq \min(d_x(m), d_x(n))$. Without loss of generality, let $\min(d_x(m), d_x(n)) = d_x(m)$. Suppose $d_x(m, n) < d_x(m)$, then there exists a positive integer c , such that $d_x(m, n) + c = d_x(m)$. Now

$$x^{d_x(m)} = x^{d_x(m, n) + c} \mid m.$$

$x^{d_x(m, n) + c} \mid n$ also because $d_x(n) \geq d_x(m)$. This implies $(m, n)x^c \mid m$ and $(m, n)x^c \mid n$, which is a contradiction since (m, n) is the greatest common divisor.

We have ruled out all other possibilities, so $d_x(m, n) = d_x(m) = \min(d_x(m), d_x(n))$.

- (f) Fix prime x . Use property (b) and taking $a = d_x(m)$, we have $x^{d_x(m)} \mid m$. By transitivity of divisibility, $x^{d_x(m)} \mid n$. Use property (b) again, we have $d_x(n) \geq d_x(m)$.
- (g) suppose $d_x(m) \leq d_x(n)$ for all primes x such that $x \mid m$. We write m in terms of its unique factorization:

$$m = q_1^{a_1} q_2^{a_2} \dots q_r^{a_r}$$

where $q_1, \dots, q_r$ are distinct primes, and $a_1, \dots, a_r$ are positive integers.

Now $a_1 = d_{q_1}(m) \leq d_{q_1}(n)$, hence $q_1^{a_1} \mid n$. Continue in this fashion, we see that $q_2^{a_2}, \dots, q_r^{a_r}$ all divide n . Therefore, $m \mid n$.

We now prove the other direction. Assume $m \mid n$, and let x be a prime such that $x \mid m$. Because $m \mid n$, there exists positive integer c such that $n = cm$. Hence

$$d_x(n) = d_x(cm) = d_x(c) + d_x(m) \geq d_x(m).$$

□

Lemma 5. If $p < m$, then there exists a prime $x > 1$ such that $x \mid m$ and $d_x(m) > d_x(p)$.

Proof. Fix p and m . Suppose for contradiction that for all $x \mid m$ and $x > 1$, we have $d_x(m) \leq d_x(p)$. Then apply property (f) above, we obtain that $m \mid p$. This means $p \geq m$, a contradiction to the assumption $p < m$. □

Proof of claim. Suppose $(k, n) = \frac{n}{m}$. Hence $(km, nm) = n$. This means that $n \mid km$.

Suppose there exists positive integer p such that $p < m$ and $n \mid kp$. Use the above lemma to choose prime x such that $x > 1$, $x \mid m$ and $d_x(m) > d_x(p)$. We claim that

$$d_x(n) = d_x(m) + d_x(k).$$

To verify this, since $(k, n) = \frac{n}{m}$, we have

$$\min(d_x(k), d_x(n)) = d_x(n) - d_x(m).$$

Do some arithmetic manipulations and obtain

$$d_x(m) = \max(0, d_x(n) - d_x(k)).$$

Since $x|m$, $d_x(m) \geq 1$, therefore

$$d_x(m) = d_x(n) - d_x(k).$$

Now

$$d_x(pk) = d_x(p) + d_x(k) < d_x(m) + d_x(k) = d_x(n).$$

Hence $n \nmid pk$.

For the other direction, assume that a^k has order m and hence satisfies the two conditions. We first introduce

Lemma 6. For any positive integer n, m, k , we have

$$n|mk \Rightarrow n|m(k, n).$$

Proof. Let x be any prime. Since $n|mk$,

$$d_x(n) \leq d_x(mk) = d_x(m) + d_x(k).$$

Now

$$\begin{aligned} d_x(m(k, n)) &= d_x(m) + d_x(k, n) \\ &= d_x(m) + \min(d_x(k), d_x(n)). \end{aligned}$$

When $\min(d_x(k), d_x(n)) = d_x(k)$, $d_x(m(k, n)) = d_x(m) + d_x(k) \geq d_x(n)$. When $\min(d_x(k), d_x(n)) = d_x(n)$, obviously we have $d_x(m(k, n)) = d_x(m) + d_x(n) \geq d_x(n)$. Hence in both cases, $d_x(m(k, n)) \geq d_x(n)$. Since this is true for all x , we have $n|m(k, n)$. $\square$

Then since $n|mk$, we apply the previous lemma and conclude that $n|m(k, n)$. Hence there exists positive integer c such that $cn = m(k, n)$. We claim that $c = 1$, which will conclude the proof.

To verify that $c = 1$, suppose for the sake of contradiction that $c > 1$. Now $(k, n) = c \frac{n}{m}$ since $m|n$. Hence c is a factor of both k and n , and we have

$$\left(\frac{k}{c}, \frac{n}{c} \right) = \frac{n}{m}.$$

Hence in particular

$$\frac{n}{m} \mid \frac{k}{c}.$$

Hence

$$n \mid m \frac{k}{c}$$

where $\frac{k}{c} < k$, a contradiction to the condition that k is the smallest integer such that $n|mk$. $\square$

Herstein, section 2.5, page 73 : Questions 2,6,7

8. 2. Recall that $G \simeq G'$ means that G is isomorphic to G' . Prove that for all groups G_1, G_2, G_3 :
- (a) $G_1 \simeq G_1$.
 - (b) $G_1 \simeq G_2$ implies that $G_2 \simeq G_1$.
 - (c) $G_1 \simeq G_2, G_2 \simeq G_3$ implies that $G_1 \simeq G_3$.

Proof. (a) Take the identity mapping $\iota_G : G \rightarrow G$. Now ι is clearly bijective, and

$$\iota_G(a_1 a_2) = a_1 a_2 = \iota_G(a_1) \iota_G(a_2).$$

- (b) Since $G_1 \simeq G_2$, there exists bijective mapping $\varphi : G_1 \rightarrow G_2$ such that for all $a, b \in G_1$, $\varphi(ab) = \varphi(a)\varphi(b)$. Now consider the inverse mapping φ^{-1} . We know that a bijective mapping has a bijective inverse, so φ^{-1} is bijective. For all elements $x, y \in G_2$, let $a = \varphi^{-1}(x)$ and $b = \varphi^{-1}(y)$. Now $\varphi(a) = x$ and $\varphi(b) = y$. Now

$$\begin{aligned} \varphi^{-1}(xy) &= \varphi^{-1}(\varphi(a)\varphi(b)) \\ &= \varphi^{-1}(\varphi(ab)) \\ &= ab \\ &= \varphi^{-1}(\varphi(a))\varphi^{-1}(\varphi(b)) \\ &= \varphi^{-1}(x)\varphi^{-1}(y). \end{aligned}$$

- (c) Since $G_1 \simeq G_2$, there exists bijective mapping $\varphi : G_1 \rightarrow G_2$ such that for all $a, b \in G_1$, $\varphi(ab) = \varphi(a)\varphi(b)$. Since $G_2 \simeq G_3$, there exists bijective mapping $\psi : G_2 \rightarrow G_3$ such that for all $a, b \in G_2$, $\psi(ab) = \psi(a)\psi(b)$.

Now consider the composition $\psi \circ \varphi : G_1 \rightarrow G_3$. Clearly it is bijective since it's composition of two bijections. Now

$$\begin{aligned} \psi \circ \varphi(ab) &= \psi(\varphi(ab)) \\ &= \psi(\varphi(a)\varphi(b)) \\ &= \psi(\varphi(a))\psi(\varphi(b)) \\ &= \psi \circ \varphi(a)\psi \circ \varphi(b). \end{aligned}$$

□

9. 6. Prove that if $\varphi : G \rightarrow G'$ is a homomorphism, then $\varphi(G)$, the image of G , is a subgroup of G' .

Proof. (a) (Closure) Take $x, y \in \varphi(G)$. Then there exists $a, b \in G$ such that $\varphi(a) = x$ and $\varphi(b) = y$. Now

$$xy = \varphi(a)\varphi(b) = \varphi(ab) \in \varphi(G).$$

- (b) (Associativity) Inherited from G .

(c) (Identity) Let $e \in G$ be the identity of G . Take $x \in \varphi(G)$. Now

$$\varphi(e)x = \varphi(e)\varphi(a) = \varphi(ea) = \varphi(a) = x.$$

Similarly, $x\varphi(e) = x$. Hence $\varphi(e)$ is an identity of $\varphi(G)$.

(d) (Inverse) Take $x \in \varphi(G)$. Then $x = \varphi(a)$ for some $a \in G$. Now

$$\varphi(a)(\varphi(a^{-1})) = (\varphi(a^{-1}))\varphi(a) = \varphi(aa^{-1}) = \varphi(e).$$

Hence $\varphi(a^{-1})$ is a inverse for $\varphi(a)$.

□

Proof. (Simpler). Take $\varphi(a), \varphi(b) \in \varphi(G)$. We want to show $\varphi(a)\varphi(b)^{-1} \in \varphi(G)$. It suffices to show $\varphi(b)^{-1} = \varphi(b^{-1})$, then

$$\varphi(a)\varphi(b)^{-1} = \varphi(a)\varphi(b^{-1}) = \varphi(ab^{-1}) \in \varphi(G).$$

To show this, note that

$$\varphi(b)\varphi(b^{-1}) = \varphi(bb^{-1}) = \varphi(e).$$

Also, $\varphi(e)$ is the identity for $\varphi(G)$ since for any $c \in G$,

$$\varphi(e)\varphi(c) = \varphi(ec) = \varphi(c) = \varphi(ce) = \varphi(c)\varphi(e).$$

□

10. 7. Show that $\varphi : G \rightarrow G'$, where φ is a homomorphism, is a monomorphism if and only if $\text{Ker } \varphi = (e)$.

Proof. Let e, e' be identity of G and G' respectively. If φ is a monomorphism, then the mapping is injective. For any element $a \in G$ such that $a \neq e$, we must have $\varphi(a) \neq \varphi(e) = e'$, hence $a \notin \text{Ker } \varphi$.

If φ is not a monomorphism, then there exists two distinct elements $a, b \in G$ such that $\varphi(a) = \varphi(b)$. Now $\varphi(ba^{-1}) = \varphi(b)\varphi(a^{-1}) = \varphi(a)\varphi(a^{-1}) = \varphi(aa^{-1}) = \varphi(e) = e'$. But $ba^{-1} \neq e$ since $a \neq b$. We have constructed a nonidentity element that is also in $\text{Ker } \varphi$. □

Homework 4: Homework 4 due Sep 21

Sat 17 Sep 2022 09:52

Herstein, section 2.5, page 74 : Questions 12,15,20,26,34,40,44

1. 12. Prove that if $Z(G)$ is the center of G , then $Z(G) \triangleleft G$.

Proof. $Z(G)$ is defined as

$$Z(G) = \{z \in G : zx = xz \text{ for all } x \in G\}.$$

From previous exercises we know that $Z(G)$ is a subgroup of G . It remains to verify that $Z(G)$ is a normal subgroup, i.e.,

$$\forall g \in G, g^{-1}Z(G)g \subset Z(G).$$

Take $g^{-1}zg \in g^{-1}Z(G)g$. Now

$$\begin{aligned} g^{-1}zg &= (g^{-1}z)g \\ &= (zg^{-1})g \\ &= z(g^{-1}g) \\ &= ze \\ &= z \\ &\in Z(G). \end{aligned}$$

Hence $Z(G)$ is a normal subgroup of G . $\square$

2. 15. If G is any group, $N \triangleleft G$, and $\varphi : G \rightarrow G'$ a homomorphism of G onto G' , prove that the image, $\varphi(N)$, of N is a normal subgroup of G' .

Proof. We have verified previously that $\varphi(N) \leq G'$. It remains to verify that $\varphi(N) \triangleleft G'$. We have to show that for $\forall x \in G'$, $x^{-1}\varphi(N)x \subset \varphi(N)$.

Take any $n \in N$. Fix some $x \in G'$. Because φ is onto, there exists $g \in G$ such that $\varphi(g) = x$. Now

$$\begin{aligned} x^{-1}\varphi(n)x &= \varphi(g)^{-1}\varphi(n)\varphi(g) \\ &= \varphi(g^{-1})\varphi(n)\varphi(g) \\ &= \varphi(g^{-1}n)\varphi(g) \\ &= \varphi/ng^{-1})\varphi(g) \\ &= \varphi(n)\varphi(g^{-1})\varphi(g) \\ &= \varphi(n)\varphi(g)^{-1}\varphi(g) \\ &= \varphi(n)e' \\ &= \varphi(n) \\ &\in \varphi(N). \end{aligned}$$

Thus for $\forall n \in N$ and $\forall x \in G'$, we have $x^{-1}\varphi(n)x \in \varphi(N)$. Therefore $x^{-1}\varphi(N)x \subset \varphi(N)$, as desired. $\square$

Correction

Note: Normal subgroup does not commute with the group elements of G element-wise. It does commute as a whole group. In particular $Ng = gN$, but it is not true in general that $ng = gn$ for all n .

$$\begin{aligned} x^{-1}\varphi(n)x &= \varphi(g)^{-1}\varphi(n)\varphi(g) \\ &= \varphi(g^{-1})\varphi(n)\varphi(g) \\ &= \varphi(g^{-1}n)\varphi(g) \\ &= \varphi(g^{-1}ng) \\ &\in \varphi(g^{-1}Ng) \\ &= \varphi(N). \end{aligned}$$

3. 20. If $M \triangleleft G$, $N \triangleleft G$, and $M \cap N = \{e\}$, show that for $m \in M, n \in N, mn = nm$.

Proof. $m^{-1}nm \in m^{-1}Nm \subset N$, hence $m^{-1}nm \in N$, therefore $m^{-1}nmn^{-1} \in N$.

$nmn^{-1} = (n^{-1})^{-1}mn^{-1} \in (n^{-1})^{-1}Mn^{-1} \subset M$, hence $nmn^{-1} \in M$, therefore $m^{-1}nmn^{-1} \in M$.

Therefore $m^{-1}nmn^{-1} \in N \cap M$. Because it is assumed that $N \cap M = \{e\}$, we have $m^{-1}nmn^{-1} = e$. It follows that $nm = mn$. $\square$

4. 26. If G is a group and $a \in G$, define $\sigma_a : G \rightarrow G$ by $\sigma_a(g) = aga^{-1}$. We saw in Example 9 of this section that σ_a is an isomorphism of G onto itself, so $\sigma_a \in A(G)$, the group of all 1-1 mappings of G (as a set) onto itself. Define $\psi : G \rightarrow A(G)$ by $\psi(a) = \sigma_a$ for all $a \in G$. Prove that:
(a) ψ is a homomorphism of G into $A(G)$.

Proof. We only need to verify

$$\varphi(ab) = \varphi(a)\varphi(b)$$

for all $a, b \in G$. Take any $g \in G$. Now

$$\begin{aligned} \varphi(ab)g &= \sigma_{ab}(g) \\ &= (ab)g(ab)^{-1} \\ &= a(bgb^{-1})a^{-1} \\ &= \sigma_a(bgb^{-1}) \\ &= \sigma_a(\sigma_b(g)) \\ &= (\sigma_a \circ \sigma_b)(g) \\ &= (\varphi(a)\varphi(b))(g). \end{aligned}$$

Since this is true for all $g \in G$, we have $\varphi(ab) = \varphi(a)\varphi(b)$. $\square$

- (b) $\text{Ker } \psi = Z(G)$, the center of G .

Proof.

$$\begin{aligned} g \in Z(G) &\Leftrightarrow \forall x \in G, xg = gx \\ &\Leftrightarrow \forall x \in G, gxg^{-1} = x \\ &\Leftrightarrow \forall x \in G, \sigma_g(x) = x \\ &\Leftrightarrow \sigma_g = \iota_G \\ &\Leftrightarrow \varphi(g) = \iota_G \\ &\Leftrightarrow g \in \text{ker } \varphi. \end{aligned}$$

Where ι_G is the identity map, which serves as the identity element in $A(G)$. $\square$

5. 34. Let G be a group, $\mathcal{A}(G)$ the group of all automorphisms of G . (See Problem 28.) Let $I(G) = \{\sigma_a \mid a \in G\}$, where σ_a is as defined in Problem 26. Prove that $I(G) \triangleleft \mathcal{A}(G)$.

Proof. We first prove $I(G) \leq \mathcal{A}(G)$. We verify that $(\sigma_b)^{-1} = \sigma_{b^{-1}}$.

$$\begin{aligned} (\sigma_b \sigma_{b^{-1}})g &= \sigma_b(b^{-1}g(b^{-1})^{-1}) \\ &= \sigma_b(b^{-1}gb) \\ &= b(b^{-1}gb)b^{-1} \\ &= (bb^{-1})g(bb^{-1}) \\ &= ege \\ &= g. \end{aligned}$$

Hence $\sigma_b \sigma_{b^{-1}} = \iota_G$, which is the identity in $\mathcal{A}(G)$. Similarly, $\sigma_{b^{-1}} \sigma_b = \iota_G$. Therefore $\sigma_{b^{-1}}$ is the inverse of σ_b . Now

$$\begin{aligned} \sigma_a(\sigma_b)^{-1}(g) &= \sigma_a(\sigma_{b^{-1}})g \\ &= \sigma_a(b^{-1}g(b^{-1})^{-1}) \\ &= \sigma_a(b^{-1}gb) \\ &= a(b^{-1}gb)a^{-1} \\ &= (ab^{-1})g(ab^{-1})^{-1} \\ &= \sigma_{ab^{-1}}(g). \end{aligned}$$

Hence $\sigma_a(\sigma_b)^{-1} = \sigma_{ab^{-1}} \in I(G)$. By subgroup criterion, $I(G)$ is a subgroup of $\mathcal{A}(G)$.

Now we prove $I(G) \triangleleft \mathcal{A}(G)$. It suffices to show that for all $\varphi \in \mathcal{A}(G)$, $\varphi^{-1}I(G)\varphi \subset I(G)$. Take any $a \in G$ and $\varphi \in \mathcal{A}(G)$,

$$\begin{aligned} (\varphi^{-1}\sigma_a\varphi)(g) &= \varphi^{-1}(\sigma_a(\varphi(g))) \\ &= \varphi^{-1}(a\varphi(g)a^{-1}) \\ &= \varphi^{-1}(a)\varphi^{-1}(\varphi(g))\varphi^{-1}(a^{-1}) && \text{since } \varphi^{-1} \text{ is homomorphism} \\ &= \varphi^{-1}(a)g\varphi^{-1}(a^{-1}) && \text{since } \varphi^{-1}\varphi = \iota_G \\ &= \varphi^{-1}(a)g(\varphi^{-1}(a))^{-1} && \text{by property of homomorphism} \\ &= \sigma_{\varphi^{-1}(a)}(g). \end{aligned}$$

Hence $\varphi^{-1}\sigma_a\varphi = \sigma_{\varphi^{-1}(a)} \in I(G)$. Therefore $\varphi^{-1}I(G)\varphi \subset I(G)$. $\square$

6. 38. Let G be a group and H a subgroup of G . Let $S = \{Ha \mid a \in G\}$ be the set of all right cosets of H in G . Define, for $b \in G$, $T_b : S \rightarrow S$ by $T_b(Ha) = Hab^{-1}$.

Note

This exercise is proven in extra because it is used in another proof.

(a) Prove that $T_b T_c = T_{bc}$ for all $b, c \in G$ [therefore the mapping $\psi : G \rightarrow A(S)$ defined by $\psi(b) = T_b$ is a homomorphism].

Proof.

$$\begin{aligned}
 T_b T_c(Ha) &= Tb(Tc(Ha)) \\
 &= Tb((Ha)c^{-1}) \\
 &= (Hac^{-1})b^{-1} \\
 &= Ha(bc)^{-1} \\
 &= T_{bc}.
 \end{aligned}$$

□

(b) Describe $\text{Ker } \psi$, the kernel of $\psi : G \rightarrow A(S)$.

Proof. Denote the identity of $A(S)$ as ι .

$$\begin{aligned}
 a \in \ker \varphi & \\
 \Leftrightarrow \varphi(a) &= \iota \\
 \Leftrightarrow T_a &= \iota \\
 \Leftrightarrow \forall b \in G, T_a(Hb) &= Hb \\
 \Leftrightarrow \forall b \in G, Hba^{-1} &= Hb \\
 \Leftrightarrow \forall b \in G, bab^{-1} &\in H.
 \end{aligned}$$

Hence $a \in H$, otherwise we can take $b = e$ and arrive at a contradiction. Also, $\ker \varphi$ has a normal-like property, that any group conjugation cannot make it larger than H . We put this precisely in the next proof. □

(c) Show that $\text{Ker } \psi$ is the largest normal subgroup of G lying in H [largest in the sense that if $N \triangleleft G$ and $N \subset H$, then $N \subset \text{Ker } \psi$].

Proof. We already know that $\ker \varphi \subset H$ and $\ker \varphi \leq G$. We first show that $\ker \varphi \triangleleft G$. Take $a \in \ker \varphi$ and $g \in G$. Now

$$\begin{aligned}
 g^{-1}ag \in \ker \varphi & \\
 \Leftrightarrow \varphi(g^{-1}ag) &= \iota \\
 \Leftrightarrow \forall b \in G, T_{g^{-1}ag}(Hb) &= Hb \\
 \Leftrightarrow \forall b \in G, Hb(g^{-1}a^{-1}g) &= Hb \\
 \Leftrightarrow \forall b \in G, Hbg^{-1}a^{-1} &= Hbg^{-1} \\
 \Leftrightarrow \forall b \in G, T_a(Hbg^{-1}) &= Hbg^{-1} \\
 \Leftrightarrow \forall b \in G, \iota(Hbg^{-1}) &= Hbg^{-1}.
 \end{aligned}$$

The last statement is true because ι is the identity in S .

Now suppose $N \triangleleft G$ and $N \subset H$. Take $n \in N$, and we need to show $n \in \ker \varphi$.

$$\begin{aligned}
 n \in \ker \varphi & \\
 \Leftrightarrow \varphi(n) = T_n &= \iota \\
 \Leftrightarrow \forall b, Hbn^{-1} &= Hb \\
 \Leftrightarrow \forall b, Hb &= Hbn \\
 \Leftrightarrow \forall b, Hb &= nHb && n \text{ commutes with } Hb \text{ since } N \text{ is normal} \\
 \Leftrightarrow \forall b, Hb &= Hb && nH = H \text{ since } n \in H.
 \end{aligned}$$

The last statement is true, hence $n \in \ker \varphi$. $\square$

7. 40. If G is a finite group, H a subgroup of G such that $n \nmid i_G(H)!$ where $n = |G|$, prove that there is a normal subgroup $N \neq (e)$ of G contained in H .

Proof. We use everything from exercise 38, and define S, φ and $A(S)$ in the same way. We have established that $\ker \varphi$ is a normal subgroup of G lying in H . Let $N = \ker \varphi$. It remains to show that $N \neq (e)$.

Suppose the opposite, that $N = (e)$. Then this means that φ is injective, so $|\varphi(G)| = |G| = n$. Now by Lagrange's Theorem, $|\varphi(G)| \mid |A(S)|$. But note that $|A(S)| = |S|! = i_G(H)!$. So we have

$$n \mid i_G(H)!$$

which is a contradiction with the assumption. Therefore we must have $N \neq (e)$. $\square$

8. 44. Prove that a group of order p^2 , p a prime, has a normal subgroup of order p . [\[\[review \]\]](#)

Proof. Let the group be G . We claim that there exists a subgroup of G with order p . Take $a \in G$ where $a \neq e$. By Lagrange theorem, we have $o(a) \mid |G|$. Since $|G| = p^2$ where p prime, we have $o(a) = 1$, $o(a) = p$, or $o(a) = p^2$. Since $a \neq e$, $o(a) \neq 1$.

Now assume $o(a) = p^2$, then G is cyclic. Use exercise 37 from Chapter 2.4 taking $m = p$, we have that there are exactly $\varphi(p) = p$ elements in G of order p . Take any of these elements and it will generate a cyclic subgroup of order p . But since G is cyclic, it is abelian, so every subgroup is a normal subgroup, so we are done.

If $o(a) = p$ then $\langle a \rangle \leq G$. We may assume in addition that there are no other element with order p^2 , otherwise it will fall to the case we just proved. We now prove that $\langle a \rangle \triangleleft G$. Now apply Exercise 40, and let $H = \langle a \rangle$. Observe that

$$p^2 \nmid p! = (p^2/p)! = (|G|/|H|)! = (|G/H|)! = i_G(H).$$

Hence there is a nontrivial normal subgroup N of G contained in H . Hence $|N| \mid |G|$ and $1 < |N| \leq |H| = p$. But this forces $|N| = p$.

□

Herstein, section 2.6, page 83: Questions 7,11,13,14,15

9. 7. If G is a cyclic group and N is a subgroup of G , show that G/N is a cyclic group.

Proof. Take some generator a of G . We claim that $G/N = \langle Na \rangle$.

We first prove $G/N \supset \langle Na \rangle$. For any integer m ,

$$(Na)^m = N(a^m).$$

And since $a^m \in G$, $N(a^m)$ is a right coset of N , hence $N(a^m) \in G/N$.

Next we prove $G/N \subset \langle Na \rangle$. Take any $Nb \in G/N$. Since G is cyclic, there exists some integer m such that $b = a^m$. Now

$$Nb = N(a^m) = (Na)^m \in \langle Na \rangle.$$

□

10. 11. If G is a group and $Z(G)$ the center of G , show that if $G/Z(G)$ is cyclic, then G is abelian.

Proof. Take some generator of $G/Z(G)$, say $Z(G)a$. Therefore, for all $b \in G$, there exists some integer m such that $(Z(G)a)^m = Z(G)a^m = Z(G)b$. This is to say that for all $b \in G$, there exists integer m such that $ba^{-m} \in Z(G)$.

Now take arbitrary $x, y \in G$. Then there exists integers n_x and n_y such that $xa^{-n_x} \in Z(G)$ and $ya^{-n_y} \in Z(G)$. We have

$$\begin{aligned} xy &= ((xa^{-n_x})a^{n_x})((ya^{-n_y})a^{n_y}) \\ &= (xa^{-n_x})a^{n_x}(ya^{-n_y})a^{n_y} \\ &= (xa^{-n_x})a^{n_x}a^{n_y}(ya^{-n_y}) \\ &= (xa^{-n_x})a^{n_x+n_y}(ya^{-n_y}) \\ &= (ya^{-n_y})a^{n_y+n_x}(xa^{-n_x}) \\ &= (ya^{-n_y})a^{n_y}a^{n_x}(xa^{-n_x}) \\ &= (ya^{-n_y})a^{n_y}(xa^{-n_x})a^{n_x} \\ &= ((ya^{-n_y})a^{n_y})((xa^{-n_x})a^{n_x}) \\ &= yx. \end{aligned}$$

□

11. 13. If G is a group and $N \triangleleft G$ is such that

$$aba^{-1}b^{-1} \in N$$

for all $a, b \in G$, prove that G/N is abelian.

Proof. Take arbitrary $a, b \in G$. Now

$$\begin{aligned} aba^{-1}b^{-1} &\in N \\ \Leftrightarrow ab(ba)^{-1} &\in N \\ \Leftrightarrow N(ab) &= N(ba) \\ \Leftrightarrow NaNb &= NbNa. \end{aligned}$$

□

12. 14. If G is an abelian group of order $p_1 p_2 \cdots p_k$, where $p_1, p_2, \dots, p_k$ are distinct primes, prove that G is cyclic. (See Problem 15.)

Proof. Since $p_1 \mid |G|$ and p_1 prime, by Cauchy's Theorem, there exists an element $a_1 \in G$ such that $o(a_1) = p_1$. Similarly, we can choose $a_2, \dots, a_k \in G$ such that $o(a_2) = p_2, \dots, o(a_k) = p_k$.

Now use exercise 15. Since p_1 and p_2 are relative prime, there exists an element $a_{12} \in G$ such that $o(a_{12}) = p_1 p_2$. Now $p_1 p_2$ is relatively prime to p_3 , so apply exercise 15 again we obtain an element $a_{123} \in G$ such that $o(a_{123}) = (p_1 p_2) p_3$. Continue in this fashion and we can obtain an element with order $p_1 p_2 \cdots p_k = |G|$, showing G cyclic. □

13. 15. If G is an abelian group and if G has an element of order m and one of order n , where m and n are relatively prime, prove that G has an element of order mn .

Proof. Let $a, b \in G$ such that $o(a) = m$ and $o(b) = n$. We claim that $o(ab) = mn$.

Suppose $(ab)^k = a^k b^k = e$ for some $k \in \mathbb{Z}$. Then $a^k = (b^k)^{-1}$. However we know that two orbits are either coincident or almost disjoint (either $\langle a \rangle = \langle b \rangle$ or $\langle a \rangle \cap \langle b \rangle = \{e\}$). Since $\langle a \rangle$ and $\langle b \rangle$ have different order, they cannot be coincident. Hence their intersection contains only e . From $a^k = (b^k)^{-1}$ we know that $a^k \in \langle a \rangle \cap \langle b \rangle$, so $a^k = e$, whence $b^k = e$. Now $m \mid k$ and $n \mid k$, so $mn \mid k$.

Now we verify $(ab)^{mn} = e$. This follows easily since $(ab)^{mn} = (a^m)^n (b^n)^m = e^n e^m = e$. Also, for any positive integer $k < mn$, we must have $mn \nmid k$, therefore $(ab)^k = e$ cannot be true. Hence mn is the smallest positive integer k such that $(ab)^k = e$. By definition, $o(ab) = mn$. □

Homework 5: Week 5 due Sep 28

Mon 19 Sep 2022 16:11

Herstein, section 2.6, page 84: Questions 16,18

1. 16. Let G be an abelian group of order $p^n m$, where p is a prime and $p \nmid m$. Let $P = \{a \in G \mid a^{p^k} = e \text{ for some } k \text{ depending on } a\}$. Prove that: (a) P is a subgroup of G .

Proof. Associativity is inherited from G . We first verify closure of P . Take $a, b \in P$. Then there exists integers k, l such that $a^{p^k} = b^{p^l} = e$. Now

$$\begin{aligned} (ab)^{p^{k+l}} &= a^{p^{k+l}} b^{p^{k+l}} \\ &= a^{p^k p^l} b^{p^k p^l} \\ &= (a^{p^k})^{p^l} (b^{p^l})^{p^k} \\ &= e^{p^l} e^{p^k} \\ &= e. \end{aligned}$$

Hence $ab \in P$. Next we verify identity. Take arbitrary k and we see that

$$e^{p^k} = e.$$

Hence $e \in P$. Finally, for any element $a \in P$, we verify that $a^{-1} \in P$. Let k be the integer such that $a^{p^k} = e$. Now

$$\begin{aligned} (a^{-1})^{p^k} &= a^{-p^k} \\ &= (a^{p^k})^{-1} \\ &= e^{-1} \\ &= e. \end{aligned}$$

Hence $a^{-1} \in P$. We have verified all group properties in the definition, so P is a subgroup of G . $\square$

(b) G/P has no elements of order p .

Proof. Suppose there exists $a \in G$ such that $o(Pa) = p$. Then $(Pa)^p = P(a^p) = P$, hence $a^p \in P$. By definition of P , there exists integer k such that $(a^p)^{p^k} = e$. But this means $a^{p^{k+1}} = e$, whence $a \in P$. Therefore $Pa = P$, meaning $o(Pa) = 1$, so $p = 1$. However it is assumed that $p \nmid m$, but $1 \mid m$, so we have a contradiction. Therefore G/P does not have element of order p . $\square$

(c) $|P| = p^n$.

Proof. Suppose $p \nmid |G/P|$. G/P is abelian and finite (since G is abelian), so there exists an element of order p by Cauchy's Theorem, a contradiction to the last statement. Hence one must have $p^n \mid |P|$.

Now suppose $m \nmid |P|$. We have $|P| = p^n m = |G|$, so $P = G$. Take some prime divisor $q > 0$ such that $q \mid m$. Thus $q \mid |P|$. Then there exists an element $a \in G$ of order q by Cauchy's Theorem. Also, there exists k such that $a^{p^k} = e$. Hence $q \mid p^k$. But it is assumed that $p \nmid m$, so we cannot have $q \mid p^k$ (otherwise $q = p$, and $q \mid m$, so $p \mid m$), a contradiction. Therefore $m \nmid |P|$. Combining all we have, we conclude $|P| = p^n$. $\square$

2. 18. Let G be an abelian group (possibly infinite) and let the set $T = \{a \in G \mid a^m = e, m > 1 \text{ depending on } a\}$. Prove that: (a) T is a subgroup of G .

Proof. Associativity is inherited from G . We first verify closure of T . Take $a, b \in T$. Then there exists positive integers m, n such that $a^m = b^n = e$. Now

$$(ab)^{mn} = a^{mn} b^{mn} = (a^m)^n (b^n)^m = e^n e^m = e.$$

Hence $ab \in T$, since $mn > 1$.

Clearly the identity $e \in T$ since $e^2 = e$. For inverse, note that

$$(a^{-1})^m = a^{-m} = (a^m)^{-1} = e^{-1} = e.$$

Hence $a^{-1} \in T$.

We have verified all group properties, so T is a subgroup of G . $\square$

(b) G/T has no element other than its identity element of finite order.

Proof. Suppose $a \in G$ such that Ta has order $n > 1$ in G/T . Then

$$(Ta)^n = Ta^n = T.$$

Hence $a^n \in T$. Thus there exists positive integer m such that $(a^n)^m = a^{nm} = e$. But this implies $a \in T$, hence $Ta = T$, so $o(Ta) = 1$, a contradiction. $\square$

Herstein, section 2.7, page 87: Questions 4,6,7

3. 4. If G_1, G_2 are two groups and $G = G_1 \times G_2 = \{(a, b) \mid a \in G_1, b \in G_2\}$, where we define $(a, b)(c, d) = (ac, bd)$, show that: (a) $N = \{(a, e_2) \mid a \in G_1\}$, where e_2 is the unit element of G_2 , is a normal subgroup of G

Proof. I use the established result that direct product of two groups is a group.

Define the mapping

$$\begin{aligned}\varphi : G_1 &\longrightarrow G \\ a &\longmapsto \varphi(a) = (a, e_2).\end{aligned}$$

And note that

$$\varphi(a)\varphi(b) = (a, e_2)(b, e_2) = (ab, e_2e_2) = (ab, e_2) = \varphi(ab).$$

Hence φ is a homomorphism. Note that $N = \varphi(G_1)$, so N is a subgroup of G . To check normality, let $g = (a, b) \in G$ and $n = (c, e_2) \in N$. Now

$$\begin{aligned}g^{-1}ng &= (a, b)^{-1}(c, e_2)(a, b) \\ &= (a^{-1}, b^{-1})(c, e_2)(a, b) \\ &= (a^{-1}ca, b^{-1}e_2b) \\ &= (a^{-1}ca, b^{-1}b) \\ &= (a^{-1}ca, e_2) \\ &\in N.\end{aligned}$$

□

(b) $N \simeq G_1$.

Proof. Define the mapping

$$\begin{aligned}\varphi : G_1 &\longrightarrow N \\ a &\longmapsto \varphi(a) = (a, e_2).\end{aligned}$$

To check well-definedness, note that for any $a \in G$, the element (a, e_2) is in N as per definition of N .

To check that φ is a homomorphism, note that

$$\varphi(a)\varphi(b) = (a, e_2)(b, e_2) = (ab, e_2e_2) = (ab, e_2) = \varphi(ab).$$

To check that φ is injective, assume $\varphi(a) = \varphi(b)$. Now $(a, e_2) = (b, e_2)$. But they are equal if and only if $a = b$.

To check that φ is surjective, take any $(a, e_2) \in N$. Then $\varphi(a) = (a, e_2)$.

Hence φ is an isomorphism from G_1 to N . Hence $G_1 \cong N$. □

(c) $G/N \simeq G_2$.

Proof. Define the mapping

$$\begin{aligned}\varphi : G_2 &\longrightarrow G/N \\ b &\longmapsto \varphi(b) = N(e_1, b).\end{aligned}$$

It is clear that φ is well-defined. To check that φ is a homomorphism, note

$$\varphi(a)\varphi(b) = N(e_1, a)N(e_1, b) = N(e_1, a)(e_1, b) = N(e_1, ab) = \varphi(ab).$$

To check that φ is injective, assume $\varphi(a) = \varphi(b)$. Hence $N(e_1, a) = N(e_1, b)$. Hence $(e, ab^{-1}) \in N$. But this means $ab^{-1} = e_2$ by definition of N . Therefore $a = b$.

To check that φ is surjective, let $(a, b) \in G$ and we claim that $\varphi(b) = N(a, b)$. But note that

$$N(a, b) = N(a, e_2)(e_1, b) = N(e_1, b)$$

where the second inequality holds true since $(a, e_2) \in N$. Hence $\varphi(b) = N(e_1, b)$ from definition of φ .

Now we have found a bijective homomorphism $\varphi : G_2 \rightarrow G/N$, so $G_2 \cong G/N$. Therefore $G/N \cong G_2$. $\square$

4. *6. If G is a group and $N \triangleleft G$, show that if $a \in G$ has finite order $o(a)$, then Na in G/N has finite order m , where $m \mid o(a)$. (Prove this by using the homomorphism of G onto G/N .)

Proof. Define

$$\begin{aligned}\varphi : G &\longrightarrow G/N \\ a &\longmapsto \varphi(a) = Na.\end{aligned}$$

We have verified that φ is a surjective homomorphism. Since $o(a) = n$ is finite,

$$(Na)^n = \varphi(a)^n = \varphi(a^n) = \varphi(e) = N.$$

Hence $o(Na)$ is finite and $o(Na) \mid n$.

The last statement uses homework 3 problem 31, saying that for any group G and $a \in G$, if $a^s = e$ then $o(a) \mid s$. $\square$

5. 7. If φ is a homomorphism of G onto G' and $N \triangleleft G$, show that $\varphi(N) \triangleleft G'$.

Proof. We have already verified in previous lectures that homomorphisms send subgroup to subgroup, i.e. $\varphi(N) \leq G'$.

Now take arbitrary $\varphi(n) \in \varphi(N)$ and $x \in G'$. We want to prove $\varphi(n)^{-1}x\varphi(n) \in G'$.

Since φ is surjective, there exists $g \in G$ such that $\varphi(g) = x$. Now

$$\begin{aligned}\varphi(n)^{-1}x\varphi(n) &= \varphi(n)^{-1}\varphi(g)\varphi(n) \\ &= \varphi(n^{-1})\varphi(g)\varphi(n) \\ &= \varphi(n^{-1}gn) \\ &\in \varphi(n^{-1}Gn) \\ &= \varphi(G) \\ &= G' .\end{aligned}$$

$\square$

Herstein, section 3.2, page 117 , Questions 3,9,11,14,17,20,23

6. 3. Express as the product of disjoint cycles and find the order. (a)
 $\begin{pmatrix} 1 & 2 & 3 & 5 & 7 \\ & 2 & 4 & 7 & 6 \end{pmatrix}$.

Solution. $(1, 2, 4)(3, 5, 7, 6)$. The order is 12. □

- (b) $\begin{pmatrix} 1 & 2 \\ & 1 & 3 \end{pmatrix} \begin{pmatrix} 1 & 4 \end{pmatrix}$.

Solution. $(1, 4, 3, 2)$. The order is 4. □

- (c) $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ & 1 & 2 & 3 & 4 & 6 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 & 4 & 7 \end{pmatrix}$.

Solution. $(1, 4, 7, 3, 6, 2, 5)$. It has order 7. □

- (d) $\begin{pmatrix} 1 & 2 & 3 \\ & 1 & 3 & 2 \end{pmatrix}$.

Solution. $(1)(2)(3)$. It has order 1. □

- (e) $\begin{pmatrix} 1 & 2 & 3 \\ & 3 & 5 & 7 & 9 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \end{pmatrix}^{-1}$.

Solution. $(1, 5, 7, 9)$. It has order 4. □

- (f) $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ & & & & \end{pmatrix}^3$.

Solution. $(1, 4, 2, 5, 3)$. It has order 5. □

7. 9. Given the two transpositions $\begin{pmatrix} 1 & 2 \\ & 1 & 3 \end{pmatrix}$ and $\begin{pmatrix} 1 & 3 \end{pmatrix}$, find a permutation σ such that $\sigma \begin{pmatrix} 1 & 2 \\ & 1 & 3 \end{pmatrix} \sigma^{-1} = \begin{pmatrix} 1 & 3 \end{pmatrix}$.

Proof. Let $\sigma = (2, 3)$. Now $\sigma(1, 2)\sigma^{-1} = \sigma(1, 2)\sigma = \sigma((1, 2)\sigma) = \sigma(1, 2, 3) = (1, 3)$. □

8. 11. Prove that there is a permutation σ such that $\sigma \begin{pmatrix} 1 & 2 & 3 \\ & 4 & 5 & 6 \end{pmatrix} \sigma^{-1} =$

Proof. Let $\sigma = (1, 4)(2, 5)(3, 6)$. Now

$$\begin{aligned} \sigma(1, 2, 3)\sigma^{-1} &= \sigma(1, 2, 3)\sigma \\ &= \sigma((1, 2, 3)(1, 4)(2, 5)(3, 6)) \\ &= \sigma(1, 4, 2, 5, 3, 6) \\ &= (1, 4)(2, 5)(3, 6)(1, 4, 2, 5, 3, 6) \\ &= (4, 5, 6). \end{aligned}$$

□

9. 14. Prove that for any permutation $\sigma, \sigma\tau\sigma^{-1}$ is a transposition if τ is a transposition.

Proof. Let $\sigma_1, \dots, \sigma_k$ be the cycle decomposition of σ . Then $\sigma_1, \dots, \sigma_k$ are disjoint. Let $\tau = (x_1, x_2)$. We have two cases:

Case 1: x_1, x_2 are in σ_1 . It is done by the following lemma:

Lemma 7. If σ is a permutation, $\tau = (x_1, x_2)$ is a transposition, and σ contains x_1 and x_2 , then $\sigma\tau\sigma^{-1}$ is a transposition.

Proof. Write $z_1 = \sigma(x_1)$ and $z_2 = \sigma(x_2)$. First assume that $z_1 \neq x_2$ and $z_2 \neq x_1$. Now

$$\sigma\tau\sigma^{-1}(z_1) = \sigma\tau(x_1) = \sigma(x_2) = z_2$$

, and

$$\sigma\tau\sigma^{-1}(z_2) = \sigma\tau(x_2) = \sigma(x_1) = z_1.$$

Also, for any other element x distinct from z_1 and z_2 , we have $\sigma^{-1}(x) \notin \{x_1, x_2\}$, so $\sigma^{-1}(x)$ is not affected by τ , whence $\sigma\tau\sigma^{-1}(x) = \sigma\sigma^{-1}(x) = x$. Therefore,

$$\sigma\tau\sigma^{-1} = (z_1, z_2).$$

Now assume, without loss of generality, that $\sigma(x_1) = x_2$ and $\sigma(x_2) = z_2$. Now

$$\sigma\tau\sigma^{-1}(z_2) = \sigma\tau(x_2) = \sigma(x_1) = x_2$$

,and

$$\sigma\tau\sigma^{-1}(x_2) = \sigma\tau(x_1) = \sigma(x_2) = z_2.$$

From a similar argument, all other elements remain unchanged. Thus

$$\sigma\tau\sigma^{-1} = (x_2, z_2).$$

□

Now consider cycle decompositions:

$$\begin{aligned} \sigma\tau\sigma^{-1} &= (\sigma_1 \dots \sigma_k)\tau(\sigma_1 \dots \sigma_k)^{-1} \\ &= \sigma_1\tau\sigma_1^{-1}. \end{aligned}$$

By the previous lemma, $\sigma\tau\sigma^{-1}$ is a transposition.

Case 2: x_1 in σ_1 and x_2 in σ_2 . First we prove a lemma.

Lemma 8. If σ is a permutation, $\tau = (x_1, x_2)$ is a transposition, and σ contains x_1 but not x_2 , then $\sigma\tau\sigma^{-1}$ is a transposition.

Proof. Write $\sigma = (x_1, y_1, \dots, y_p)$ where $y_1, \dots, y_p$ are elements distinct from x_1 and x_2 .

$$\begin{aligned}\sigma\tau\sigma^{-1} &= \sigma(x_1, x_2)\sigma^{-1} \\ &= (x_1, y_1, y_2, \dots, y_p)(x_1, x_2)\sigma^{-1} \\ &= (x_1, x_2, y_1, y_2, \dots, y_p)\sigma^{-1} \\ &= (x_1, x_2, y_1, y_2, \dots, y_p)(y_p, \dots, y_2, y_1, x_1) \\ &= (x_2, y_1).\end{aligned}$$

Thus $\sigma\tau\sigma^{-1}$ is a transposition. $\square$

$$\begin{aligned}\sigma\tau\sigma^{-1} &= (\sigma_1 \dots \sigma_k)\tau(\sigma_1 \dots \sigma_k)^{-1} \\ &= (\sigma_1\sigma_2)\tau(\sigma_1^{-1}\sigma_2^{-1})(\sigma_3 \dots \sigma_k)(\sigma_3 \dots \sigma_k)^{-1} \\ &= (\sigma_1\sigma_2)\tau(\sigma_1^{-1}\sigma_2^{-1}) \\ &= (\sigma_1\sigma_2)\tau(\sigma_1^{-1}\sigma_2^{-1}) \\ &= \sigma_2(\sigma_1\tau\sigma_1^{-1})\sigma_2^{-1} \\ &= \sigma_2(\sigma_1\tau\sigma_1^{-1})\sigma_2^{-1}.\end{aligned}$$

Apply the lemma twice and we see it is a transposition. $\square$

10. 17. Let $(1 \ 2)$ and $(1 \ 2 \ 3 \ \dots \ n)$ be in S_n . Show that any subgroup of S_n that contains both of these must be all of S_n (so these two permutations generate S_n)

Proof. We show that all transpositions in S_n can be generated by $(1, 2)$ and $(1, 2, 3, \dots, n)$. Write $\sigma_0 = (1, 2, 3, \dots, n)$. Then S_n is generated by all the transpositions.

First, we show that $\sigma_0(a, b)\sigma_0^{-1} = (a + 1, b + 1)$, where the $+$ operation is addition modulo n . Also, we use 1 to denote the element 0 in $\mathbb{Z}_n$, 2 to denote 1 in $\mathbb{Z}_n$, etc.

Take any x . If $\sigma^{-1}(x) = x - 1 \notin \{a, b\}$, then $\sigma_0(a, b)\sigma_0^{-1}(x) = \sigma_0\sigma_0^{-1}(x) = x$. If $x = a + 1$, then $\sigma_0(a, b)\sigma_0^{-1}(x) = \sigma_0(a, b)(a) = \sigma(b) = b + 1$. Similarly, $\sigma_0(a, b)\sigma_0^{-1}(b + 1) = a + 1$. Hence

$$\sigma_0(a, b)\sigma_0^{-1} = (a + 1, b + 1).$$

Now by repetitively using the last result, we obtain n transpositions: $(1, 2), (2, 3), \dots, (n - 1, n), (n, 1)$.

Next we build up the remaining transpositions from what we have. We verify that

$$(b, c)(a, b)(b, c) = (a, c).$$

Now suppose we want a transposition (x, y) . We first build the transposition $(x, x+2)$ from $(x, x+1)$ and $(x+1, x+2)$. Then we combine it with $(x+2, x+3)$ to obtain $(x, x+3)$. Repeat in this fashion until we obtain (x, y) .

With all transpositions in hand, it will be easy to construct all the permutations. We have verified in class that for any $\sigma = (a_1, a_2, \dots, a_k)$,

$$\sigma = (a_1, a_k) \dots (a_1, a_3)(a_1, a_2).$$

□

11. 20. If τ_1, τ_2 are distinct transpositions, show that $\tau_1\tau_2$ is of order 2 or 3.

Proof. If τ_1 and τ_2 are disjoint, then

$$o(\tau_1\tau_2) = \text{LCM}(2, 2) = 2.$$

If τ_1 and τ_2 are coincident, then

$$o(\tau_1\tau_2) = o(\tau_1\tau_1) = o(e) = 1.$$

If τ_1 and τ_2 share 1 element, then we write $\tau_1 = (a, b)$ and $\tau_2 = (b, c)$, and

$$o(\tau_1\tau_2) = o((a, b)(b, c)) = o((b, c, a)) = 3.$$

Thus if τ_1 and τ_2 are not coincident, i.e. distinct, their product is of order 2 or 3. □

12. 23. Let σ, τ be two permutations such that they both have decompositions into disjoint cycles of lengths $m_1, m_2, \dots, m_k$. (We say that they have similar decompositions into disjoint cycles.) Prove that for some permutation $\rho, \tau = \rho\sigma\rho^{-1}$.

Proof. We assume that the underlying set is S . Without loss of generality, assume S is finite, and all the 1-cycles are also included in the decomposition. This implies that $m_1 + m_2 + \dots + m_k = |S|$.

We write $\sigma = \sigma_1\sigma_2 \dots \sigma_k$, where σ_i is the i -th cycle in the cycle decomposition with length m_i . Similarly write $\tau = \tau_1\tau_2 \dots \tau_k$.

We write $\sigma_i = (x_1^i, x_2^i, \dots, x_{m_i}^i)$. Similarly, we write $\tau_i = (y_1^i, \dots, y_{m_i}^i)$.

Now since the cycle decomposition is disjoint, the set $X = \{x_j^i : 1 \leq i \leq k, 1 \leq j \leq m_i\}$ has exactly $|S|$ elements, so is the set $Y = \{y_j^i : 1 \leq i \leq k, 1 \leq j \leq m_i\}$. We define the map

$$\begin{aligned} \rho : S &\longrightarrow S \\ x_j^i &\longmapsto \varphi(x_j^i) = y_j^i. \end{aligned}$$

We check the map is well-defined. First, as we have verified, each element in S has a unique representation x_j^i . Also, each element in S has a unique representation y_j^i , where $\rho(x_j^i) = y_j^i$. Thus ρ is surjective. Injectivity is also easy, since $x_j^i \neq x_{j'}^{i'} \Leftrightarrow (j, i) \neq (j', i') \Leftrightarrow y_j^i \neq y_{j'}^{i'}$. Thus ρ defines a permutation on S .

Now

$$\tau(y_j^i) = y_{j+1}^i = \rho(x_{j+1}^i) = \rho(\sigma(x_j^i)) = \rho\sigma\rho^{-1}(y_j^i).$$

Hence $\tau = \rho\sigma\rho^{-1}$. $\square$

Homework 6

Mon 10 Oct 2022 20:22

Herstein, section 3.3, page 123: Questions 2, 3, 7

1. 2. If σ is a k -cycle, show that σ is an odd permutation if k is even, and is an even permutation if k is odd.

Proof. We verify that

$$(1, 2, \dots, k) = (1, 2)(2, 3) \dots (k-1, k).$$

Thus a k -cycle can be written as product of $k-1$ transpositions. k and $k-1$ always have opposite parity, so the claim is proved. $\square$

2. 3. Prove that σ and $\tau^{-1}\sigma\tau$, for any $\sigma, \tau \in S_n$, are of the same parity.

Proof. Suppose the parity of σ is m and the parity of τ is n . Then the parity of $\tau^{-1}\sigma\tau$ is $n+m+n \equiv 2n+m \equiv m \pmod{2}$. $\square$

3. 7. Show that every element in A_n is a product of n -cycles.

Proof. Using the identity

$$(1, 2, \dots, k)(k, k-1, \dots, 3, 1, 2) = (1, 3, 2)$$

, we can write any 3-cycle as a product of two k -cycles where k is larger or equal to 3.

Since A_n is generated by 3-cycles, we can write each of the 3-cycle as two n -cycles, and obtain the desired representation.

We have just proved the result for $n \geq 3$. For $n = 1$ or $n = 2$, A_n become trivial, containing only the identity element which can be represented as $(1, 2, \dots, n)(n, n-1, \dots, 1)$ for any $n \geq 1$. $\square$

Herstein, section 6.1, page 221: Questions 1, 2, 8

4. 1. Prove that if $n > 2$, the center of S_n is (e) .

Proof. Suppose for the sake of contradiction that there exists $\sigma \in Z(S_n)$ such that $\sigma \neq e$. Let

$$A := \{x \in [n] : \sigma(x) \neq x\}.$$

Since $\sigma \neq e$, $|A| \geq 2$. Take $a, b \in A$ where $a \neq b$. We claim that $\sigma(a, b) \neq (a, b)\sigma$.

Suppose $\sigma(a, b) = (a, b)\sigma$, then $(a, b)^{-1}\sigma(a, b) = \sigma$. Thus if we interchange a and b in the cycle notation of σ , σ remains unchanged. This can happen only when $\sigma = (a, b)\tau$ where τ is disjoint from (a, b) . Suppose τ is

nontrivial, and let c be an element changed by τ . Now $\sigma(a, c) = (a, c)\sigma$, so

$$\begin{aligned}(a, b)\tau(a, c) &= (a, c)(a, b)\tau \\ \Leftrightarrow (a, b)(a, c)(a, b)\tau(a, c) &= \tau \\ \Leftrightarrow (a, b)(a, c)\tau(a, b)(a, c) &= \tau \\ \Leftrightarrow (a, c, b)\tau(a, c, b) &= \tau.\end{aligned}$$

Now $(a, c, b)\tau(a, c, b)(b) = (a, c, b)\tau(a) = (a, c, b)(a) = c$, but this is a contradiction since this implies $\tau(b) = c$, but $\tau(b) = b$. Therefore τ is trivial, and $\sigma = (a, b)$.

However, we have $(a, b)(a, c) \neq (a, c)(a, b)$, so $\sigma \notin Z(S_n)$, a contradiction. Hence in all cases $\sigma \notin Z(S_n)$, so $Z(S_n)$ having an element other than e leads to contradiction, so $Z(S_n) = (e)$. $\square$

5. 2. Prove that if $n > 3$, the center of A_n is (e) .

Proof. Suppose for contradiction that center of A_n is not (e) . Take non-trivial $\sigma \in Z(A_n)$. Take a, b such that $\sigma(a) \neq a$ and $\sigma(b) \neq b$. Because $\sigma(a, b) = (a, b)\sigma$, we have $(a, b)^{-1}\sigma(a, b) = \sigma$, so σ must be in the form $(a, b)\tau$ where τ is disjoint from (a, b) . Since $\sigma \in A_n$, σ must be of the form $(a, b)(c, d)\tau$ where $\tau \in A_n$. Now

$$\begin{aligned}(a, b)(c, d)\tau(a, b, c) &= (a, b, c)(a, b)(c, d)\tau \\ \Leftrightarrow (a, b, d)\tau(a, b, c) &= \tau\end{aligned}$$

But now $(a, b, d)\tau(a, b, c)(a) = (a, b, d)(b) = d$, which means $\tau(a) \neq a$, a contradiction since τ is disjoint from a .

We have arrived at a contradiction, so the center of A_n must be (e) . $\square$

Correction

The red part is incorrect since $(a, b) \notin A_n$ (it is in fact in S_n), and σ only commutes with elements in A_n .

Take a, b, c, d such that $\sigma(a) = b$ and $\sigma(c) = d$. Now $(b, a, c)\sigma(a) = a$ but $\sigma(b, a, c)(a) = d$, a contradiction.

6. 8. If $M \triangleleft N$ and $N \triangleleft G$, show that aMa^{-1} is normal in N for every $a \in G$.

Proof. Take arbitrary $n \in N$. Now $a^{-1}n^{-1}aMa^{-1}na = (a^{-1}na)^{-1}M(a^{-1}na)$. Since $N \triangleleft G$, we have $a^{-1}na \in N$. Since $M \triangleleft N$, $(a^{-1}na)^{-1}M(a^{-1}na) = M$.

Thus $a^{-1}n^{-1}aMa^{-1}na = M$, hence $n^{-1}(aMa^{-1})n = aMa^{-1}$, therefore $aMa^{-1} \triangleleft N$. $\square$

Homework 7

Sat 15 Oct 2022 18:57

Herstein, section 2.8, page 91, Questions 2, 4, 5, 8, 9, 10, 12

1. 2. Prove that a group of order 35 is cyclic.

Proof. Apply theorem 2.8.5. Note that G has order pq where $p = 7$ and $q = 5$, and p, q are primes. We also have $p > q$ and $q = 5 \nmid 6 = p - 1$. Thus G must be cyclic. $\square$

2. 4. Construct a nonabelian group of order 21. (Hint: Assume that $a^3 = e$, $b^7 = e$ and find some i such that $a^{-1}ba = b^i \neq b$, which is consistent with the relations $a^3 = b^7 = e$.)

Proof. Let G be the group

$$G = \langle a, b \mid a^3 = b^7 = e, a^{-1}ba = b^2 \rangle.$$

The group is nonabelian by definition.

Using the relation $a^{-1}ba = b^2$, we can simplify any finite product of a and b to the form $a^i b^j$. Also, any two elements $a^i b^j$ and $a^k b^l$ such that $i \not\equiv k \pmod{3}$ and $j \not\equiv l \pmod{7}$ cannot be simplified into each other by the rule $a^{-1}ba = b^2$, and are viewed as distinct elements. Therefore there are $3 \cdot 7 = 21$ elements in the group. $\square$

3. 5. Let G be a group of order $p^n m$, where p is prime and $p \nmid m$. Suppose that G has a normal subgroup P of order p^n . Prove that $\theta(P) = P$ for every automorphism θ of G .

Proof. Since $P \triangleleft G$ and θ is a homomorphism, we have $\theta(P) \triangleleft \theta(G) = G$. Since θ is an isomorphism, $|\theta(P)| = |P| = p^n$.

Now we have

$$|P\theta(P)| = \frac{|P||\theta(P)|}{|P \cap \theta(P)|} = \frac{p^{2n}}{|P \cap \theta(P)|} \mid |G| = p^n m.$$

Since $p \nmid m$, this can happen only when $p^n \mid |P \cap \theta(P)|$. Thus $|P \cap \theta(P)| \geq p^n$. But we also have $|P \cap \theta(P)| \leq |P| = p^n$, so $|P \cap \theta(P)| = p^n = |P|$, so $\theta(P) = P$. $\square$

4. 8. Prove that a group of order 99 has a nontrivial normal subgroup.

Proof. By Cauchy Theorem, there is an element $a \in G$ such that $o(a) = 11$. We claim that $\langle a \rangle \triangleleft G$.

Take arbitrary element $g \in G$. We claim that $g^{-1}\langle a \rangle g = \langle a \rangle$. If they are not equal, then $(g^{-1}\langle a \rangle g) \cap \langle a \rangle$ has order 1, because it is a proper subgroup of $\langle a \rangle$, and $o(a) = 11$ is prime.

But then

$$|(g^{-1}\langle a \rangle g) \langle a \rangle| = \frac{|g^{-1}\langle a \rangle g| |\langle a \rangle|}{|(g^{-1}\langle a \rangle g) \cap \langle a \rangle|} = 11^2 > 99.$$

Which is a contradiction, since $|(g^{-1}\langle a \rangle g) \langle a \rangle| \leq |G| = 99$. $\square$

9. Prove that a group of order 42 has a nontrivial normal subgroup.

Proof. By Cauchy Theorem, there exists an element $a \in G$ such that $o(a) = 7$. We claim that $\langle a \rangle \triangleleft G$.

Take arbitrary $g \in G$. We must have either $g^{-1}\langle a \rangle g = \langle a \rangle$ or $g^{-1}\langle a \rangle g \neq \langle a \rangle$. In the first case we are done; in the second case we must have $(g^{-1}\langle a \rangle g) \cap \langle a \rangle = \{e\}$, since its order must divide $o(a)$ and be strictly less than $o(a)$ which is a prime. Now using the counting property,

$$|(g^{-1}\langle a \rangle g) \langle a \rangle| = \frac{|g^{-1}\langle a \rangle g| |\langle a \rangle|}{|(g^{-1}\langle a \rangle g) \cap \langle a \rangle|} = 7^2 > 42.$$

Which yields a contradiction since the order of $(g^{-1}\langle a \rangle g) \langle a \rangle$ must be less than or equal to 42.

Since $g^{-1}\langle a \rangle g = \langle a \rangle$ for all $g \in G$. Hence $\langle a \rangle \triangleleft G$ as desired. $\square$

10. From the result of Problem 9, prove that a group of order 42 has a normal subgroup of order 21.

Proof. We already know G has a normal subgroup of order 7, say $\langle a \rangle \triangleleft G$.

Now $|G/\langle a \rangle| = 6$. By Lemma 2.8.3, a group of order 6 has a normal subgroup of order 3. Call this subgroup N' .

Define the homomorphism

$$\begin{aligned} \varphi : G &\longrightarrow G/\langle a \rangle \\ g &\longmapsto \varphi(g) = \langle a \rangle g. \end{aligned}$$

We have verified that φ is a surjective homomorphism when we were studying homomorphism theorems. Let $K = \langle a \rangle$ be the kernel of φ .

By the Correspondence Theorem, $N := \varphi^{-1}(N')$ is a normal subgroup of G , $K \subset N$, and $N/K \cong N'$. Thus

$$|(G/\langle a \rangle) / (N/\langle a \rangle)| = 6/3 = 2.$$

By Third Homomorphism theorem,

$$G/N \cong (G/K)/(N/K).$$

So $|G/N| = 2$. Thus $|N| = |G|/2 = 21$.

$\square$

5. 12. Prove that any two nonabelian groups of order 21 are isomorphic. (See Problem 4.)

Proof. By Cauchy Theorem, there is an element $b \in G$ with order 7. Using Lemma 2.8.3, $\langle b \rangle \triangleleft G$. Also by Cauchy Theorem, there is an element $a \in G$ with order 3. Now we have $a^{-1}ba = b^i$ for some i such that $0 \leq i < 7$. Therefore, any element $g \in G$ can be uniquely written as $g = a^j b^k$ where $0 \leq j \leq 2, 0 \leq k \leq 6$.

First of all, $i \neq 1$ and $i \neq 0$, otherwise we have $ab = ba$ or $b = e$, which are contradictory to our assumptions.

Also, if we conjugate b by a for 3 times,

$$b = a^{-3}ba^3 = b^{i^3}.$$

Thus $i^3 \cong 1 \pmod{7}$. The possible solutions of i are 2 and 4. We now want to show that the definitions of G by setting $i = 2$ or $i = 4$ are equivalent.

Assume $a^{-1}ba = b^4$. Now

$$aba^{-1} = a^2(a^{-1}ba)a^{-2} = a^2b^4a^{-2} = a^{-1}b^4a = (b^4)^4 = b^{16} = b^2.$$

Assume $a^{-1}ba = b^2$. Now

$$aba^{-1} = a^2(a^{-1}ba)a^{-2} = a^2b^2a^{-2} = a^{-1}b^2a = (b^2)^2 = b^4.$$

Therefore we can replace a^{-1} by a to go from one definition to another, and the two definitions are identical. $\square$

Correction

We explicitly construct an isomorphism. Let

$$G_2 = \{a_2^m b_2^n : 0 \leq m \leq 2, 0 \leq n \leq 6\}.$$

$$G_4 = \{a_4^m b_4^n : 0 \leq m \leq 2, 0 \leq n \leq 6\}.$$

We need φ is a homomorphism:

$$\varphi(a_2^{m_1} b_2^{n_1} a_4^{m_2} b_4^{n_2}) = \varphi(a_2^{m_1} b_2^{n_1}) \varphi(a_4^{m_2} b_4^{n_2}).$$

Compute the lhs:

$$\varphi(a_2^{m_1} b_2^{n_1} a_4^{m_2} b_4^{n_2}) = \varphi(a_2^{m_1+m_2} b_2^{n_1 2^{m_2} + n_2}).$$

We define

$$\begin{aligned} \varphi : G_2 &\longrightarrow G_4 \\ a_2^m b_2^n &\longmapsto \varphi(a_2^m b_2^n) = a_4^{2m} b_4^{4n}. \end{aligned}$$

Herstein, section 2.9, page 96, Questions 1, 2, 3, 5, 6

6. 1. If G_1 and G_2 are groups, prove that $G_1 \times G_2 \simeq G_2 \times G_1$.

Proof. Construct a map

$$\begin{aligned} \varphi : G_1 \times G_2 &\longrightarrow G_2 \times G_1 \\ (a, b) &\longmapsto \varphi((a, b)) = (b, a). \end{aligned}$$

This map is clearly bijective. Also, for any pairs (a_1, b_1) and (a_2, b_2) in $G_1 \times G_2$,

$$\begin{aligned} \varphi((a_1, b_1)(a_2, b_2)) &= \varphi(a_1 a_2, b_1 b_2) \\ &= (b_1 b_2, a_1 a_2) = (b_1, a_1)(b_2, a_2) \\ &= \varphi(a_1, b_1) \varphi(a_2, b_2). \end{aligned}$$

Hence φ is an isomorphism. Hence $G_1 \times G_2 \cong G_2 \times G_1$. $\square$

7. 2. If G_1 and G_2 are cyclic groups of orders m and n , respectively, prove that $G_1 \times G_2$ is cyclic if and only if m and n are relatively prime.

Proof. Take $a \in G_1$ and $b \in G_2$ such that $G_1 = \langle a \rangle$ and $G_2 = \langle b \rangle$, so $o(a) = m$, $o(b) = n$. Now for any $s \in \mathbb{Z}$, $(a, b)^s = (a^s, b^s)$ which is equal to (e_1, e_2) if and only if $a^s = e_1$ and $b^s = e_2$. This is true if and only if $m|s$ and $n|s$.

Assume m and n are relatively prime. Now $(a, b)^s = (e_1, e_2)$ if and only if $mn|s$. Thus there is no integer t smaller than mn such that $(a, b)^t = (e_1, e_2)$. Hence $o((a, b)) = mn = |G_1 \times G_2|$, so $G_1 \times G_2 = \langle (a, b) \rangle$.

Assume m and n are not relatively prime. Then there exists some $c > 1$ such that $c|m$ and $c|n$. Now $(a, b)^s = (e_1, e_2)$ if $\frac{mn}{c}|s$. Hence mn is not the smallest integer t such that $(a, b)^t = (e_1, e_2)$. Hence $o(a, b) < mn$, hence $G_1 \times G_2 \neq \langle (a, b) \rangle$. We chose our a, b as arbitrary generators of G_1 and G_2 , and any generator of $G_1 \times G_2$ must contain a generator of G_1 and a generator of G_2 . So no such generator exists, so $G_1 \times G_2$ cannot be cyclic. $\square$

8. 3. Let G be a group, $A = G \times G$. In A let $T = \{(g, g) \mid g \in G\}$. (a) Prove that $T \simeq G$.

Proof. Define

$$\begin{aligned}\varphi : G &\longrightarrow T \\ g &\longmapsto \varphi(g) = (g, g).\end{aligned}$$

This map is clearly well-defined and injective. To check that it is surjective, note that for any $(g, g) \in T$, we have $\varphi(g) = (g, g)$. To check that it is a homomorphism, note

$$\varphi(g_1)\varphi(g_2) = (g_1, g_1)(g_2, g_2) = (g_1g_2, g_1g_2) = \varphi(g_1g_2).$$

Thus φ is an isomorphism, therefore $T \cong G$. $\square$

- (b) Prove that $T \triangleleft A$ if and only if G is abelian.

Proof.

$$\begin{aligned}T &\triangleleft A \\ \Leftrightarrow \forall (g_1, g_2) \in A, (g_1, g_2)^{-1}T(g_1, g_2) &= T \\ \Leftrightarrow \forall (g_1, g_2) \in A, \forall g \in G, (g_1, g_2)^{-1}(g, g)(g_1, g_2) &\in T \\ \Leftrightarrow \forall (g_1, g_2) \in A, \forall g \in G, g_1^{-1}gg_1 &= g_2^{-1}gg_2 \\ \Leftrightarrow \forall g_1, g_2, g \in G, (g_2g_1^{-1})g &= g(g_2g_1^{-1}) \\ \Leftrightarrow \forall g_1, g_2, g \in G, (g_2g_1^{-1})g &= g(g_2g_1^{-1}) \\ \Leftrightarrow \forall h, g \in G, hg &= gh \\ &= G \text{ is abelian.}\end{aligned}$$

$\square$

9. 5. Let G be a finite group, $N_1, N_2, \dots, N_k$ normal subgroups of G such that $G = N_1 N_2 \cdots N_k$ and $|G| = |N_1| |N_2| \cdots |N_k|$. Prove that G is the direct product of $N_1, N_2, \dots, N_k$.

Proof. Define

$$\begin{aligned} \varphi : N_1 \times N_2 \times \cdots \times N_k &\longrightarrow G \\ (n_1, n_2, \dots, n_k) &\longmapsto \varphi((n_1, n_2, \dots, n_k)) = n_1 n_2 \cdots n_k. \end{aligned}$$

By assumptions, obviously this map is well-defined and surjective. Also it is assumed that $|G| = |N_1| |N_2| \cdots |N_k|$, so φ is bijective. This means that each element in G is *uniquely* represented by a product of $n_1 n_2 \cdots n_k$ where $n_i \in N_i$ for $1 \leq i \leq k$. Thus G is a direct product of $N_1, \dots, N_k$. $\square$

10. 6. Let G be a group, $N_1, N_2, \dots, N_k$ normal subgroups of G such that: 1. $G = N_1 N_2 \cdots N_k$. 2. For each i , $N_i \cap (N_1 N_2 \cdots N_{i-1} N_{i+1} \cdots N_k) = (e)$. Prove that G is the direct product of $N_1, N_2, \dots, N_k$.

Proof. By assumption 1, each element of G can be represented as product of elements in $N_1, \dots, N_k$. We need to show such representation is unique.

Let $(n_1, n_2, \dots, n_k)$ and $(n'_1, n'_2, \dots, n'_k)$ be elements in $N_1 \times N_2 \times \cdots \times N_k$. Fix any i such that $1 \leq i \leq k$. Now

$$n_1 n_2 \cdots n_i \cdots n_k = n'_1 n'_2 \cdots n'_i \cdots n'_k.$$

Rearrange:

$$n_i = (n_{i-1}^{-1} n_{i-2}^{-1} \cdots n_1^{-1}) n'_1 n'_2 \cdots n'_i \cdots n'_k (n_k^{-1} n_{k-1}^{-1} \cdots n_{i+1}^{-1}).$$

We can group $n_1 n'_1$ and $n'_k n_k$ to obtain

$$n_i \in (n_{i-1}^{-1} n_{i-2}^{-1} \cdots n_2^{-1}) N_1 (n'_2 \cdots n'_i \cdots n'_{k-1}) N_k (n_{k-1}^{-1} \cdots n_{i+1}^{-1}).$$

Since N_1 and N_k are normal subgroups of G , they commute with group items. Hence

$$n_i \in N_1 (n_{i-1}^{-1} n_{i-2}^{-1} \cdots n_2^{-1}) (n'_2 \cdots n'_i \cdots n'_{k-1}) (n_{k-1}^{-1} \cdots n_{i+1}^{-1}) N_k.$$

Now we may continue grouping $n_2 n'_2$ and $n'_{k-1} n_{k-1}$. Continue in this fashion, we finally obtain

$$n_i \in N_1 N_2 \cdots N_{i-1} n'_i N_{i+1} \cdots N_k.$$

Hence

$$n_i (n'_i)^{-1} \in N_1 N_2 \cdots N_{i-1} N_{i+1} \cdots N_k.$$

But we also have $n_i (n'_i)^{-1} \in N_i$. Therefore by assumption 2,

$$n_i = n'_i.$$

Since i is arbitrary, we conclude that any two representations of some same element in G must be identical. Hence G is the direct product of $N_1, \dots, N_k$. $\square$

- (a) 13. Using the fact that any group of order 9 is abelian, prove that any group of order 99 is abelian.

Proof. By Cauchy Theorem, there exists an element $a \in G$ such that $o(a) = 11$. By using a counting argument, we conclude that $\langle a \rangle \triangleleft G$, and $G/\langle a \rangle$ has order 9.

Now take any $g \in G$. We know that $g^9 = a^k$ and $g^{-1}ag = a^i$ for some i and k . Now

$$a = a^{-k}aa^k = g^{-9}ag^9 = a^{i^9}.$$

Thus $i^9 \equiv 1 \pmod{11}$. Surprisingly, the only solution is $i = 1$. Thus $\langle a \rangle \leq Z(G)$.

Hence $G/Z(G)$ has order at most 9. If it has order 3 or 1, it is cyclic, and hence G is abelian. If it has order 9, then $G/Z(G)$ is abelian. Then it is either isomorphic to $\mathbb{Z}_9$ or isomorphic to $\mathbb{Z}_3 \times \mathbb{Z}_3$. In the first case we are done.

In the second case,

There exists $a \in G$ with order 11. So $\langle a \rangle \subset G$. Then $\langle a \rangle \triangleleft G$ by a counting argument.

Alternative method: use Sylow Third Theorem: There is only one Sylow 11-subgroup of G .

$|G/\langle a \rangle| = 9$, Hence $G/\langle a \rangle$ which is abelian.

Suppose $b \in G/\langle a \rangle$. Such an element exists of order 3.

Since $\langle a \rangle \triangleleft G$ then $b^{-1}ab = a^i$ for some $1 \leq i \leq 10$. Now using conjugacy argument, $i^3 \equiv 1 \pmod{11}$. Thus $i = 1$. Hence $ab = ba$. Then $|Z(G)| \geq 33$. $\square$

- (b) 14. Let $p > q$ be two primes such that $q \mid p-1$. Prove that there exists a nonabelian group of order pq . (Hint: Use the result of Problem 40 of Section 4, namely that U_p is cyclic if p is a prime, and the idea needed to do Problem 4 above.)

Proof. Let G be the group

$$G = \langle a, b \mid a^q = b^p = e, a^{-1}ba = b^i \rangle$$

where i is to be determined later.

Now

$$a^{-q}ba^q = b = b^{i^q}.$$

Thus $i^q \equiv 1 \pmod{p}$. We demonstrate that this equation has a solution such that $i > 1$. $\square$

- (c) 15. Prove that if $p > q$ are two primes such that $q \mid p-1$, then any two nonabelian groups of order pq are isomorphic.

Homework 8

Sun 23 Oct 2022 14:00

Herstein, section 2.10, page 101, Questions 1,3.

- 1. Let A be a normal subgroup of a group G , and suppose that $b \in G$ is an element of prime order p , and that $b \notin A$. Show that $A \cap \langle b \rangle = \{e\}$.

Proof. Suppose $A \cap \langle b \rangle$ is nontrivial, so we can pick $b^i \in A \cap \langle b \rangle$ where $b^i \neq e$ and $b^i \in A$. Now we want to prove $b \in A$.

Since $A \triangleleft G$ and $b^i \in A$, for all $g \in G$ we have $g^{-1}b^ig \in A$. Since $o(b) = p$ and p prime, we have $(i, p) = 1$. By Euclidean Algorithm, there exists integers k, m such that $ki + mp = 1$, so $ki \equiv 1 \pmod{p}$. Therefore

$$(g^{-1}b^ig)^k = g^{-1}b^{ik}g = g^{-1}bg \in A.$$

Hence $bg \in gA$. But $gA = Ag$, so $bg \in Ag$. This implies $b \in A$. $\square$

- 3. Let G be a finite group, with $N \triangleleft G$ and $a \in G$. Prove that:
 - (a) The order of aN in G/N divides the order of a in G , that is, $o(aN) \mid o(a)$.

Proof. We have

$$(aN)^{o(a)} = a^{o(a)}N = eN = N.$$

This $o(aN) \mid o(a)$. This follows from a past exercise saying that if $a^s = e$ then $o(a) \mid s$. $\square$

(b) If $a \in N$, then $o(aN) = o(a)$.

Proof. Since $(aN)^{o(aN)} = a^{o(aN)}N = N$, we have $a^{o(aN)} \in \langle a \rangle \cap N = \{e\}$. Thus $o(a) \mid o(aN)$. Thus $o(a) = o(aN)$. $\square$

- Additional question (2.10.A): Give a representative for every isomorphism class of abelian groups of order 240.

Proof. Factorize: $240 = 2^4 3^1 5^1$. The representatives are

- $\mathbb{Z}_{30} \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$
- $\mathbb{Z}_{60} \times \mathbb{Z}_2 \times \mathbb{Z}_2$
- $\mathbb{Z}_{60} \times \mathbb{Z}_4$
- $\mathbb{Z}_{120} \times \mathbb{Z}_2$
- $\mathbb{Z}_{240}$

 $\square$

- Additional question (2.10.B): Give a representative for every isomorphism class of abelian groups of order 540. In each case, make sure that your answer is a direct product involving no more than four factors.

Proof. Factorize: $540 = 2^2 3^3 5^1$. The representatives are

- $\mathbb{Z}_{30} \times \mathbb{Z}_3 \times \mathbb{Z}_3 \times \mathbb{Z}_2 \cong \mathbb{Z}_{30} \times \mathbb{Z}_6 \times \mathbb{Z}_3$
- $\mathbb{Z}_{90} \times \mathbb{Z}_3 \times \mathbb{Z}_2 \cong \mathbb{Z}_{90} \times \mathbb{Z}_6$

- $\mathbb{Z}_{270} \times \mathbb{Z}_2$
- $\mathbb{Z}_{60} \times \mathbb{Z}_3 \times \mathbb{Z}_3$
- $\mathbb{Z}_{180} \times \mathbb{Z}_3$
- $\mathbb{Z}_{540}$

□

Herstein, section 2.11, page 106, Questions 1, 3, 4, 5, 9, 13.

- 1. In S_3 , the symmetric group of degree 3, find all the conjugacy classes, and check the validity of the class equation by determining the orders of the centralizers of the elements of S_3 .

Proof. $S_3 = \{e, (1, 2), (1, 3), (2, 3), (1, 2, 3), (3, 2, 1)\}$

- $\text{cl}(e) = \{e\}$
- $\text{cl}((1, 2)) = \{(1, 2), (2, 3), (1, 3)\}$
- $\text{cl}((1, 2, 3)) = \{(1, 2, 3), (3, 2, 1)\}$

Also,

- $C(e) = S_3$, so $i_{S_3}(C(e)) = 1$
- $C((1, 2)) = \{e, (1, 2)\}$, so $i_{S_3}(C(e)) = \frac{6}{2} = 3$
- $C((1, 2, 3)) = \{e, (1, 2, 3), (3, 2, 1)\}$, so $i_{S_3}(C(e)) = \frac{6}{3} = 2$

We verify that

- $|\text{cl}(e)| = 1 = i_{S_3}(C(e))$
- $|\text{cl}((1, 2))| = 3 = i_{S_3}(C((1, 2)))$
- $|\text{cl}((1, 2, 3))| = 2 = i_{S_3}(C((1, 2, 3)))$

Since $1 + 3 + 2 = 6 = 3! = |S_3|$, we have verified the class equation. $\square$

- 3. If $a \in G$, show that $C(x^{-1}ax) = x^{-1}C(a)x$.

Proof. Take any $g \in G$.

$$\begin{aligned}
 g \in C(x^{-1}ax) & \\
 \Leftrightarrow gx^{-1}ax &= x^{-1}axg \\
 \Leftrightarrow g^{-1}x^{-1}axg &= x^{-1}ax \\
 \Leftrightarrow x^{-1}axgx^{-1}a^{-1}x &= g \\
 \Leftrightarrow axgx^{-1}a^{-1} &= xgx^{-1} \\
 \Leftrightarrow axgx^{-1} &= xgx^{-1}a \\
 \Leftrightarrow xgx^{-1} &\in C(a) \\
 \Leftrightarrow g &\in x^{-1}C(a)x.
 \end{aligned}$$

$\square$

- 4. If φ is an automorphism of G , show that $C(\varphi(a)) = \varphi(C(a))$ for $a \in G$.

Proof. Take any $g \in G$. Now

$$\begin{aligned}
 g &\in C(\varphi(a)) \\
 &\Leftrightarrow g\varphi(a) = \varphi(a)g \\
 &\Leftrightarrow \varphi(\varphi^{-1}(g))\varphi(a) = \varphi(a)\varphi(\varphi^{-1}(g)) && \text{since } \varphi \text{ is bijective} \\
 &\Leftrightarrow \varphi(\varphi^{-1}(g)a) = \varphi(a\varphi^{-1}(g)) && \text{since } \varphi \text{ is homomorphism} \\
 &\Leftrightarrow \varphi^{-1}(g)a = a\varphi^{-1}(g) && \text{since } \varphi \text{ is bijective} \\
 &\Leftrightarrow \varphi^{-1}(g) \in C(a) \\
 &\Leftrightarrow g \in \varphi(C(a)).
 \end{aligned}$$

□

- 5. If $|G| = p^3$ and $|Z(G)| \geq p^2$, prove that G is abelian.

Proof. We assume p is prime. Now $|G/Z(G)| \leq p$, so $|G/Z(G)| = 1$ or p . If $|G/Z(G)| = 1$ then $|Z(G)| = |G|$, so $Z(G) = G$, hence G abelian. If $|G/Z(G)| = p$, then $Z(G) = \{e\}$. However this cannot happen since by theorem 2.11.4, the center of G is nontrivial.

If p is not prime, this statement is not always true. We show this by giving a counterexample. Let A be an abelian group of order p^2 and B be a non-abelian group of order p (such group can sometimes exist since p is not prime). Then the group $A \times B$ is clearly nonabelian; but it has order p^3 and $|Z(A \times B)| \geq |A| \times |Z(B)| \geq |A| = p^2$. □

- 9. Prove that $N(H)$ is a subgroup of G , $H \subset N(H)$ and in fact $H \triangleleft N(H)$.

Proof. Let $a, b \in N(H)$. We want to show $ab^{-1} \in N(H)$.

$$\begin{aligned}
 a, b &\in N(H) \\
 &\Leftrightarrow aH = Ha, bH = Hb \\
 &\Rightarrow aH = ab^{-1}bH = ab^{-1}Hb \\
 &\Leftrightarrow ab^{-1}H = aHb^{-1} = Hab^{-1} \\
 &= ab^{-1} \in N(H).
 \end{aligned}$$

Hence by subgroup criterion, $N(H) \leq G$.

To see $H \subset N(H)$, note that for any $h \in H$, $hH = H = Hh$.

To see that $H \triangleleft N(H)$, take any element $n \in N(H)$, and we want to show $nH = Hn$. But this directly follows from definition of $N(H)$. □

- 13. Prove that if G is a finite group and H is a subgroup of G , then the number of distinct subgroups $x^{-1}Hx$ of G equals $i_G(N(H))$.

Proof. Define φ to be the mapping $x^{-1}Hx \mapsto N(H)x$. Observe that for any $x, y \in G$,

$$\begin{aligned}
 x^{-1}Hx &= y^{-1}Hy \\
 &\Leftrightarrow (xy^{-1})^{-1}H(xy^{-1}) = H \\
 &\Leftrightarrow xy^{-1} \in N(H) \\
 &\Leftrightarrow N(H)x = N(H)y.
 \end{aligned}$$

So φ is a bijection from all subgroups $x^{-1}Hx$ to $G/N(H)$. So the number of distinct $x^{-1}Hx$ is exactly equal to $|G/N(H)| = i_G(N(H))$. $\square$

Homework 9

Mon 31 Oct 2022 16:12

Herstein, section 2.11, page 106, Questions 15, 16, 18, 20, 23, 24.

15. If $N \triangleleft G$, let $B(N) = \{x \in G \mid xa = ax \text{ for all } a \in N\}$. Prove that $B(N) \triangleleft G$.

Proof. Take $b \in B(N)$. Now for arbitrary $n \in N$, we consider the product

$$ng^{-1}bgn^{-1}.$$

Since $N \triangleleft G$, we have $Ng^{-1} = g^{-1}N$, so there exists $n' \in N$ such that $ng^{-1} = g^{-1}n'$. Hence the product is equal to

$$g^{-1}n'bn'^{-1}g.$$

Now by definition of $B(N)$, we know that $n'bn'^{-1} = b$. So the product is equal to

$$g^{-1}bg.$$

But we see that $g^{-1}bg$ is invariant under conjugation by elements from N . Hence $g^{-1}bg \in B(N)$ by definition. Since g and b are arbitrary, we conclude that $B(N) \triangleleft G$. $\square$

16. Show that a group of order 36 has a normal subgroup of order 3 or 9. (Hint: See Problem 40 of Section 5.)

Proof. By Sylow's Theorem, since $|G| = 36 = 3^2 \cdot 4$ where 3 is prime and $3 \nmid 4$, G has a subgroup of order 9, say H .

Now by Problem 40 of section 5, since $H \leq G$ and $36 \nmid 4! = i_G(H)!$, there exists nontrivial N contained in H such that $N \triangleleft G$, thus in particular $N \leq H$. By Lagrange Theorem, N has order 1, 3, or 9. Since N is nontrivial, N has order 3 or 9. $\square$

18. If P is a p -Sylow subgroup of G , show that $N(N(P)) = N(P)$.

Proof. Obviously $N(P) \subset N(N(P))$, so we only prove $N(N(P)) \subset N(P)$.

We have, by assumption, that $P \leq N(P) \leq N(N(P))$. Take $a \in N(N(P))$. Then we can take conjugates on the equation and obtain $a^{-1}Pa \leq a^{-1}N(P)a$. Since $a \in N(N(P))$, $a^{-1}N(P)a = N(P)$, so $a^{-1}Pa \leq N(P)$.

Now $a^{-1}Pa$ is a group of order $|P|$. Since P is a p -Sylow subgroup of G , it is a p -Sylow subgroup of $N(P)$. Also, $a^{-1}Pa$ is a p -Sylow subgroup of $N(P)$.

Then by the Sylow Theorem, P and $a^{-1}Pa$ are conjugate in $N(P)$. Hence there exists some $b \in N(P)$ such that $b^{-1}Pb = a^{-1}Pa$. Thus $ab^{-1} \in N(P)$. Therefore $a \in N(P)$ and we are done showing $N(N(P)) \subset N(P)$. $\square$

20. If p^m divides $|G|$, show that G has a subgroup of order p^m .

Proof. Let H be a Sylow p -subgroup of G of order p^n . We have $m \leq n$. It suffices to show that there exists a subgroup M of H of order p^m .

We claim the following: For any group H with order p^n and any $m \leq n$, there exists a subgroup M of H with order p^m . We proceed by induction on n . When $n = 2$, m can be either 0, 1, or 2. The cases $m = 0$ and $m = 2$ are trivial, and the case $m = 1$ follows directly from Cauchy's Theorem.

Now assume that the statement is true for all groups of order p^k where $k < n$. We may also assume without loss of generality that $m > 1$ (since $m = 1$ is trivial). Since H is of prime-power order, it has nontrivial center. Hence by Cauchy's Theorem we can choose $a \in H$ such that $o(a) = p$. Since a is in the center of H , we have $\langle a \rangle \triangleleft H$. Now the quotient group $H/\langle a \rangle$ has order p^{n-1} , so the induction hypothesis applies. We may choose a subgroup M' of $H/\langle a \rangle$ with order $m - 1$.

Now we construct the projection homomorphism

$$\begin{aligned}\varphi : H &\longrightarrow H/\langle a \rangle \\ h &\longmapsto \varphi(h) = \langle a \rangle h.\end{aligned}$$

By Correspondence Theorem, there exists $M \leq H$ and $M/\langle a \rangle \cong M'$. Therefore $|M| = |M'| |\langle a \rangle| = p^{m-1} p = p^m$. $\square$

23. Let G be a group, H a subgroup of G . Define for $a, b \in G$, $a \sim b$ if $b = h^{-1}ah$ for some $h \in H$. Prove that
(a) this defines an equivalence relation on G .

Proof. Reflexivity: For any $a \in G$, $e^{-1}ae = a$ where $e \in H$, hence $a \sim a$.

Symmetry: If $a \sim b$, then $b = h^{-1}ah$ for some $h \in H$. Hence $a = (h^{-1})^{-1}bh^{-1}$ where $h^{-1} \in H$. Hence $b \sim a$.

Transitivity: If $a \sim b$ and $b \sim c$, then $b = h^{-1}ah$ and $c = k^{-1}bk$ for $h, k \in H$. Then $c = k^{-1}(h^{-1}ah)k = (hk)^{-1}a(hk)$ where $hk \in H$, so $a \sim c$. $\square$

(b) If $[a]$ is the equivalence class of a , show that if G is a finite group, then $[a]$ has m elements where m is the index of $H \cap C(a)$ in H .

Proof. Let $b \in [a]$, so $b = h_1^{-1}ah_1$ for some $h_1 \in H$. For each $h \in C(a) \cap H$, we also have $b = (h_1^{-1}h^{-1})a(hh_1)$, so b can be expressed as conjugation by at least $|C(a) \cap H|$ different items on a . On the other hand, if $b = h_1^{-1}ah_1 = h_2^{-1}ah_2$ where $h_1, h_2 \in H$, then we have $h_1h_2^{-1} \in C(a) \cap H$, so each item b can be obtained by conjugating on a exactly $|C(a) \cap H|$ items in H .

Therefore the number of different conjugates is

$$|[a]| = m = |H|/|C(a) \cap H| = i_H(H \cap C(a)).$$

$\square$

24. If G is a group, H a subgroup of G , define a relation $B \sim A$ for subgroups A, B of G by the condition that $B = h^{-1}Ah$ for some $h \in H$. (a) Prove that this defines an equivalence relation on the set of subgroups of G .

Proof. Reflexivity: $A = e^{-1}Ae$.

Symmetry: If $A \sim B$ then $B = h^{-1}Ah$ for some $h \in H$. Now $A = (h^{-1})^{-1}B(h^{-1})$ hence $B \sim A$.

Transitivity: If $A \sim B$ and $B \sim C$ then $B = h^{-1}Ah$ and $C = k^{-1}Bk$ where $h, k \in H$. Now $C = k^{-1}(h^{-1}Ah)k = (hk)^{-1}A(hk)$ where $hk \in H$, hence $A \sim C$. $\square$

(b) If G is finite, show that the number of distinct subgroups equivalent to A equals the index of $N(A) \cap H$ in H .

Proof. For every element B in $[A]$, we have $B = h^{-1}Ah$ for some $h \in H$. For every element k in $N(A) \cap H$, we have $A = k^{-1}Ak$, so $B = (kh)^{-1}A(kh)$. Hence each element B in the conjugacy class of A is associated with at least $|N(A) \cap H|$ elements from H .

Now suppose $B = h^{-1}Ah = k^{-1}Ak$ where $h, k \in H$. Now $hk^{-1} \in N(A) \cap H$. Hence each element B in $[A]$ is associated with exactly $|N(A) \cap H|$ elements from H .

Thus the size of $[A]$ is $|H|/|N(A) \cap H| = i_H(N(A) \cap H)$. $\square$

Small groups (and the Sylow theorems): SG.A In this problem, the letter p denotes a prime and n a natural number.

(a) Show that when $n \geq 2$ and G is a group of order p^n , then G is not simple.

Proof. Since G has prime-power order, it has nontrivial center. Hence we can choose $a \in Z(G)$ such that $o(a) = p$. Hence $\langle a \rangle \triangleleft G$. Therefore G is not simple since it has a nontrivial normal subgroup. $\square$

(b) Show that when $p \neq 2$, then the group G cannot be simple if it has order $2p^n$ or $3p^n$ or $5p^n$.

Proof. We may assume without loss of generality that G is not a group of prime-power order, because we have already completed this special case.

By Sylow's Theorem we can choose a Sylow p -subgroup with order p^n , say H . By Third Sylow Theorem, the number of Sylow p -subgroups must be the form $kp + 1$ and divide $|G|$. Suppose there is more than one p -subgroup. Since $kp + 1$ does not divide p when $k \neq 0$, we must have $kp + 1$ divides 2, 3, or 5. However all three cases are impossible. Therefore H is the only Sylow p -subgroup.

Hence $H \triangleleft G$ and G is not simple. $\square$

(c) Show that when G is a group of order $4p^n$ ($p \neq 3$) then G is not simple.

Proof. Let H be a Sylow p -subgroup of G . By Sylow's Third Theorem, the number of distinct Sylow p -groups must be equal to $kp + 1$ and divide 4. The only solution is $p = 3$ and $k = 1$, in this case $kp + 1 = 4 \nmid 4$. However it is assumed that $p \neq 3$, so H is the only Sylow p -subgroup of G . Therefore $H \triangleleft G$ and G is not simple. $\square$

SG.B Let G be a group of order 56. (a) Show that G has either 1 or 8 Sylow 7-subgroups, and either 1 or 7 Sylow 2-subgroups.

Proof. Let H be a Sylow 7-subgroup of order 7. Now by Sylow's Third Theorem, the number of Sylow 7-subgroups must be $7k + 1$ and divide 8. Solving this and we have $k = 0$ or $k = 1$. When $k = 0$, there are 1 Sylow 7-subgroup; when $k = 1$, there are 8 Sylow 7-subgroups.

Now let H be a Sylow 2-subgroup of order 8. Now by Sylow's Third Theorem, the number of Sylow 2-subgroups must be $2k + 1$ and divide 7. We can only have $2k + 1 = 1$ or $2k + 1 = 7$. Thus there are 1 or 7 Sylow 2-subgroups. $\square$

(b) By showing that, if there are 2 distinct Sylow 7-subgroups, they necessarily have trivial intersection, show that G cannot be simple.

Proof. Take H and H' be Sylow 7-subgroups. Then we have $H \cap H' \leq H$, hence by Lagrange Theorem, $|H \cap H'|$ is 1 or 7. Since they are distinct, the intersection is trivial.

If all 8 Sylow 7-subgroups have trivial intersection, they contain $8 \cdot (7 - 1) = 48$ nontrivial elements of G , hence the set of remaining 8 elements, say K , must form a normal subgroup of G . To see this, first note that any conjugation of a nonidentity element of a Sylow 7-subgroup will only send it to an element in another Sylow 7-subgroup and never into K . Also, any nonidentity element $x \in K$ won't be sent to a nonidentity element in any Sylow 7-subgroup. Hence K must be closed under conjugation.

Now that $K \triangleleft G$ and K nontrivial so G is not simple. $\square$

SG.C By considering Sylow p -subgroups with p equal to 3 and 5, show that no group of order 30 is simple.

Proof. The number n_3 of Sylow 3-subgroups satisfies $n_3 = 3k + 1 \mid 10$, hence $n_3 = 1$ or 10. The number n_5 of Sylow 5-subgroups satisfies $n_5 = 5k + 1 \mid 6$, hence $n_5 = 1$ or 6.

If $n_3 = 1$ or $n_5 = 1$ then the group has a normal Sylow group. Hence we can assume $n_3 = 10$ and $n_5 = 6$.

Any Sylow 3-subgroup and any Sylow 5-subgroup must be trivial intersection. To see this, note that the intersection is a subgroup of the Sylow 3-group, and apply Lagrange's Theorem.

Hence if G has both 10 Sylow 3-groups and 6 Sylow 5-subgroups, then these subgroups contain $10 \cdot (3 - 1) + 6 \cdot (5 - 1) = 44$ distinct nontrivial elements, which is impossible because G has only 30 elements.

Hence either $n_3 = 1$ and $n_5 = 1$, so we are done. $\square$

SG.D (a) Show that no group of order 12, 24, 36 or 48 can be simple. [Hint: Recall homework question 16 from section 2.11].

Proof. We treat each case differently

order 12 Apply Sylow Third Theorem, the possible number of Sylow 2-subgroups are $n_2 = 1$ and $n_2 = 3$. If $n_2 = 1$ then we are done. If $n_2 = 3$, then let H, H' be two distinct Sylow 2-groups with nontrivial intersection (We have justified that if all such intersections are trivial then G is not simple).

$$|HH'| = \frac{16}{|H \cap H'|}.$$

Hence $|H \cap H'|$ is greater than 1, divides 4, and less than 4. Hence it can only be 2. Now since H and H' are of order 2^2 , they are abelian, thus $N(H \cap H')$ contains H and H' . Therefore $N(H \cap H')$ divides 12, is divisible by 4, and contains at least 8 elements. Hence $|N(H \cap H')| = 12$. Thus $H \cap H' \triangleleft G$ by definition of normalizer.

order 24 Apply Sylow Third Theorem and obtain $n_3 = 1$ or 4. If $n_3 = 1$ then we are done; now assume $n_3 = 4$. But for any two Sylow 3-subgroups H and H' , we have either $H_1 = H_2$ or $H_1 \cap H_2 = (e)$. Thus the previous argument applies: all Sylow p -subgroups pairwise intersect trivially implies the group is not simple.

order 36 Apply Sylow Third Theorem and obtain $n_3 = 1$ or 4. Consider the nontrivial case $n_3 = 4$. Then pick two distinct Sylow 3-subgroups that intersect nontrivially, and use the counting argument,

$$|HH'| = \frac{81}{|H \cap H'|} \leq |G| = 36.$$

Thus $H \cap H'$ has at least 3 elements (from counting argument), with order dividing 9 and between 1 and 9. Hence $|H \cap H'| = 3$. Since $|H| = |H'| = 3^2$, they are abelian. Thus $N(H \cap H')$ contains both H and H' , so $N(H \cap H')$ has at least 27 elements, with order dividing 36 and divisible by 9. Hence $|N(H \cap H')| = 36$. This implies $H \cap H' \triangleleft G$.

order 48 Apply Sylow Third Theorem and obtain $n_3 = 1$ or 4 or 16. But any two Sylow 3-groups with order 3 are either coincident or intersect trivially, so the previous argument applies: All distinct Sylow p -subgroups intersect trivially implies the group is not simple.

□

(b) Let G be a simple group of order n , where $2 \leq n \leq 59$. Show that n is a prime number p and G is cyclic of order p .

Proof. We perform Sylow test on every integer. A number n passes the "Sylow test" if it does not have a prime factor p such that the p -subgroups of a group of order n is unique. In other words, for all prime factor p of n , we write $n = p^r m$ such that $p \nmid m$. Then try to find nontrivial solution of $kp + 1 \mid m$.

This test will eliminate all orders of prime power (where p -subgroup is the group itself), and non-simple groups (where p -subgroup is never the group itself). We have proved that groups of prime power order are simple iff the power is 1.

Running a Python program and sifting through all numbers, the following orders pass the test, which means they could be simple:

12, 24, 30, 36, 48, 56.

But we have ruled out each of them in previous exercises.

Hence the only simple groups with order less than 60 and greater than 1 are those of prime orders.

We have proven earlier that prime-order groups are cyclic. $\square$

```

1 import sympy
2 orders = set(range(1, 60)) #possible nonsimple orders
3 for n in range(1,60):
4     factors = sympy.factorint(n)
5     for p, rp in factors.items():
6         mp = n / p**rp
7         sols = []
8         for k in range(0,100):
9             if k * p + 1 > mp:
10                 break
11             if mp % (k * p + 1) == 0:
12                 sols.append(k)
13         if len(sols) == 1:
14             orders.remove(n)
15         break
16 print(orders)

```

Listing 1: Python code

Homework 10

Fri 04 Nov 2022 13:52

Herstein, section 4.1, page 133 , Questions 1, 2, 3, 13, 19, 20, 31.

- 1. Find all the elements in $\mathbb{Z}_{24}$ that are invertible (i.e., have a multiplicative inverse) in $\mathbb{Z}_{24}$.

Proof. We find all solution $p, q \in \{0, 1, 2, \dots, 23\}$ such that $pq \equiv 1 \pmod{24}$.

$$\begin{aligned}
 1 \times 1 &= 24 \times 0 + 1 \\
 5 \times 5 &= 24 \times 1 + 1 \\
 7 \times 7 &= 24 \times 2 + 1 \\
 11 \times 11 &= 24 \times 5 + 1 \\
 13 \times 13 &= 24 \times 7 + 1 \\
 17 \times 17 &= 24 \times 12 + 1 \\
 19 \times 19 &= 24 \times 15 + 1 \\
 23 \times 23 &= 24 \times 22 + 1.
 \end{aligned}$$

Thus the only invertible elements are 1, 5, 7, 11, 13, 17, 19, 23. $\square$

- 2. Show that any field is an integral domain.

Proof. Assume R is a field, in particular it is commutative. Hence it suffices to show that R is a domain.

Assume $a \cdot b = 0$ and $a \neq 0$. We have to show $b = 0$. Now because R is a division ring (part of definition of a field), there exists $a^{-1} \in R$ such that $aa^{-1} = a^{-1}a = 1$. Thus

$$a^{-1} \cdot (a \cdot b) = a^{-1} \cdot 0.$$

Use associative law in the left and use the identity $x \cdot 0 = 0$ on the right, we obtain

$$(a^{-1} \cdot a) \cdot b = 1 \cdot b = b = 0.$$

□

- 3. Show that $\mathbb{Z}_n$ is a field if and only if n is a prime.

Proof. Assume $n = p$ is a prime and $m < p$. Then since $(m, p) = 1$, there exist integers k and l such that $km + lp = 1$. Then we have $km \equiv 1 \pmod{p}$. Apply Euclidean algorithm we obtain $k = sp + r$ where $s \in \mathbb{Z}$ and $0 \leq r < p$. Thus $sm \equiv 1 \pmod{p}$, so $s \in \mathbb{Z}_n$ and $s \cdot m = 1$ in $\mathbb{Z}_n$. Hence $\mathbb{Z}_n$ is a division ring. Since multiplication is commutative, $\mathbb{Z}_n$ is a field.

Assume n is not prime. Then there exists some $a, b < n$, where $a, b > 1$ such that $n = ab$. But then $ab = 0$ modulo n , but $a, b \neq 0$. Hence $\mathbb{Z}_n$ is not a domain. Hence it cannot be a field. □

- 13. Find the following products of quaternions. (a) $(i + j)(i - j)$.

Solution.

$$(i + j)(i - j) = i^2 - ij + ji - j^2 = -1 - k - k + 1 = -2k.$$

□

(b) $(1 - i + 2j - 2k)(1 + 2i - 4j + 6k)$.

Solution.

$$\begin{aligned} (1 - i + 2j - 2k)(1 + 2i - 4j + 6k) \\ = (1 + 2 + 8 + 12) + (2 - 1 + 12 - 8)i + (-4 + 2 - 4 + 6)j + (6 - 2 + 4 - 4)k \\ = 23 + 5i + 4k. \end{aligned}$$

□

(c) $(2i - 3j + 4k)^2$.

Solution.

$$\begin{aligned} (2i - 3j + 4k)^2 &= (2i - 3j + 4k)(2i - 3j + 4k) \\ &= (-4 - 9 - 16) + (-12 + 12)i + (8 - 8)j + (-6 + 6)k \\ &= -29 \end{aligned}$$

□

(d) $i(\alpha_0 + \alpha_1 i + \alpha_2 j + \alpha_3 k) - (\alpha_0 + \alpha_1 i + \alpha_2 j + \alpha_3 k)i$.

Solution.

$$\begin{aligned}
 & i(\alpha_0 + \alpha_1 i + \alpha_2 j + \alpha_3 k) - (\alpha_0 + \alpha_1 i + \alpha_2 j + \alpha_3 k)i \\
 &= (-\alpha_1 + \alpha_0 i - \alpha_3 j + \alpha_2 k) - (-\alpha_1 + \alpha_0 i + \alpha_3 j - \alpha_2 k) \\
 &= -2\alpha_3 j + 2\alpha_2 k.
 \end{aligned}$$

□

- 19. Show that there is an infinite number of solutions to $x^2 = -1$ in the quaternions.

Proof. Observe that

$$(ai + bj + ck)^2 = -(a^2 + b^2 + c^2).$$

And since there are infinitely many 3-tuples (a, b, c) such that $(a^2 + b^2 + c^2 = 1)$, we have infinitely many solutions to $x^2 = -1$ by letting $x = ai + bj + ck$. □

- 20. In the quaternions, consider the following set G having eight elements: $G = \{\pm 1, \pm i, \pm j, \pm k\}$.

(a) Prove that G is a group (under multiplication).

Proof. The multiplication table for $1, i, j, k$ is

$\cdot$	1	i	j	k
1	1	i	j	k
i	i	-1	k	$-j$
j	j	$-k$	-1	i
k	k	j	$-i$	-1

We can see that closure is satisfied in G , multiplication is associative, and 1 acts as identity element. Moreover, the inverse exists for each element. If $a = 1$ then a itself is the inverse; if $a = i, j$, or k , then $a(-a) = -a^2 = -(-1) = 1$, so $-a$ is the multiplicative inverse of a .

We have verified all group axioms so G is a group under multiplication. □

(b) List all subgroups of G .

Solution. — $\{1\}$

— $\{1, -1\}$

— $\{1, -1, i, -i\}$

— $\{1, -1, j, -j\}$

— $\{1, -1, k, -k\}$

— $\{1, -1, \pm i, \pm j, \pm k\}$

□

(c) What is the center of G ?

Proof. The center of G is $\{1, -1\}$. □

(d) Show that G is a nonabelian group all of whose subgroups are normal.

Proof. To show non-abelian, note that $ij = k \neq -k = ji$.

The whole group and the center are normal. The trivial subgroup is normal. Hence we need to show the other three subgroups of order 4 are normal. By symmetry, we only need to show $A := \{1, -1, i, -i\}$ is normal (since the role of i, j, k are entirely symmetrical in definition of quaternions).

We only need to show that conjugates of elements in $A \setminus Z(G)$ by elements in $G \setminus Z(G)$ remain inside A . Thus we only need to show conjugate of $\pm i$ by $\pm i, \pm j, \pm k$ lie in A . By since the product between different elements of i, j, k are anticommutative ($ij = -ji, ik = -ki, jk = -kj$), we may obtain

$$j^{-1}ij = j^{-1}(-ji) = -(j^{-1}j)i = -i \in A.$$

And the same for $-j, k, -k$. Finally,

$$i^{-1}ii = (i^{-1}i)i = i.$$

and

$$(-i)^{-1}i(-i) = (-i)^{-1}(-i)i = i.$$

Thus A is normal in G . □

- 31. Let R be the ring of all 2×2 matrices over $\mathbb{Z}_p, p$ a prime. Show that if $\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc \neq 0$, then $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is invertible in R .

Proof. Define

$$\begin{aligned} \varphi : \mathbb{Z}_p &\longrightarrow R \\ a &\longmapsto \varphi(a) = \begin{pmatrix} a & 0 \\ 0 & a \end{pmatrix}. \end{aligned}$$

Then φ is a homomorphism by rule of matrix multiplication; and φ is injective since for $a \neq b$, the two matrices $\varphi(a)$ and $\varphi(b)$ differ in their diagonal entries.

Then we have an isomorphism $\mathbb{Z}_p \cong \varphi(\mathbb{Z}_p)$.

Since $|\mathbb{Z}_p| = p$ is prime, $\mathbb{Z}_p$ is a field (so is $\varphi(\mathbb{Z}_p)$).

Whenever $ad - bc \neq 0$, due to field property, there exists $(ad - bc)^{-1} \in \mathbb{Z}_p$.

Now

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \varphi((ad - bc)^{-1}).$$

□

Herstein, section 4.2, page 139, Questions 1, 2, 3.

- 1. Let R be a ring: since R is an abelian group under $+$, na has meaning for us for $n \in \mathbb{Z}, a \in R$. Show that $(na)(mb) = (nm)(ab)$ if n, m are integers and $a, b \in R$.

Proof. We fix n and induct on m . Note that if the desired equation $(na)(mb) = (nm)(ab)$ holds for some m , it automatically holds for $-m$ since $(-m)b = -(mb)$ and $n(-m) = -(nm)$. Hence we can assume $m \geq 0$. The base case is trivial since both sides becomes 0.

Now assume the equation is true for m and we want to prove for $m + 1$. Note that

$$\begin{aligned} (na)((m+1)b) &= (na)(mb + b) \\ &= (na)(mb) + (na)b \\ &= (nm)(ab) + n(ab) \\ &= (nm + n)(ab) \\ &= (n(m+1))(ab). \end{aligned}$$

This completes the induction, thus the equation is true for all $m \in \mathbb{N}$, hence true for all $m \in \mathbb{Z}$. n is arbitrary, so we are done. $\square$

- 2. If R is an integral domain and $ab = ac$ for $a \neq 0, b, c \in R$, show that $b = c$.

Proof. $ab = ac$ iff $ab - ac = 0$, iff $a(b - c) = 0$. By property of integral domain, either $a = 0$ or $b - c = 0$. Since $a \neq 0$, we have $b - c = 0$, hence $b = c$. $\square$

- 3. If R is a finite integral domain, show that R is a field.

Proof. An integral domain is a commutative domain, and a field is a commutative division ring. Hence it suffices to show that R is a division ring.

First, let $A = R \setminus \{0\}$. Consider arbitrary $a \in A$ and consider the map

$$\begin{aligned} \varphi_a : A &\longrightarrow A \\ x &\longmapsto \varphi_a(x) = ax. \end{aligned}$$

Now we know $a \neq 0$, so $ax = ay$ implies $x = y$ so φ_a is injective. Since R is finite, A is finite. Therefore φ_a is bijective. Therefore there exists $1_a \in A$ such that $a \cdot 1_a = a$.

Next we show that $1_a = 1_b$ for any $a, b \in A$, which implies existence and uniqueness of multiplicative identity, which we denote as 1. To verify $1_a = 1_b$, note that (we suppress $\cdot$ sign)

$$(ab)1_a = (ba)1_a = b(a1_a) = ba = ab = a(b1_b) = (ab)1_b.$$

Now consider any $a \in A$. Since the map φ_a is bijective, there exists unique element x such that $ax = 1$. Since R is commutative, $ax = xa = 1$. Hence R is a division ring. Therefore R is a field. $\square$

Homework 11

Sun 20 Nov 2022 22:21

Herstein, section 4.2, page 139, Question 8. Herstein, section 4.3, page 146, Questions 1, 2, 3, 4, 18, 20, 21, 22, 23, 24.

8. If F is a finite field, show that: (a) There exists a prime p such that $pa = 0$ for all $a \in F$.

Proof. Let p be the smallest nonzero natural number such that $p1 = 0$. We know that such p exists, since F is finite, there must exist two distinct natural numbers $n, m \in \{1, 2, \dots, |F| + 1\}$ such that $n1 = m1$. Then assuming $n < m$, we have $(m - n)1 = 0$.

p must be prime. If $p = mn$ where $m, n > 0$ are integers, then $p1 = (m1)(n1) = 0$, so either $(m1) = 0$ or $(n1) = 0$, a contradiction to the fact that p is minimal.

Now for any $a \in F$, we have $pa = p(a1) = a(p1) = a0 = 0$. $\square$

- (b) If F has q elements, then $q = p^n$ for some integer n . (Hint: Cauchy's Theorem)

Proof. If there were another element with order s , s prime and $s \neq p$, we would have an element in F , say a , of order s (by Cauchy's Theorem). But then we can apply Euclid's Algorithm to p and s , and have $pa = (ms + r)a = ra$ where $0 < r < s$, so $pa \neq 0$, a contradiction. Hence $|F| = p^n$. $\square$

- 1. If R is a commutative ring and $a \in R$, let $L(a) = \{x \in R \mid xa = 0\}$. Prove that $L(a)$ is an ideal of R .

Proof. Take $x, y \in L(a)$. Hence $xa = 0$ and $ya = 0$. Now

$$(x + (-y))a = xa + (-y)a = xa - ya = 0 - 0 = 0.$$

Hence $(x + (-y))a \in L(a)$. Thus $L(a)$ is a subgroup of the additive abelian group R . Now let $x \in L(a)$. Since R commutative, for any $r \in R$,

$$(xr)a = (rx)a = r(xa) = r0 = 0.$$

Hence $xr = rx \in L(a)$. Hence $L(a) \triangleleft R$. $\square$

- 2. If R is a commutative ring with 1 and R has no ideals other than (0) and itself, prove that R is a field. (Hint: Look at Example 9.)

Proof. Let $a \in R$ nonzero. Then by example 9, the set $(a) = \{xa : x \in R\}$ is an ideal of R . Clearly $(a) \neq (0)$ since $a = 1a \in (a)$. Thus $(a) = R$. Hence the operation

$$\begin{aligned} \varphi : R &\longrightarrow R \\ x &\longmapsto \varphi(x) = xa. \end{aligned}$$

is a bijection. Injectivity comes from cancellation law while surjectivity comes from counting.

Hence there exists b such that $\varphi(b) = ba = 1$. Hence R is a division ring, thus a field. $\square$

- *3. If $\varphi : R \rightarrow R'$ is a homomorphism of R onto R' and R has a unit element, 1, show that $\varphi(1)$ is the unit element of R' .

Proof. Since φ is onto, for any element $r \in R'$, there exists $a \in R$ such that $\varphi(a) = r$. Observe that

$$\varphi(1)\varphi(a) = \varphi(1a) = \varphi(a) = \varphi(a1) = \varphi(a)\varphi(1).$$

Hence $\varphi(1)$ is indeed the multiplicative identity of R' . $\square$

- 4. If I, J are ideals of R , define $I + J$ by $I + J = \{i + j \mid i \in I, j \in J\}$. Prove that $I + J$ is an ideal of R .

Proof. First, take $i + j, i' + j' \in I + J$, and note

$$(i + j) \pm (i' + j') = i + j \pm i' \pm j' = i \pm i' + j \pm j' = (i \pm i') + (j \pm j') \in I + J.$$

Hence $I + J$ is a subgroup of the additive abelian group. Now let $r \in R$. Note that

$$r(i + j) = ri + rj \in I + J.$$

and

$$(i + j)r = (ir) + (jr) \in I + J.$$

By using the fact that I, J are each an ideal of R . Hence $I + J \triangleleft R$. $\square$

- 18. Show that $R \oplus S$ is a ring and that the subrings $\{(r, 0) \mid r \in R\}$ and $\{(0, s) \mid s \in S\}$ are ideals of $R \oplus S$ isomorphic to R and S , respectively.

Proof. $R \oplus S$ is a additive abelian group, by relevant result for groups. Now we verify the ring properties. Let (r, s) and $(r', s') \in R \oplus S$.

$$\text{M0 } (r, s)(r', s') = (rr', ss') \in R \oplus S.$$

M1 associativity is inherited from R and S .

$$\begin{aligned} \text{D1 } (r, s)((r', s') + (r'', s'')) &= (r, s)(r' + r'', s' + s'') = (r(r' + r''), s(s' + s'')) \\ &= (rr' + rr'', ss' + ss'') = (rr', ss') + (rr'', ss'') = (r, s)(r', s') + (r, s)(r'', s''). \end{aligned}$$

D2 Similar to D1.

Now construct mapping

$$\begin{aligned} \varphi : R &\longrightarrow R \oplus S \\ r &\longmapsto \varphi(r) = (r, 0). \end{aligned}$$

To verify it is a homomorphism, note that

$$\varphi(rr') = (rr', 0) = (r, 0)(r', 0) = \varphi(r)\varphi(r').$$

To verify injectivity, note that for any $r \neq r'$, obviously $(r, 0) \neq (r', 0)$.

To verify surjectivity onto $\{(r, 0) : r \in R\}$, fix any $(r, 0)$, and note that $\varphi(r) = (r, 0)$.

To verify it is an ideal, it is a normal subgroup of the abelian additive group by previous knowledge. Now fix any $(r, s) \in R \oplus S$. Now

$$(r', 0)(r, s) = (r'r, 0s) = (r'r, 0) \in \{(r, 0) : r \in R\}.$$

The other side is similar. Hence an isomorphism. The proof for S is similar. $\square$

- 20. If I, J are ideals of R , let $R_1 = R/I$ and $R_2 = R/J$. Show that $\varphi : R \rightarrow R_1 \oplus R_2$ defined by $\varphi(r) = (r + I, r + J)$ is a homomorphism of R into $R_1 \oplus R_2$ such that $\text{Ker } \varphi = I \cap J$.

Proof. To verify homomorphism,

$$\begin{aligned}
 & \varphi(rr') \\
 &= (rr' + I, rr' + J) \\
 &= (rr' + rI + r'I + II, rr' + rJ + r'J + JJ) \\
 &= ((r + I)(r' + I), (r + J)(r' + J)) \\
 &= (r + I, r + J)(r' + I, r' + J) \\
 &= \varphi(r)\varphi(r').
 \end{aligned}$$

Now we investigate the kernel.

$$\begin{aligned}
 r &\in \text{Ker } \varphi \\
 &\Leftrightarrow \varphi(r) = (I, J) \\
 &\Leftrightarrow (r + I, r + J) = (I, J) \\
 &\Leftrightarrow r \in I \wedge r \in J \\
 &\Leftrightarrow r \in I \cap J.
 \end{aligned}$$

□

- 21. Let $\mathbb{Z}_{15}$ be the ring of integers mod 15. Show that $\mathbb{Z}_{15} \simeq \mathbb{Z}_3 \oplus \mathbb{Z}_5$.

Proof. By Chinese Remainder Theorem, for each $m \in \mathbb{Z}_5$, $n \in \mathbb{Z}_3$, the system of equations $x \equiv m \pmod{5}$, $x \equiv n \pmod{3}$ has unique solution modulo $5 \cdot 3 = 15$. Hence we may define the mapping

$$\begin{aligned}
 \varphi : \mathbb{Z}_5 \times \mathbb{Z}_3 &\longrightarrow \mathbb{Z}_{15} \\
 (m, n) &\longmapsto \varphi((m, n)) = p_{mn}.
 \end{aligned}$$

where p_{mn} is the uniquely determined solution of congruence equations.

Moreover, φ is surjective, since each element in $\mathbb{Z}_{15}$ has a remainder when divided by 5 or divided by 3. Hence φ is a bijection by counting.

Now we verify φ is a homomorphism. Let $m, m' \in \mathbb{Z}_5$ and $n, n' \in \mathbb{Z}_3$. Then by properties of congruence,

$$\begin{aligned}
 p_{mn} + p_{m'n'} &\equiv m + m' \pmod{5} \\
 p_{mn} + p_{m'n'} &\equiv n + n' \pmod{3}.
 \end{aligned}$$

Also,

$$\begin{aligned}
 p_{mn}p_{m'n'} &\equiv mm' \pmod{5} \\
 p_{mn}p_{m'n'} &\equiv nn' \pmod{3}.
 \end{aligned}$$

Hence $\varphi((m + m', n + n')) = p_{mn} + p_{m'n'}$, and $\varphi((mm', nn')) = p_{mn}p_{m'n'}$. Thus a homomorphism. Because it is bijective, it is isomorphism. □

- 22. Let $\mathbb{Z}$ be the ring of integers and m, n two relatively prime integers, I_m the multiples of m in $\mathbb{Z}$, and I_n the multiples of n in $\mathbb{Z}$. (a) What is $I_m \cap I_n$?

Solution. It is I_p where p is the least common multiple of m and n . In case m, n are relatively prime, $p = mn$. $\square$

(b) Use the result of Problem 20 to show that there is a one-to-one homomorphism from $\mathbb{Z}/I_{mn}$ to $\mathbb{Z}/I_m \oplus \mathbb{Z}/I_n$.

Proof. Define $\varphi : \mathbb{Z}/I_{mn} \rightarrow \mathbb{Z}/I_m \oplus \mathbb{Z}/I_n$. It is straightforward to verify that $\mathbb{Z}/I_m, \mathbb{Z}/I_n$ are ideals of $\mathbb{Z}/I_{mn}$. Also, m, n coprime implies $I_m \cap I_n = I_{mn}$ which is identity in $\mathbb{Z}/I_{mn}$, so φ is injective. $\square$

- 23. If m, n are relatively prime, prove that $\mathbb{Z}_{mn} \simeq \mathbb{Z}_m \oplus \mathbb{Z}_n$. (Hint: Use a counting argument to show that the homomorphism of Problem 22(b) is onto.)

Proof. We have $|\mathbb{Z}/I_{mn}| = mn$, $\mathbb{Z}/I_m$ has order m and $\mathbb{Z}/I_n$ has order n . Hence by counting, φ is an isomorphism. $\square$

- 24. Use the result of Problem 23 to prove the Chinese Remainder Theorem, which asserts that if m and n are relatively prime integers and a, b any integers, we can find an integer x such that $x \equiv a \pmod{m}$ and $x \equiv b \pmod{n}$ simultaneously.

Proof. Fix m, n . Consider an isomorphism $\varphi : \mathbb{Z}/I_{mn} \rightarrow \mathbb{Z}/I_m \oplus \mathbb{Z}/I_n$. By surjective, there exists $c_1, c_2 \in \mathbb{Z}_{mn}$ such that $\varphi(c_1) = (1, 0)$ and $\varphi(c_2) = (0, 1)$.

Let a_0, b_0 be element in $\mathbb{Z}_m, \mathbb{Z}_n$ such that $a_0 \equiv a \pmod{m}$ and $b_0 \equiv b \pmod{n}$.

Now let $x = ac_1 + bc_2 \in \mathbb{Z}_{mn}$. $\square$

Homework 12

Mon 21 Nov 2022 20:15

Herstein, section 4.4, page 151, Questions 9, 10, 11, 12, 13.

- 9. Show that $(p-1)/2$ of the numbers $1, 2, \dots, p-1$ are quadratic residues and $(p-1)/2$ are quadratic nonresidues mod p . [Hint: Show that $\{x^2 \mid x \neq 0 \in \mathbb{Z}_p\}$ forms a group of order $(p-1)/2$.]

Proof. p has to be a prime, to ensure that $\mathbb{Z}_p^\times$ is a multiplicative group mod p . Consider the mapping

$$\begin{aligned} \varphi : \mathbb{Z}_p^\times &\longrightarrow \mathbb{Z}_p^\times \\ x &\longmapsto \varphi(x) = x^2. \end{aligned}$$

We show that this map is a homomorphism between two groups of multiplication modulo p . Let $x, y \in \mathbb{Z}_p^\times$. We have

$$\varphi(x)\varphi(y) = x^2y^2 = (xy)^2 = \varphi(xy).$$

Hence the image of φ , the set of quadratic residues, form a subgroup of $\mathbb{Z}_p^\times$. By first homomorphism theorem,

$$\mathbb{Z}_p^\times / \ker \varphi \cong \varphi(\mathbb{Z}_p^\times).$$

So what is the size of kernel? Consider $x^2 \equiv 1 \pmod{p}$. This has exactly two solutions, $x \equiv 1 \pmod{p}$, or $x \equiv -1 \pmod{p}$. Thus $|\ker \varphi| = 2$. Also, $|\mathbb{Z}_p^\times| = p - 1$ since it is $\mathbb{Z}_p$ with 0 removed. Hence

$$|\varphi(\mathbb{Z}_p^\times)| = \frac{|\mathbb{Z}_p^\times|}{|\ker \varphi|} = \frac{p-1}{2}.$$

□

10. Let $m > 0$ be in $\mathbb{Z}$, and suppose that m is not a square in $\mathbb{Z}$. Let $R = \{a + \sqrt{m}b \mid a, b \in \mathbb{Z}\}$. Prove that under the operations of sum and product of real numbers R is a ring.

Proof. Since $(\sqrt{m})^2 = m \in \mathbb{Z}$, R is equal to

$$R = \{f(\sqrt{m}) : f \in \mathbb{Z}[x]\}.$$

And all ring properties are inherited from the ring $\mathbb{Z}[x]$.

We could justify this by constructing the homomorphism

$$\begin{aligned} \varphi : \mathbb{Z}[x] &\longrightarrow \mathbb{R} \\ f &\longmapsto \varphi(f) = f(\sqrt{m}). \end{aligned}$$

Now φ is a homomorphism since

$$\varphi(f + g) = (f + g)(\sqrt{m}) = f(\sqrt{m}) + g(\sqrt{m}) = \varphi(f) + \varphi(g).$$

and

$$\varphi(fg) = (fg)(\sqrt{m}) = f(\sqrt{m})g(\sqrt{m}) = \varphi(f)\varphi(g).$$

Hence $R = \varphi(\mathbb{Z}[x])$ is a subring of $\mathbb{R}$. Hence R is a ring. □

11. If p is an odd prime, let us set $I_p = \{a + \sqrt{m}b \mid p \mid a \text{ and } p \mid b\}$, where $a + \sqrt{m}b \in R$, the ring in Problem 10. Show that I_p is an ideal of R .

Proof. $I_p = pR$ is an ideal of R . To verify,

$$pR = \{pf(\sqrt{m}) : f \in \mathbb{Z}[x]\} = \{f(\sqrt{m}) : f \in p\mathbb{Z}[x]\}.$$

Now $p\mathbb{Z}[x]$ is an ideal of $\mathbb{Z}[x]$, so pR is an ideal of R .

To justify, consider the surjective homomorphism

$$\begin{aligned} \varphi : \mathbb{Z}[x] &\longrightarrow R \\ f &\longmapsto \varphi(f) = f(\sqrt{m}). \end{aligned}$$

We note that $pR = \varphi(p\mathbb{Z}[x])$. Hence $p\mathbb{Z}[x] \triangleleft \mathbb{Z}[x] \Rightarrow pR \triangleleft R$, since image of an ideal under surjective homomorphism is an ideal. □

12. If m is a quadratic nonresidue mod p , show that the ideal I_p in Problem 11 is a maximal ideal of R .

Proof. Define

$$\begin{aligned}\psi : \mathbb{Z}_p[x] &\longrightarrow R/pR \\ f &\longmapsto \psi(f) = f(\sqrt{m}) + pR.\end{aligned}$$

We first verify the map is surjective. Let $a + b\sqrt{m} + pR \in R/pR$. We may assume $a, b \in \mathbb{Z}_p$ due to the term pR . Hence we have $a + bx \in \mathbb{Z}_p[x]$, and

$$\psi(a + bx) = a + b\sqrt{m} + pR.$$

Next we verify the map is a homomorphism. Let $f_1, f_2 \in \mathbb{Z}_p[x]$. Then

$$\psi(f_1) + \psi(f_2) = f_1(\sqrt{m}) + pR + f_2(\sqrt{m}) + pR = (f_1 + f_2)(\sqrt{m}) + pR = \psi(f_1 + f_2).$$

Proof for multiplication is identical.

Now $\ker \psi = \{f \in \mathbb{Z}_p[x] : f(\sqrt{m}) = 0\} = (x^2 - m)$. Now since $x^2 - m$ is irreducible in $\mathbb{Z}_p[x]$ and $\mathbb{Z}_p$ is a field, $(x^2 - m)$ is a maximal ideal of $\mathbb{Z}_p[x]$. Using correspondence theorem, we see that 0 is a maximal ideal of R/pR . Hence R/pR is a field. Hence pR is a maximal ideal of R . $\square$

13. In Problem 12 show that R/I_p is a field having p^2 elements.

Proof. From the last problem, we have the surjective homomorphism $\psi : \mathbb{Z}_p[x] \rightarrow R/pR$. By first homomorphism theorem,

$$\mathbb{Z}_p[x]/(x^2 - m) \cong R/pR.$$

How many elements are in $A := \mathbb{Z}_p[x]/(x^2 - m)$? By division algorithm, each element in A can be identified as a polynomial in $\mathbb{Z}_p[x]$ with degree at most 1, i.e. $A = \{a + bx + (x^2 - m) : a, b \in \mathbb{Z}_p\}$. Also, for each choice of a, b , we get distinct elements from A . Since there are p^2 choices for (a, b) , there are p^2 elements in A . Therefore there are p^2 elements in R/pR . $\square$

Herstein, section 4.5, page 163, Questions 3, 5, 12, 18.

- 3. Find the greatest common divisor of the following polynomials over $\mathbb{Q}$, the field of rational numbers.

(a) $x^3 - 6x + 7$ and $x + 4$.

Solution. The gcd is 1. $\square$

(b) $x^2 - 1$ and $2x^7 - 4x^5 + 2$.

Solution. The gcd is $(x - 1)$. $\square$

(c) $3x^2 + 1$ and $x^6 + x^4 + x + 1$.

Solution. The gcd is 1. $\square$

(d) $x^3 - 1$ and $x^7 - x^4 + x^3 - 1$.

Solution. The gcd is $(x^3 - 1)$. $\square$

5. In Problem 3, let $I = \{f(x)a(x) + g(x)b(x)\}$, where $f(x), g(x)$ run over $\mathbb{Q}[x]$ and $a(x)$ is the first polynomial and $b(x)$ the second one in each part of the problem. Find $d(x)$, so that $I = (d(x))$ for Parts (a), (b), (c), and (d).

Proof. (a) Let $d(x) = 1$. Since $a(x)$ and $b(x)$ are coprime, there exists $u, v \in \mathbb{Z}$ such that $au + bv = 1$. Hence for any polynomial f in $\mathbb{Q}[x]$, we have $f = (fu)a + (fv)b$.

We prove the result in general, so we don't have to repeat the procedure for b, c, d . Let $d(x) = (a(x), b(x))$. Let $a'(x) = a(x)/d(x)$ and $b'(x) = b(x)/d(x)$. Now $(a', b') = 1$, so there exists $u, v \in \mathbb{Z}$ such that $a'u + b'v = 1$. Hence for any polynomial $f d(x)$ in $(d(x))$, we have $f d(x) = (fu)a'd(x) + (fv)b'd(x) = (fu)a + (fv)b$. $\square$

12. If $F \subset K$ are two fields and $f(x), g(x) \in F[x]$ are relatively prime in $F[x]$, show that they are relatively prime in $K[x]$.

Proof. Since f and g are coprime in $F[x]$, there exists $a, b \in F[x]$ such that $af + bg = 1$. Then suppose that we have $u \in K[x]$ such that $u|f$ and $u|g$. Then $u|af + bg = 1$, so $u = 1$. This shows that $(f, g) = 1$ in $K[x]$. $\square$

18. Let F be a finite field. Show that $F[x]$ contains irreducible polynomials of arbitrarily high degree. (Hint: Try to imitate Euclid's proof that there is an infinity of prime numbers.)

Proof. Suppose for sake of contradiction that there exists $f \in F[x]$ which is irreducible and has highest degree. Since F is finite, the number of possible polynomials with degree $\leq \deg f$ is finite, so in particular it has finitely many irreducible polynomials. Let $f_1, \dots, f_n$ be all irreducible polynomials where $f = f_1$. We now define

$$g = f_1 \dots f_n + 1.$$

Suppose there exists some $c \in F[x]$ such that $c|g$. Then we may assume without loss of generality that c is irreducible, otherwise we can keep considering factors of c until we obtain an irreducible polynomial. Hence $c = f_i$ for some i . Then $c|1$, so $c = 1$. This implies g is irreducible. But $\deg g > \deg f_1$, a contradiction. So $F[x]$ contains irreducible polynomials of arbitrarily large degree. $\square$

Homework 13

Wed 30 Nov 2022 13:51

Herstein, section 4.5, page 164, Questions 10 (but use your knowledge from section 4.6).

10. Show that the following polynomials are irreducible over the field F indicated.

(a) $x^2 + 7$ over $F = \text{real field} = \mathbb{R}$.

Proof. If $p(x) = x^2 + 7 \in F[x]$ is reducible, then it be factored into the form $p(x) = (x - a)(x - b)$. Thus $p(x) = 0$ for some $x \in F$. However from our knowledge of the reals, $x^2 + 7 \geq 7 > 0$. Hence $p(x)$ is irreducible. $\square$

(b) $x^3 - 3x + 3$ over $F = \text{rational field} = \mathbb{Q}$.

Proof. Observe that 3 is prime, $3 \nmid a_3 = 1$, $3 \mid a_2 = 0$, $3 \mid a_1 = -3$, $3 \mid a_0 = 3$, and $3^2 \nmid a_0 = 3$. Hence Eisenstein Criterion applies, so $p(x)$ irreducible over $\mathbb{Q}$. $\square$

(c) $x^2 + x + 1$ over $F = \mathbb{Z}_2$.

Proof. Since $p(x)$ has degree 2, if it is reducible then it has a root. However $p(0) = 1$ and $p(1) = 1$, so p does not have a root. Hence p irreducible. $\square$

(d) $x^2 + 1$ over $F = \mathbb{Z}_{19}$.

Proof. Since $p(x)$ has degree 2, in order to be reducible it has to have a root. Suppose $p(a) = 0$. Then $a^2 \equiv -1 \pmod{19}$. However this congruence equation does not have a solution. $\square$

(e) $x^3 - 9$ over $F = \mathbb{Z}_{13}$.

Proof. Since $p(x)$ has degree 3, in order to be reducible it has to have a root. Suppose $p(a) = 0$, then $a^3 - 9 \equiv 0 \pmod{13}$. However this congruence equation does not have a solution. $\square$

(f) $x^4 + 2x^2 + 2$ over $F = \mathbb{Q}$.

Proof. Pick $p = 2$, and Eisenstein Criterion applies. $\square$

Herstein, section 4.6, page 171, Questions 2, 3, 6, 10 .

2. Prove that $f(x) = x^3 + 3x + 2$ is irreducible in $\mathbb{Q}[x]$.

Proof. $f(x)$ irreducible iff $f(x+1) = 6 + 6x + 3x^2 + x^3$ is irreducible. By picking $p = 3$ we see that Eisenstein Criterion applies to $f(x+1)$, so f irreducible in $\mathbb{Q}[x]$. $\square$

3. Show that there is an infinite number of integers a such that $f(x) = x^7 + 15x^2 - 30x + a$ is irreducible in $\mathbb{Q}[x]$. What a 's do you suggest?

Proof. Pick $p = 5$ and apply Eisenstein Criterion. For the criterion to apply we only need $p|a$ and $p^2 \nmid a$. Therefore, it suffices to let $a = 5q$ where q is prime and $q \neq 5$. $\square$

6. Let F be the field and φ an automorphism of $F[x]$ such that $\varphi(a) = a$ for every $a \in F$. If $f(x) \in F[x]$, prove that $f(x)$ is irreducible in $F[x]$ if and only if $g(x) = \varphi(f(x))$ is.

Proof. Use exercise 10 below (which is proved without using this exercise), we conclude that φ preserves degree of all $f(x) \in F[x]$.

If $f(x)$ is reducible, say $f(x) = a(x)b(x)$ where a, b have degree at least 1, then

$$\varphi(f(x)) = \varphi(a(x)b(x)) = \varphi(a(x))\varphi(b(x)).$$

where $a(x)$ and $b(x)$ both have degree at least 1. Thus $\varphi(f(x))$ reducible. Conversely, considering φ^{-1} , we see that $\varphi(f(x))$ reducible implies $f(x)$ reducible. Take contrapositives and we have the claim. $\square$

10. Let φ be an automorphism of $F[x]$, where F is a field, such that $\varphi(a) = a$ for every $a \in F$. Prove that if $f(x) \in F[x]$, then $\deg \varphi(f(x)) = \deg f(x)$.

Proof. We proceed by strong induction on $\deg f$. The statement is clearly true when $\deg f = 0$, where $f(x) = a \in F$. Now assume $\deg \varphi(f(x)) = \deg f(x)$ whenever $\deg f < n$ for some $n \in \mathbb{N}$. Now let $g(x) \in F[x]$ and $\deg g(x) = n$. Let $h(x) \in F[x]$ and $1 \leq \deg h < \deg g$.

Now since F is a field, $F[x]$ is a Euclidean Domain. Hence we may apply the division algorithm: there exists $a, r \in F[x]$ such that $g = ah + r$, $\deg r < \deg h$. Hence $\deg g = \deg a + \deg h$, so $\deg a < n$. Now by the induction hypothesis, φ preserves degree of a, h and r . Hence

$$\begin{aligned} \deg \varphi(g) &= \deg (\varphi(a)\varphi(h) + \varphi(r)) \\ &= \max\{\deg \varphi(a) + \deg \varphi(h), \deg \varphi(r)\} \\ &= \max\{\deg a + \deg h, \deg r\} \\ &= \deg(ah + r) \\ &= \deg g. \end{aligned}$$

This concludes the induction, so φ preserves degree of all elements in $F[x]$. $\square$

Herstein, section 5.1, page 179, Questions 4, 7, 8, 9, 10 (not to be handed in - but do these at your leisure and compare against the solutions).

4. If D is an integral domain, show that $D[x, y]$ is an integral domain.

Proof. We observe that

$$D[x, y] = (D[x])[y].$$

To justify, first note that each polynomial $p(x, y)$ in $D[x, y]$ can be rearranged to the form $q_n(x)y^n + \dots + q_1(x)y + q_0(x)$, so $D[x, y] \subset D[x][y]$. Conversely, for each polynomial $p(x, y) \in (D[x])[y]$, we may simply expand it, so $(D[x])[y] \subset D[x, y]$.

Hence this follows from using this fact twice: If D is integral domain then $D[x]$ is integral domain. $\square$

7. If F is a field of characteristic $p \neq 0$, show that $(a + b)^p = a^p + b^p$ for all $a, b \in F$. (Hint: Use the binomial theorem and the fact that p is a prime.)

Proof. By binomial theorem,

$$(a + b)^p = \sum_{k=0}^p \binom{p}{k} a^k b^{p-k}.$$

But for $k = 1, \dots, p-1$, we have $p \mid \binom{p}{k}$. Thus only the terms $k = 0$ and $k = p$ survive. Hence

$$(a + b)^p = a^p + b^p.$$

$\square$

8. If F is a field of characteristic $p \neq 0$, show that $(a + b)^m = a^m + b^m$, where $m = p^n$, for all a, b in F and any positive integer n .

Proof. Almost same as last exercise. $\square$

9. Let F be a field of characteristic $p \neq 0$ and let $\varphi : F \rightarrow F$ be defined by $\varphi(a) = a^p$ for all $a \in F$. (a) Show that φ defines a monomorphism of F into itself.

Proof.

$$\ker \varphi = \{a \in F : a^p = 0\} = \{0\}.$$

Since F is a field (so it has no zero divisor). Hence φ injective. $\square$

(b) Give an example of a field F where φ is not onto. (Very hard.)

Proof. $\mathbb{Z}_2(x)$ is such an example. The characteristic is 2. Also, there does not exist a such that $a^2 = x \in \mathbb{Z}_2(x)$. $\square$

10. If F is a finite field of characteristic p , show that the mapping φ defined above is onto, hence is an automorphism of F .

Proof. Since φ is a monomorphism, and F is finite, φ is onto by counting, therefore an automorphism. $\square$

Lecture 1: lecture

Sat 27 Aug 2022 12:14

1 Prerequisite

1.1 Sets and Functions

1.2 Arithmetics of Integers

Notation. $\mathbb{N} = \{1, 2, 3, \dots\}$

Theorem 1 (Well-ordering). Any nonempty set of nonnegative integers has a smallest member.

Theorem 2 (Euclid's Algorithm). For $m, n \in \mathbb{Z}$ and $n \neq 0$, there exists $q, r \in \mathbb{Z}$ such that $0 \leq r < n$ and $m = nq + r$.

A *Division Algorithm* computes the result of Euclid's Algorithm. Division algorithms allow us to compute division efficiently, and are implemented in modern computers.

Definition 2 (Divisibility).

- We say m *divides* n , writing $m \mid n$, if $n = cm$ for some $c \in \mathbb{Z}$.
- A *proper divisor* m of n satisfies $m \mid n$ and $1 \leq m < |n|$.
- We write $a^k \parallel b : \Leftrightarrow a^k \mid b$ but $a^{k+1} \nmid b$.

Lemma 9 (Basic properties of divisibility).

1. $1 \mid n$ for all n , and $n \mid 0$ for all $n \neq 0$.
2. (transitivity) $a \mid b \wedge b \mid c \Rightarrow a \mid c$.
3. (linear combination) $a \mid b \wedge a \mid c \Rightarrow \forall x, y \in \mathbb{Z}, a \mid (bx + cy)$.
4. (antisymmetry) $a \mid b \wedge b \mid a \Rightarrow a = \pm b$.
5. (ordering) $a \mid b, a > 0, b > 0 \Rightarrow a \leq b$.
6. (multiplication) $m \neq 0 \Rightarrow (a \mid b \Leftrightarrow ma \mid mb)$.

Definition 3 (greatest common divisor). Given $a, b \in \mathbb{Z}$, not both 0, $c \in \mathbb{Z}$ is their greatest common divisor if:

- $c > 0$,
- $c \mid a \wedge c \mid b$, and
- if $d \mid a$ and $d \mid b$, then $d \mid c$.

Notation. GCD of two integers a, b is written as (a, b) . This needs to be

disambiguated from the 2-tuple (a, b) by context.

Theorem 3. For any two integers $a, b \in \mathbb{Z}$, their GCD is a linear combination. In other words, there exists $m, n \in \mathbb{Z}$ such that $(a, b) = ma + nb$.

Note also, that (a, b) is the least positive linear combination of a and b .

Definition 4 (relative prime). $a, b \in \mathbb{Z}$ are relatively prime if $(a, b) = 1$.

Definition 5 (prime). p is prime if

1. $p > 1$
2. $d|p \Rightarrow d = 1 \vee d = p$.

Theorem 4 (1.5.7). Every positive integer is prime or product of primes.

Theorem 5 (Fundamental Theorem of Arithmetics). There is a unique factorization of each positive integer into primes.

Remark. The FTA may not be generally true for rings, since rings may not have a well-ordered structure.

Lecture 2: Groups

Tue 30 Aug 2022 19:05

Corollary 1 (Euclid's Theorem). There are infinitely many primes.

Proposition 1. Let $p_1 = 2, p_2 = 3, \dots$ be an enumeration of primes. Then the smallest divisor of

$$(p_1 \dots p_n)^{(p_1 \dots p_n)^{(p_1 \dots p_n)}}$$

is p_{n+1} .

2 Groups

2.1 Definition and Examples

Definition 6. A *binary operation* ψ over a set G is a function

$$\psi : G \times G \rightarrow G.$$

Definition 7 (Group). A *group* is a set G equipped with a binary operation ψ such that

1. (G0)(closure) ψ is closed in G ,
2. (G1)(associativity) ψ is associative,
3. (G2)(identity) Exists an identity $e \in G$ such that $ae = ea = a$ for $\forall a \in G$, and
4. (G3)(inverse) For $\forall a \in G$, there exists $a^{-1} \in G$ such that $aa^{-1} = a^{-1}a = e$.

Example. The *trivial group* $\{e\}$. ◇

Example. The *symmetry group* $A(S)$ with *composition* $\circ$ as the binary operation. ◇

Example. $(\mathbb{Z}, +)$, $(\mathbb{R}^\times, \times)$ are addition group over integers and multiplication group over nonzero reals. ◇

Example. $(\mathbb{Z}_n, +_n)$ where $\mathbb{Z}_n = \{0, 1, \dots, n-1\}$ is the group of addition mod n . ◇

Example. $(\text{SL}_2(\mathbb{R}), *)$ is the *special linear group* of 2×2 matrices with determinant 1. This is precisely the positive isometries. ◇

Example. $(D_{2n}, *)$ is the *dihedral group* of symmetry of regular n -sided polygons.

This group can be constructed explicitly by the n th roots of unity. The group elements are then

$$\sigma : e^{2\pi i \frac{a}{n}} \mapsto e^{2\pi i \frac{a+1}{n}}$$

the counterclockwise rotation by $e^{\frac{2}{n}i\pi}$, and

$$\tau : e^{2\pi i \frac{a}{n}} \mapsto e^{-2\pi i \frac{a}{n}}$$

the reflection over real axis.

The mappings $1, \sigma, \sigma^2, \dots, \sigma^{n-1}$ are distinct. The mappings $\tau, \tau\sigma, \dots, \tau\sigma^{n-1}$ are all distinct and distinct from $1, \sigma, \dots, \sigma^{n-1}$.

Check: $\sigma\tau = \tau\sigma^{-1}$.

$$\sigma\tau \left(e^{2\pi i \frac{a}{n}} \right) = \sigma \left(e^{-2\pi i \frac{a}{n}} \right) = e^{-2\pi i \frac{1-a}{n}}.$$

$$\tau\sigma^{-1} \left(e^{2\pi i \frac{a}{n}} \right) = \tau \left(e^{2\pi i \frac{a-1}{n}} \right) = e^{2\pi i \frac{a-1}{n}}.$$

◇

Remark. Geometers often refer to the group D_{2n} as D_n .

Definition 8. A group $(G, *)$ is said to be a finite group if $\text{card } G < \infty$, in this case we say the *order* of group G is $|G| = \text{card } G$.

- Example.**
1. $|A(S)| = n!$ where $|S| = n$
 2. $|\mathbb{Z}| = +\infty$
 3. $|\mathbb{Z}_n| = n$
 4. $|\mathbb{R}^\times| = +\infty$
 5. $|\text{SL}_2(\mathbb{R})| = +\infty$
 6. (harder) one can consider $\text{SL}_2(\mathbb{Z}_p)$ has cardinality $(p-1)p(p+1)$
 7. $|D_{2n}| = 2n$

◇

Definition 9 (Abelian Group). A group $(G, *)$ is called *abelian* if $*$ is commutative, i.e., for all $a, b \in G$, we have $a * b = b * a$.

- Example.**
1. In general $A(S)$ is non-abelian
 2. $(\mathbb{Z}, +)$ is abelian
 3. $(\mathbb{Z}_n, +)$ is abelian
 4. $(\mathbb{R}^\times, \times)$ is abelian
 5. $\text{SL}_2(\mathbb{R})$ is non-abelian (matrices do not commute in general)
 6. D_{2n} is non-abelian when $n > 1$

◇

2.2 The most basic properties

Lemma 10. (2.2.1) Suppose that G is a group, then

1. the identity element $e \in G$ is unique;
2. for every $a \in G$, the inverse $a^{-1} \in G$ is unique;
3. for all $(a^{-1})^{-1} = a$;
4. for all $a, b \in G$, $(ab)^{-1} = b^{-1}a^{-1}$.

Proof. 1. Suppose $e_1, e_2 \in G$ satisfy identity axiom. Then

$$e_1 e_2 = e_1 = e_2$$

where we applied the identity property twice.

2. Suppose $b_1, b_2 \in G$ are both inverses for $a \in G$. Then if e is the identity of G ,

$$\begin{aligned} b_1 a b_2 &= b_1 (a b_2) = b_1 e = b_1 \\ &= (b_1 a) b_2 = e b_2 = b_2. \end{aligned}$$

3.

$$a^{-1} a = e = a a^{-1}.$$

So a is the inverse of a^{-1} .

4. Let $c = (ab)^{-1}$. Then $(ab)c = e$. Then $b^{-1}a^{-1}(ab)c = b^{-1}a^{-1}$. Use associativity, inverse and identity twice, we simplify the LHS to c .

□

Lemma 11 (cancellation rule). Suppose G is a group, and $a, b, c \in G$. Then

1. whenever $ab = ac$, one has $b = c$;
2. whenever $ba = ca$, one has $b = c$.

Proof. 1. Since $ab = ac$, $a^{-1}ab = a^{-1}ac$. Use associativity, inverse and identity axiom, and we simplify to $b = c$.

2. Similar.

□

2.3 Subgroups

Definition 10 (subgroup). Suppose H is a non-empty subset of a group G equipped with a binary operation. If H forms a group with respect to the same binary operation, then we call H a *subgroup* of G and write $H \leq G$.

Observation. Suppose G is a group with identity element e , then there are two *trivial subgroups*

- $G \leq G$
- $\{e\} \leq G$

Any subgroup H of G that is nontrivial is called a *proper subgroup*.

Lemma 12 (subgroup criterion). Suppose that G is a group, and $\emptyset \neq H \subset G$. Then $H \leq G$ iff for all $a, b \in H$ one has $ab^{-1} \in H$.

Proof. If $H \leq G$ then for all $b \in H$, one has $b^{-1} \in H$, so by closure we have for all $a, b \in H$, $ab^{-1} \in H$.

Let $a \in H$ with $\emptyset \neq H \subset G$, we may assume that for $b \in H$ we have $ab^{-1} \in H$. So in particular $aa^{-1} = e \in H$. Also, since $e \in H$, we have $ea^{-1} \in H$, so

$a^{-1} \in H$. For associativity, this is inherited from G . Finally, to check closure, assume that whenever $a, b \in H$, $b^{-1} \in H$, so $ab = a(b^{-1})^{-1} \in H$. $\square$

Example. 1. Let $n \in \mathbb{N}$ (think $n \geq 2$). Consider $n\mathbb{Z}$ as a subset of $\mathbb{Z}$ with $(\mathbb{Z}, +)$ a group. Then $n\mathbb{Z} \leq \mathbb{Z}$. Proof: If $a, b \in n\mathbb{Z}$, then $n|a$ and $n|b$, so $n|(a - b)$, where $a - b \in n\mathbb{Z}$. So subgroup criterion applies.

2. Let $(\mathbb{C}^\times, \times)$ be the group of non-zero complex numbers with ordinary multiplication. Consider

$$\Pi = \{z \in \mathbb{C} : |z| = 1\}.$$

Check that $\Pi \leq \mathbb{C}^\times$.

3. Consider $\mu_n = \{e^{2\pi i \frac{a}{n}} : 0 \leq a < n, a \in \mathbb{Z}\}$. Now

$$\mu_n \leq \Pi \leq \mathbb{C}^\times.$$

4. Let $(G, \cdot)$ be a group and $a \in G$. Then $\langle a \rangle = \{a^n : n \in \mathbb{Z}\}$ forms a subgroup of G .

Definition 11 (cyclic group). A group G generated by a single element a so that $G = \langle a \rangle$ is called *cyclic*.

5. We have $\langle 2 \rangle \leq \mathbb{Z}_6 = \langle 1 \rangle$. $\langle 3 \rangle \leq \mathbb{Z}_6$. $\mathbb{Z}_7$ does not have any proper subgroups.

6. (Centralizer) When $a \in G$, define

$$C(a) := \{g \in G : ga = ag\}.$$

Theorem 6. For each $a \in G$, one has $C(a) \leq G$.

Proof. Observe that whenever $h \in C(a)$, then

$$\begin{aligned} ha &= ah \\ a &= h^{-1}ha = h^{-1}ah \\ ah^{-1} &= h^{-1}ahh^{-1} = h^{-1}a \end{aligned}$$

. Hence $h^{-1}ha = h^{-1}ah$, where $h^{-1} \in C(a)$. Thus, whenever $g, h \in C(a)$, one has

$$a(gh^{-1}) = (ag)h^{-1} = (ga)h^{-1} = g(ah^{-1}) = g(h^{-1}a) = (gh^{-1})a.$$

So $gh^{-1} \in C(a)$. Subgroup criterion applies. $\square$

7. (Center)

$$Z(G) = \{z \in G : zx = xz \text{ for all } x \in G\}.$$

$Z(G)$ is a subgroup of G .

Proof. Observe that $Z(G) = \bigcap_{a \in G} C(a)$. Note: if K and H are both subgroups of G , then $K \cap H \leq G$. (In case of uncountable groups we need axiom of choice to settle this down.) $\square$

◇

Lecture 3

Wed 07 Sep 2022 10:31

11. (Conjugate Subgroup) Suppose $H \leq G$. Then, for each $g \in G$, the subset

$$g^{-1}Hg = \{g^{-1}hg : h \in H\}$$

is a subgroup of G .

12. (Normalizer) When $A \subset G$, define

$$N(A) := \{g \in G : gA = Ag\}$$

is a subgroup of G .

2.4 Lagrange's Theorem

Theorem 7 (Lagrange's Theorem). If G is finite and $H \leq G$, then

$$|H| \mid |G|.$$

Definition 12. A *relation* R on a set S is a statement concerning a pair of elements $a, b \in S$ of the shape aRb , which may be true or false. We write aRb to denote that the statement is true.

$$R : S \times S \rightarrow \{T, F\}.$$

Example. 1. On $\mathbb{Z}$, one has relation $=, \leq, >$

◇

Definition 13 (Equivalence Relation). An *equivalence relation* $\sim$ on a set S is a relation satisfying:

1. (reflexive) $a \sim a$ for $\forall a \in S$
2. (symmetry) if $a \sim b$ then $b \sim a$
3. (transitivity) if $a \sim b$ and $b \sim c$ then $a \sim c$.

Example. 1. The usual $=$ is an equivalence relation on any set S . Equivalence relation is a **generalization** of the notion of *equality*.

2. (Congruence modulo n) The set $\mathbb{Z}$ and “ $\equiv \pmod{n}$ ”.

- $a \equiv a \pmod{n}$ since $n|(a - a)$

- if $a \equiv b$ then $b \equiv a$ since $n|(a-b) \Rightarrow n|(b-a)$
 - if $a \equiv b$ and $b \equiv c$ then $a \equiv c$, since if $n|(a-b)$ and $n|(b-c)$ then $n|(a-b+b-c)$
3. Let G be a group and suppose $H \leq G$. Then when $a, b \in G$, define $a \sim b$ when $ab^{-1} \in H$. Check properties of equivalence relation:
- We have $a \sim a$ if and only if $aa^{-1} = e \in H$.
 - If $a \sim b$, then $ab^{-1} \in H$. Hence $(ab^{-1})^{-1} = ba^{-1} \in H$. Hence $b \sim a$.
 - If $a \sim b$ and $b \sim c$, then $ab^{-1} \in H$, and $bc^{-1} \in H$. Hence $(ab^{-1})(bc^{-1}) = (ac^{-1}) \in H$. Hence $a \sim c$.

Remark. Congruence modulo n is a **special case** of the equivalence defined here: Let $G = (\mathbb{Z}, +)$ and $H = n\mathbb{Z}$.

4. Define $a \sim b$ if there exists x such that $b = x^{-1}ax$. This equivalence relation is called *conjugacy*.

Intuition. In linear algebra, two linear maps that are identical up to a change of base are conjugate. Note how the definition looks like the change of base formulae.

◇

Definition 14 (Equivalence Class). The equivalence class of elements in a set S equipped with an equivalence relation $\sim$ is

$$[a] := \{b \in S : b \sim a\}.$$

Lemma 13. Equivalence classes are either coincidental or disjoint.

Theorem 8. Suppose that $\sim$ is an equivalence relation on S , then $\sim$ partitions S into equivalence classes. Thus

$$S = \cup [a]$$

where the union is over representatives of equivalence classes.

Proof. Since $a \in [a]$ for all $a \in S$, since $a \sim a$, so $S = \bigcup_{a \in S} \{a\} \subset \bigcup_{a \in S} [a]$.

Also, $[a] \subset S$, then $\bigcup_{a \in S} [a] \subset S$, and thus $S = \bigcup_{a \in S} [a]$. But in view of above note, we have that S is the union over representatives of each equivalence class. $\square$

Definition 15 (Coset). Consider now $\sim$ defined by $a \sim b$ iff $ab^{-1} \in H$. $[a] = \{a \in G : ba^{-1} \in H\} = \{b \in G : b \in Ha\} = Ha$. The set Ha (equivalence class) is called a *right coset* of H .

Left coset is defined in a symmetric manner.

Because G can be partitioned into equivalence classes and Ha is a equivalence class,

$$G = \bigcup_{a \in G, \text{representatives}} Ha$$

where the union is taken over representatives from each equivalence class.

Claim. When G is finite, for any $a \in G$, each right coset Ha contains exactly $|H|$ elements.

Proof. To confirm this, construct a bijection:

$$\begin{aligned} \phi : H &\longrightarrow Ha \\ h &\longmapsto \phi(h) = ha. \end{aligned}$$

Injective: $\phi(h_1) = \phi(h_2) \Leftrightarrow h_1a = h_2a \Leftrightarrow h_1 = h_2$

Surjective: If $g \in Ha$, then $g = ha$ for some $h \in H$. Now $\phi(h) = ha = g$.

Hence ϕ is a bijection and $|H| = |Ha|$. □

Now $|G| = \sum_{\text{rep}} |Ha| = \sum_{\text{rep}} |H| = n|H|$ for some $n =$ number of equivalence classes. We have proved

Theorem 9 (Lagrange's Theorem). Suppose $H \leq G$, then

$$|H| \mid |G|.$$

Lecture 4: Consequences of Lagrange's Theorem

Fri 09 Sep 2022 10:29

Corollary 2. (Theorem 2.4.3) Any group of prime order has no proper subgroups, and in particular is cyclic.

Proof. Say $|G| = p$ is some prime number, suppose $H \leq G$. Then $|H| \mid p$, so $|H| = 1$ or p , whence H must be G or $\{e\}$.

But if $H = \langle a \rangle$ for some nonidentity a , then $H \leq G$. We have $H \neq \{e\}$. Hence $H = G$. Thus G is cyclic. □

Remark. This means groups of prime order are **abelian** as well.

Definition 16 (Order of an element). Let G be a group. The smallest positive integer n such that $a^n = e$, if one exists, is called the *order* of $a \in G$, and is written $o(a)$. If no such integer exists, we define $o(a) = +\infty$.

- Example.**
1. In $(\mathbb{R}^\times)$, one has $o(-1) = 2$, because $(-1)^2 = 1$.
 2. In $\mathbb{Z}$, $o(1) = +\infty$.
 3. In $\mathbb{Z}_3$, $o(1) = 3$ since $1 + 1 + 1 \equiv_3 0$.

◇

Theorem 10. When G is a finite group, we have $o(a) \mid |G|$ for all $a \in G$.

Proof. If $a \in G$, we have $o(a) = |\langle a \rangle|$. Thus, by Lagrange's Theorem, we have $o(a) \mid |G|$ since $|\langle a \rangle| \mid |G|$. □

Corollary 3. Suppose that G is a finite group. Then for all $a \in G$,

$$a^{|G|} = e.$$

Proof. Since $o(a) \mid |G|$, we have $|G| = mo(a)$. Now

$$a^{|G|} = \left(a^{o(a)}\right)^m = e^m = e.$$

□

2.4.1 Elementary Number Theory

We've seen $(\mathbb{Z}_n, +)$. We think of it as a equivalence class of integers:

$$\mathbb{Z}_n = \{[0], [1], \dots, [n-1]\}$$

where the equivalence relation is integers modulo n .

$$[a] + [b] = [a + b]$$

We can also consider multiplication (mod n).

$$[a] \times [b] = [ab].$$

- (identity) $[1]$ is identity.
- (inverse) $[2]^{-1}$ does not exist in $\mathbb{Z}_6$.

Define

$$U_n = \{[a] : (a, n) = 1\}.$$

Notation. In number theory we often write U_n as $(\mathbb{Z}/n\mathbb{Z})^\times$

Claim. U_n forms a group with multiplication mod n .

Proof.

- (closure) If $(a, n) = 1$ and $(b, n) = 1$ then $(ab, n) = 1$. Hence $[a][b] = [ab] \in U_n$.

- (identity) $[1]$ is identity.

- (associativity) Inherited from $(\mathbb{Q}^x, \times)$
- (inverse) If $(a, n) = 1$, then there exists integers $x, y \in \mathbb{Z}$ such that $ax + ny = 1$ (Euclidean Algorithm), whence $ax \equiv_n 1$.

Then $[x] = [a]^{-1}$. So $(U_n, \times)$ is indeed a group.

□

Hence if $a \in U_n$, then $a^{|U_n|} \equiv_n 1$.

Definition 17 (Euler φ -function). The Euler φ -function is defined for each $n \in \mathbb{N}$ by

$$\varphi(n) = |\{1 \leq a \leq n : (a, n) = 1\}|.$$

Fact.

$$\phi(n) = n \prod_{p^4 | n, p \text{ prime}} \left(1 - \frac{1}{p}\right)$$

Corollary 4 (Euler's Theorem). When $(a, n) = 1$ one has $a^{\phi(n)} \equiv_n 1$.

Corollary 5 (Fermat's Little Theorem). When p is prime, one has $a^p \equiv_p a$ for all $a \in \mathbb{Z}$.

Proof. Apply Euler's Theorem when $(a, p) = 1$. When $(a, p) = 1$, If $p|a$, then $a^p \equiv a \equiv 0 \pmod{p}$. □

Which U_n are cyclic?

Observe: U_n is abelian.

Fact. Every finite abelian group is "isomorphic" to a group of the shape

$$U_n^{(d)} = \{a^d : a \in U_n\}$$

for some $d, n \in \mathbb{N}$.

2.5 Homomorphisms and normal subgroups

Definition 18 (Homomorphism). Let G and G' be two groups. The mapping

$$\phi : G \rightarrow G'$$

is called a *homomorphism* when for all $a, b \in G$, one has $\phi(ab) = \phi(a)\phi(b)$.

- Injective homomorphism is called *monomorphism*.
- Surjective homomorphism is called *epimorphism*.
- Bijective homomorphism is called *isomorphism*.
- Bijective self-mapping homomorphism is called an *automorphism*.

Definition 19. We say that two groups G and G' are *isomorphic* if there exists a mapping $\phi : G \rightarrow G'$ such that ϕ is an isomorphism. We write

$$G \cong G'.$$

Lecture 5

Mon 12 Sep 2022 10:30

Nonexample. Compare $(\mathbb{Z}_6, +)$ and $D_6 = \{\text{id}, \sigma, \sigma^2, \tau, \tau\sigma, \tau\sigma^2\}$. They both have 6 elements, hence there exists a bijection from $\mathbb{Z}_6$ onto D_6 . However there is no isomorphism.

Example. 1. (Trivial homomorphism)

$$\begin{aligned}\varphi : G &\longrightarrow G' \\ g &\longmapsto \varphi(g) = e'.\end{aligned}$$

2. Consider

$$\begin{aligned}\varphi : \mathbb{Z} &\longrightarrow \mathbb{Z}_n \\ a &\longmapsto \varphi(a) = [a] = a \pmod{n}.\end{aligned}$$

Then $\varphi(a + b) = [a + b] = [a] + [b] = \varphi(a) + \varphi(b)$.

3. (Identity map is a trivial homomorphism)

$$\begin{aligned}\varphi : G &\longrightarrow G \\ g &\longmapsto \varphi(g) = g.\end{aligned}$$

4.

$$\begin{aligned}\varphi : (\mathbb{Z}, +) &\longrightarrow (\mathbb{Q}^\times, \times) \\ n &\longmapsto \varphi(n) = 2^n.\end{aligned}$$

Now $\varphi(a + b) = 2^{a+b} = 2^a 2^b = \varphi(a)\varphi(b)$.

5.

$$\begin{aligned}\varphi : (\mathbb{Q}^\times, \times) &\longrightarrow (\mathbb{Q}^\times, \times) \\ x &\longmapsto \varphi(x) = x^2.\end{aligned}$$

Now $\varphi(ab) = (ab)^2 = a^2 b^2 = \varphi(a)\varphi(b)$.

◇

Theorem 11 (Cayley's Theorem). Every group G is isomorphic to a subgroup of $A(S)$ for some set S .

Corollary 6. When G is a finite group, then G is isomorphic to a subgroup of S_n for some n .

Proof. Let G be a group, and put $S = G$. We define the set $A(G)$ to be the set of bijective self-mapping of G . Observe that whenever $a \in G$, the map

$$\begin{aligned} T_a : G &\longrightarrow G \\ g &\longmapsto T_a(g) = ag. \end{aligned}$$

belongs to $A(G)$. To check bijectivity, observe that $T_a^{-1} := T_{a^{-1}}$ (for all a , $(T_a^{-1} \circ T_a)g = T_a^{-1}(ag) = (a^{-1}a)g = g$).

Observe next that $H = \{T_a : a \in G\}$ is a subgroup of $A(G)$. For whenever $T_a, T_b \in H$, one has

$$T_a \circ T_b^{-1} = T_a \circ T_{b^{-1}} = T_{ab^{-1}} \in H.$$

Hence $H \leq A(G)$ by subgroup criterion.

To justify the last inequality, note: if $a, b \in G$, for all $g \in G$, one has

$$(T_a \circ T_b)(g) = T_a(bg) = abg = T_{ab}(g),$$

so $T_a \circ T_b = T_{ab}$.

Goal now: Show that $G \cong H \leq A(G)$. We need to construct an isomorphism $\varphi : G \rightarrow H$. Define

$$\begin{aligned} \varphi : G &\longrightarrow H \\ a &\longmapsto \varphi(a) = T_a. \end{aligned}$$

This is a homomorphism, since for $a, b \in G$, one has

$$\varphi(ab) = \varphi(a)\varphi(b) = T_{ab} = T_a T_b = \varphi(a)\varphi(b).$$

Injectivity: $\varphi(a) = \varphi(b) \Leftrightarrow T_a = T_b \Leftrightarrow \forall g \in G, ag = bg \Leftrightarrow a = b$. So $\varphi(a) = \varphi(b) \Leftrightarrow a = b$, so φ is injective.

Surjectivity: Whenever $T_a \in H$, one has $\varphi(a) = T_a$, so there exists $a \in G$ with $\varphi(a) = T_a$. So φ is a bijective homomorphism (i.e. an isomorphism from G to H). Thus $G \cong H \leq A(G)$. $\square$

Lemma 14. (2.5.2 + 2.5.3) If $\varphi : G \rightarrow G'$ is a homomorphism of groups then $\varphi(G) \leq G'$.

Proof. Homework exercise. $\square$

Definition 20. If $\varphi : G \rightarrow G'$ is a homomorphism of groups, then the *kernel* of φ is defined by $\ker(\varphi) = \{a \in G : \varphi(a) = e'\}$.

Fact. $\ker \varphi \leq G$.

Lecture 6

Wed 14 Sep 2022 10:32

Lemma 15. If $\varphi : G \rightarrow G'$ is a homomorphism of groups, then $\varphi(G) \leq G'$.

Proof. Homework exercise. □

Theorem 12. If $\varphi : G \rightarrow G'$ is a homomorphism of groups, then:

1. $\ker \varphi \leq G$
2. For all $g \in G$, one has $g^{-1}(\ker \varphi)g \subset \ker \varphi$

Proof. Suppose $a, b \in \ker \varphi$. Then

$$\varphi(ab^{-1}) = \varphi(a)\varphi(b)^{-1} = e'(e')^{-1} = e'.$$

So $ab^{-1} \in \ker \varphi$. Thus $\ker \varphi \leq G$ by subgroup criterion. Whenever $k \in \ker \varphi$, one has for all $g \in G$,

$$\varphi(g^{-1}kg) = \varphi(g)^{-1}\varphi(k)\varphi(g) = \varphi(g)^{-1}e'\varphi(g) = \varphi(g)^{-1}\varphi(g) = e'.$$

□

Remark. The kernel is **fixed under conjugation**, and forms a *normal subgroup*. Also note: we actually have $g^{-1}(\ker \varphi)g = \ker \varphi$.

Definition 21 (Normal Subgroup). Suppose $N \leq G$, then N is a *normal subgroup* of G when

$$g^{-1}Ng \subset N$$

for all $g \in G$. We then write $N \triangleleft G$.

If $g^{-1}Ng \subset N$, then $g^{-1}Ng = N$ (exercise). Then $N = gNg^{-1}$. So if these relations hold for all $g \in G$, we have

$$gN = Ng$$

for all $g \in G$.

Example. 1. If G is abelian and $H \leq G$ then $H \triangleleft G$. Observe that for all $G \in G$, and $h \in H$, one has $g^{-1}hg = hg^{-1}g = he = h$, so $g^{-1}Hg = H$ for all $g \in G$.

2. Recall $Z(G) = \{z \in G : \forall x \in G, zx = xz\}$. Thus, for all $g \in G$, one has $g^{-1}Z(G)g = Z(G)$. Hence $Z(G) \triangleleft G$.

3. Consider the group

$$G = D_{2n} = \langle \sigma, \tau : \sigma^n = \tau^2, \sigma\tau = \tau\sigma \rangle.$$

Consider $H = \langle \sigma \rangle$. We check that $H \triangleleft G$. We have

$$\sigma^{-m}(\sigma^l)\sigma^m = \sigma^l \in H..$$

$$\begin{aligned}
(\tau\sigma^m)^{-1}\sigma^l(\tau\sigma^m) &= \sigma^{-m}\tau^{-1}(\sigma^l\tau)\sigma^m \\
&= \sigma^{-m}\tau^{-1}(\tau\sigma^l)\sigma^m \\
&= \sigma^{-m}\sigma^{-l}\sigma^m \\
&= \sigma^{-l} \in H.
\end{aligned}$$

Thus $g^{-1}Hg \subset H$ for all $g \in D_{2n}$. Hence $H \triangleleft D_{2n}$.

◇

2.6 Quotient Groups (or Quotient Groups)

$$(\mathbb{Z}_n, +) \cong \mathbb{Z}/n\mathbb{Z}.$$

Recall that when $H \leq G$, we can define $a \sim b$ when $ab^{-1} \in H$. Then $[a] = Ha$.

Theorem 13. Suppose $N \triangleleft G$. Define

$$G/N = \{Na : a \in G\}.$$

Then G/N is a group relative to the group operation

$$Na \cdot Nb = N(ab).$$

Proof. • (well-defined group operation) If $Na = Na'$ and $Nb = Nb'$, then $Na \cdot Nb = Na' \cdot Nb'$.

If $Na = Na'$, then $a' = n_1a$ for some $n_1 \in N$. And if $Nb = Nb'$, then $b' = n_2b$ for some $n_2 \in N$. Thus

$$\begin{aligned}
a'b' &= n_1an_2b \\
&= n_1(an_2a^{-1})ab \\
&\in n_1(aNa^{-1})ab \\
&= n_1Nab \\
&= Nab
\end{aligned}$$

Hence $Na'b' \subset Nab$. By symmetry $Nab \subset Na'b'$, hence $Nab = Na'b'$. Thus the putative group operation on G/N is well-defined.

- (closure) follows from definition of the product.
- (associativity) inherited from the group G .

$$(Na)(NbNc) = Na(Nbc) = Na(bc) = N(ab)c = (Nab)Nc = (NaNb)Nc.$$

- (identity) $Ne = N$ serves as identity:

$$NaN e = Nae = Na = Nea = NeNa.$$

- (inverse) Na^{-1} serves as inverse of Na . Similar argument.

So G/N is indeed a group. □

Lecture 7

Fri 16 Sep 2022 10:31

Theorem 14. (2.6.2) Suppose $N \triangleleft G$. Then there exists a surjective homomorphism $\psi : G \rightarrow G/N$, such that $\ker \psi = N$.

Proof. We define the *canonical* mapping:

$$\begin{aligned}\psi : G &\longrightarrow G/N \\ a &\longmapsto \psi(a) = Na.\end{aligned}$$

Properties to check:

- (Homomorphism) For all $a, b \in G$,

$$\psi(ab) = Nab = NaNb = \psi(a)\psi(b).$$

- (surjective) Given any $Na \in G/N$, we have $\psi(a) = Na$.
- (kernel) $\ker \psi = \{a \in G : \psi(a) = Ne = N\}$. But $\psi(a) = Na = N$ if and only if $a \in N$. Thus $\ker \psi = N$.

□

Definition 22. If G is a (finite) group and $H \leq G$, the number of right cosets of H in G is called the *index* of H in G , written $i_G(H)$.

Notation. Also written as $[G : H]$

Theorem 15. (2.6.3) When G is a finite group and $N \triangleleft G$, then one has

$$|G/N| = |G|/|N|.$$

Proof. Use the proof of Lagrange's Theorem: $|G/N|$ is equal to the right cosets of N in G , which is $|G|/|N|$. □

Observation. A group may have no non-trivial normal subgroups. Example: $(\mathbb{Z}_p, +)$ for p prime.

Definition 23 (Simple Groups). A group with no non-trivial normal subgroups is called *simple*.

Remark. The classification of simple groups.

Theorem 16 (Cauchy's Theorem for Abelian Groups). (2.6.4) Suppose G is an abelian finite group of order $|G|$, and $p \mid |G|$ with p prime. Then G has an element of order p .

Proof. Proceed by induction on $n = |G|$. The conclusion is trivial for $n = 1$. We suppose $n > 1$ and that the conclusion holds for **all** abelian groups G with order $1 \leq |G| < n$. We now consider an abelian group G of order n .

Possibly, we have that G has no non-trivial subgroups. In this case, $G = \langle a \rangle$ for some $a \in G \setminus \{e\}$, and $|G|$ is prime. Then the order of a is equal to $|G|$, and G has an element of prime order.

If G has a non-trivial subgroup N . Since G is abelian, one has $N \triangleleft G$. Then

$$|G/N| \cdot |N| = |G| \equiv 0 \pmod{p}.$$

So $p \mid |N|$ or $p \mid |G/N|$. In the first case we are done by using induction hypothesis.

At this point we can assume that $N \triangleleft G$, $1 < |N| < |G|$, and $p \nmid |N|$. But $|G/N||N| = |G|$ and $p \mid |G|$, we have $p \mid |G/N|$. By the induction hypothesis, since $1 < |G/N| < |G|$, the group G/N contains an element of order p , say Na . Suppose that a has order m in G . Then

$$(Na)^m = Na^m = Ne = N,$$

so since Na has order p , we must have $p \mid m$. We write $m = kp$ with $k \in \mathbb{N}$, and observe that a^k has order p in G .

So G does indeed have an element of order p , completing the induction. $\square$

Lecture 8: homomorphism theorems

Mon 19 Sep 2022 10:46

2.7 The homomorphism theorems

Theorem 17. If $N \triangleleft G$, then there exists a surjective homomorphism $\varphi : G \rightarrow G/N$ with $N = \ker \varphi$.

Theorem 18 (First Homomorphism Theorem). (2.7.1) Let $\varphi : G \rightarrow G'$ be a surjective homomorphism of groups with kernel K . Then $G' \cong G/K$.

Proof. Define the map

$$\begin{aligned} \psi : G/K &\longrightarrow G' \\ Ka &\longmapsto \psi(Ka) = \varphi(a). \end{aligned}$$

Check this gives an isomorphism:

- (Well-defined property) We check that if $Ka = Kb$ then $\psi(Ka) = \psi(Kb)$. But $\psi(Ka) = \varphi(a)$ and $\psi(Kb) = \varphi(b)$. Then if $Ka = Kb$, then $K = Kba^{-1}$, whence $ba^{-1} \in K$, so that

$$\varphi(b)\varphi(a)^{-1} = \varphi(ba^{-1}) = e' \Rightarrow \varphi(b) = \varphi(a).$$

Thus $\psi(Ka) = \psi(Kb)$.

- (Homomorphism) Whenever $a, b \in G$,

$$\psi(Ka)\psi(Kb) = \varphi(a)\varphi(b) = \varphi(ab) = \psi(Kab) = \psi(Ka \cdot Kb).$$

- (ψ surjective) Whenever $g \in G'$, by the surjective property of φ , there exists $a \in G$ such that $\varphi(a) = g$, and $\psi(Ka) = \varphi(a)$, so there exists $Ka \in G/K$ such that $\psi(Ka) = g$.
- (ψ injective) Whenever $\psi(Ka) = \psi(Kb)$, one has $\varphi(a) = \varphi(b)$, so $\varphi(ab^{-1}) = \varphi(a)\varphi(b)^{-1} = e'$, whence $ab^{-1} \in \ker \varphi = K$. Thus $a \in Kb$, whence $Ka = Kb$. Thus ψ injective.

So, ψ is a well-defined bijective homomorphism from G/K to G' , and thus $G/K \cong G'$. $\square$

Theorem 19 (Correspondence Theorem). (2.7.2) Let $\varphi : G \rightarrow G'$ be a surjective homomorphism of groups with kernel K . Suppose $H' \leq G'$ and

$$H = \{a \in G : \varphi(a) \in H'\}.$$

Then $H \leq G$, $K \subset H$, and $H/K \cong H'$. Also, whenever $H' \triangleleft G'$ then $H \triangleleft G$.

Remark. When $H' = \{e\}$, H is the kernel. We have φ condensing information from H to H' , forgetting about the kernel. We can still work back and understand what is going on.

Proof. Check:

1. $H \leq G$: We have $e \in H$, so $H \neq \emptyset$. Also, whenever $a, b \in H$, one has $\varphi(ab^{-1}) = \varphi(a)\varphi(b)^{-1} \in H'$, and hence $ab^{-1} \in H$. So $H \leq G$ by the subgroup criterion.
2. Since $K = \ker \varphi$, we have $a \in K \Rightarrow \varphi(a) = e' \in H'$, so $a \in H$, so $K \subset H$.
3. Next, since $K \triangleleft G$, we have $K \triangleleft H \subset G$. We consider the restricted map

$$\begin{aligned} \varphi|_H : H &\longrightarrow H' \\ h &\longmapsto \varphi|_H(h) = \varphi(h). \end{aligned}$$

So this map is surjective from H to H' , with $\ker \varphi|_H = K$. Then by First Homomorphism Theorem, we have

$$H/K \cong H'.$$

4. When $H' \triangleleft G'$, then whenever, $g \in G$,

$$\varphi(g^{-1}Hg) = \varphi(g)^{-1}\varphi(H)\varphi(g) = \varphi(g)^{-1}H'\varphi(g) \subset H'.$$

So $g^{-1}Hg \subset H$. Hence $H \triangleleft G$.

$\square$

When $H \leq G$ and $N \triangleleft G$, consider

$$HN = \{hn \in G : h \in H, n \in N\}.$$

Claim. $HN \leq G$.

Proof. Observe that whenever $h_1, h_2 \in H$ and $n_1, n_2 \in N$, one has

$$\begin{aligned} (h_1 n_1)(h_2 n_2)^{-1} &= h_1 h_2^{-1} (h_2 n_1 n_2^{-1} h_2^{-1}) \\ &\in H(h_2 N h_2^{-1}) \\ &\subset HN. \end{aligned}$$

□

Lecture 9: second homomorphism theorem

Wed 21 Sep 2022 10:43

See the subgroup lattice:

Theorem 20 (Second Isomorphism Theorem). Suppose $H \leq G$ and $N \triangleleft G$. Then

$$H/(H \cap N) \cong (HN)/N.$$

Proof. Observe that $N \subset HN \subset G$, hence $N \triangleleft HN$. We define a homomorphism:

$$\begin{aligned} \varphi : H &\longrightarrow (HN)/N \\ a &\longmapsto \varphi(a) = Na. \end{aligned}$$

This map is well defined, and $\varphi(ab) = N(ab) = NaNb = \varphi(a)\varphi(b)$ for all $a, b \in H$. So φ is a homomorphism, and surjective by construction. (why?)

We apply the first homomorphism theorem,

$$(HN)/N \cong H/\ker \varphi.$$

and have $\ker \varphi = \{a \in H : aN = N\} = \{a \in H : a \in N\} = H \cap N$. □

Lecture 10

Fri 23 Sep 2022 10:38

Theorem 21 (Third Homomorphism Theorem). Suppose $\varphi : G \rightarrow G'$ is a surjective homomorphism of groups, with $K = \ker \varphi$, and $N' \triangleleft G'$. Put $N = \{a \in G : \varphi(a) \in N'\}$. Then one has $G/N \cong G'/N'$ and hence $G/N \cong (G/K)/(N/K)$.

Proof. Define

$$\begin{aligned} \psi : G &\longrightarrow G'/N' \\ a &\longmapsto \psi(a) = N'\varphi(a). \end{aligned}$$

Since φ is surjective, each element of G'/N' has the form $N'b'$ for some $b' \in G'$, hence there exists $a \in G$ with $\varphi(a) = b'$, so $N'b' = N'\varphi(a) = \psi(a)$. So ψ is surjective.

When $a, b \in G$, one has

$$\psi(ab) = N'\varphi(ab) = N'\varphi(a)\varphi(b) = N'\varphi(a)N'\varphi(b) = \psi(a)\psi(b).$$

So ψ is a homomorphism. Moreover,

$$\begin{aligned}\ker \psi &= \{a \in G : \psi(a) = N'\} \\ &= \{a \in G : N'\varphi(a) = N'\} \\ &= \{a \in G : \varphi(a) \in N'\} \\ &= N.\end{aligned}$$

Now by the first homomorphism theorem, since ψ is a surjective homomorphism with kernel N ,

$$G/N \cong G'/N'.$$

Also by the first homomorphism theorem,

$$G/K = G/\ker \varphi \cong G'.$$

and by the Correspondence Theorem,

$$N/K = N/\ker \varphi \cong N'.$$

Thus

$$G/N \cong (G/K)/(N/K).$$

□

Remark. The compatibility is guaranteed by Correspondence Theorem.

Theorem 22 (Zassenhaus Theorem). Suppose

$$\begin{aligned}H &\leq G, & K &\leq G \\ H^* &\triangleleft H, & K^* &\triangleleft K.\end{aligned}$$

Then

- $H^*(H \cap K^*) \triangleleft H^*(H \cap K)$
- $K^*(H^* \cap K) \triangleleft K^*(H \cap K)$
-

$$\begin{aligned}H^*(H \cap K)/H^*(H \cap K) \\ &\cong K^*(H \cap K)/K^*(H^* \cap K) \\ &\cong H \cap K / ((H^* \cap K)(H \cap K^*))\end{aligned}$$

Also called [\[\[Butterfly Lemma \]\]](#) or [\[\[Fourth Homomorphism Theorem \]\]](#).

3 Permutation Groups

3.1 The symmetric group

Notation. If $\sigma \in S_n$, then we write

$$\sigma = \begin{pmatrix} x_1 & x_2 & x_3 & \cdots & x_n \\ \sigma(x_1) & \sigma(x_2) & \sigma(x_3) & \cdots & \sigma(x_n) \end{pmatrix} \quad (4)$$

where $\sigma(j) = a_j$ for each j , and the a_j are distinct and lie in $\{1, 2, \dots, n\}$.

See how to combine permutations.

Notation (Cycle Notation). When $i_1, \dots, i_k$ are k distinct integers lying in $\{1, 2, \dots, n\}$, we use $(i_1, i_2, \dots, i_k)$ to denote $\sigma \in S_n$ with the property given by

$$\sigma(i_1) = i_2, \sigma(i_2) = i_3, \dots, \sigma(i_{k-1}) = i_k, \sigma(i_k) = i_1$$

, and $\sigma(s) = s$ for $s \in \{1, 2, \dots, n\} \setminus \{i_1, \dots, i_k\}$.

Also called [\[\[Cycle Decomposition \]\]](#).

Definition 24 (Transposition). A permutation interchanging 2 elements only, such as (i_1, i_2) , is called a *transposition*, or *2-cycle*.

An *m-cycle* is a permutation of shape

$$(i_1, i_2, \dots, i_m).$$

Two cycles are disjoint if they contain no common element, e.g. $(1, 4, 2)$ and $(4, 5)$ are disjoint.

Note: any two disjoint cycles commute.

Lecture 11: Cycle Decomposition

Mon 26 Sep 2022 10:30

3.2 Cycle Decomposition

Theorem 23. (3.2.2) Every permutation in S_n can be written as a product of disjoint cycles.

Proof. (exercise) consider the orbits of items. □

Implicitly we require a *cycle decomposition* to be disjoint cycles.

Theorem 24. (3.2.4) Let $\sigma \in S_n$ have cycle decomposition $\tau_1 \tau_2 \dots \tau_k$ where τ_j is an m_j -cycle for each $1 \leq j \leq k$. Then $o(\sigma) = [m_1, \dots, m_k]$ (the LCM).

Proof. Disjoint cycles commute, and hence if $\sigma^M = e$, then $\tau_1^M \tau_2^M \dots \tau_k^M = e$. Since these cycles commute and are disjoint, this can happen if and only if $\tau_j^M = e$ for each j (Note that each τ_j^M are still disjoint m_j -cycles). Since $o(\tau_j) = m_j$, this can happen iff $m_j | M$ for every j . This is to say that $[m_1, \dots, m_k] | M$. □

Theorem 25. (3.2.5) Every permutation in S_n is the product of transpositions.

Proof. Suffice to prove this for a k -cycle, since each $\sigma \in S_n$ has a cycle decomposition. Consider $\sigma = (i_1, i_2, \dots, i_k) \in S_n$. Observe that

$$\sigma = (i_k, i_1)(i_{k-1}, i_1) \dots (i_2, i_1)$$

is a product of transpositions. $\square$

3.3 Odd and Even permutations

Theorem 26. (3.3.1) A permutation $\sigma \in S_n$ is either odd or even, in the sense that σ is either a product of an odd number of transpositions (and never an even number) or else is an even number of transpositions (and never an odd number).

Proof. Consider the polynomial in n variables $x_1, \dots, x_n$, $f(x) = \prod_{1 \leq i < j \leq n} (x_i - x_j) = \det(x_i^{j-1})_{1 \leq i, j \leq n}$. (This is the [Vandermonde determinant](#)).

We can consider the action of $\sigma \in S_n$ on $f(x)$ by putting $\sigma(f(x)) = \prod_{1 \leq i < j \leq n} (x_{\sigma(i)} - x_{\sigma(j)}) = \det(x_{\sigma(i)}^{j-1})_{1 \leq i, j \leq n}$.

Thus σ permutes the columns of the determinant.

Suppose σ is a product of k transpositions, say $\sigma = \tau_1 \dots \tau_k$, where each τ_j is a transposition. Then

$$\sigma(f(x)) = \tau_1 \dots \tau_k(f(x)).$$

Each τ switches two columns of determinant. Then

$$\sigma(f(x)) = (-1)^k f(x).$$

If $\sigma = \tau_1 \dots \tau_k$ and $\sigma = \tau'_1 \dots \tau'_l$, are both decompositions of σ into transpositions, then

$$\sigma(f(x)) = (-1)^k f(x) = (-1)^l f(x).$$

So $(-1)^k = (-1)^l$, hence $l \equiv k \pmod{2}$, i.e. l and k has same parity. $\square$

Definition 25 (Alternating Group). The subset A_n of all even permutations of S_n is called the *alternating group* of degree n .

Implicit claim: A_n forms a subgroup of S_n .

Proof. Since e is a product of 0 transpositions, $e \in A_n$. Next, $\sigma, \tau \in A_n$, then σ and τ are each a product of an even number of transpositions, say $\tau = \omega_1 \dots \omega_k$ as a product of transpositions, then $\tau^{-1} = \omega_k^{-1} \dots \omega_1^{-1} = \omega_k \dots \omega_1 \in A_n$. Hence $\sigma\tau^{-1}$ is a product of even number of transpositions, since this is true for both σ and τ^{-1} , so $\sigma\tau^{-1} \in A_n$. Conclusion follows from subgroup criterion. $\square$

Lecture 12

Wed 28 Sep 2022 10:30

Theorem 27. One has $A_n \triangleleft S_n$.

Proof. Let $\sigma \in A_n$. Then for all $\tau \in S_n$ we have to prove $\tau^{-1}\sigma\tau \in A_n$. Thus we need to prove $\tau^{-1}A_n\tau \leq A_n$.

But if $\sigma \in A_n$, then $\sigma = \omega_1\omega_2\ldots\omega_k$, each ω_i is a transposition, and k even. Also, $\tau \in S_n$ so $\tau = \nu_1\ldots\nu_l$, with ν_i transposition.

$$\begin{aligned}\tau^{-1}\sigma\tau &= (\nu_1\ldots\nu_l)^{-1}(\omega_1\ldots\omega_k)(\nu_1\ldots\nu_l) \\ &= \nu_l\ldots\nu_1\omega_1\ldots\omega_k\nu_1\ldots\nu_l.\end{aligned}$$

This is a product of transpositions and $2l + k$ is even, so $\tau^{-1}\sigma\tau \in A_n$. Thus $A_n \triangleleft S_n$. $\square$

To check $|A_n| = \frac{1}{2}n!$ ($n \geq 2$):

Proof. Consider

$$\begin{aligned}\varphi : S_n &\longrightarrow E = \{+1, -1\} \\ \sigma &\longmapsto \varphi(\sigma) = (-1)^k.\end{aligned}$$

where k is odd if σ is odd, k is even if σ is even.

Claim. This is a homomorphism of groups, and is surjective.

If $\sigma, \tau \in S_n$, then $\varphi(\sigma\tau) = (-1)^m$, where m odd if $\sigma\tau$ is odd, m even if $\sigma\tau$ even.

$\varphi(\sigma)\varphi(\tau) = (-1)^{k+l}$ where k and l depend on parity of σ and τ . But $\sigma\tau$ is odd if σ and τ are neither not both even nor not both odd. $\sigma\tau$ is even otherwise. Then $m = k + l \pmod{2}$, so $\varphi(\sigma\tau) = \varphi(\sigma)\varphi(\tau)$.

Then by First Homomorphism Theorem,

$$S_n / \ker \varphi \cong E.$$

Also, $\ker \varphi = \{\sigma \in S_n : \varphi(\sigma) = +1\} = \{\sigma \in S_n : \sigma \text{ even}\} = A_n$.

Hence $S_n/A_n \cong E$. Hence $|S_n|/|A_n| = |E| = 2$. Hence $|A_n| = \frac{1}{2}|S_n| = \frac{1}{2}n!$. $\square$

3.4 The group A_n is simple for $n \geq 5$

Example (Simple Groups). 1. $\mathbb{Z}_p$ (cyclic group of prime order)

2. A_n ($n \geq 5$)

$\diamond$

Theorem 28. (6.1.2) A_n is generated by the 3-cycles in S_n (when $n \geq 3$).

Proof. If $\sigma \in A_n$, then σ is a product of an even number of transpositions, say $\sigma = (t_1, t_2)(t_3, t_4) \dots (t_{2k-1}, t_{2k})$, where the t_j might not be disjoint.

Consider any pair of these transpositions (when $k \geq 1$). We can relabel $(t_{2i-1}, t_{2i})(t_{2i+1}, t_{2i+2})$ as $(1, 2)(3, 4)$, or $(1, 2)(1, 2)$ or $(1, 2)(1, 3)$. But $(1, 2)(1, 2) = (1, 2, 3)(1, 3, 2)$. $(1, 2)(3, 4) = (1, 4, 2)(1, 4, 3)$.

In all cases these pairs can be written as product of 3-cycles. Because there is an even number of transpositions, we can write σ as a product of 3-cycles. $\square$

Theorem 29. (6.1.8) The group A_5 is simple.

Proof. We have $|A_5| = \frac{1}{2}5! = 60$, so all elements $\sigma \in A_5$ have orders dividing 60 (by Lagrange's Theorem). The possible orders include:

- 10, 15, 20, 30, 60 : not elements of S_5 , since cycle decomposition includes 5-cycle + extra cycles, but no room.
- 12: not element of S_5 , since cycle decomposition contain 3-cycle and 4-cycle, and not enough room for 7 elements.
- 6-cycle, by relabeling, looks like $(1, 2, 3)(4, 5)$. This is same as $(1, 3)(1, 2)(4, 5)$ which is odd, so not in A_5 .
- 4-cycle: by relabeling, looks like $(1, 2, 3, 4) = (1, 4)(1, 3)(1, 2)$ which is odd, so not in A_5 .

Then these are elements of order 1, 2, 3, 5.

Let's suppose that $N \triangleleft A_5$ and $1 < |N| < 60$, and seek a contradiction. Then N contains an element of order 2, 3, or 5. We claim that N contains an element of order 3. Three possibilities:

- N contains an element of order 3.
- N contains an element of order 2: By relabeling, it can be assumed to be $(1, 3)(2, 4)$ (note: $(1, 2)(1, 3)$ has order 3 so is not interesting). But then

$$(1, 2, 3, 4, 5)^{-1}(1, 3)(2, 4)(1, 2, 3, 4, 5) = (1, 3)(2, 5) \in N.$$

Also,

$$(1, 3)(2, 5)(1, 3)(2, 4) = (2, 4, 5) \in N.$$

- N contains an element of order 5: We can label it as $(1, 2, 3, 4, 5)$. Then

$$((1, 3)(2, 4))^{-1}(1, 2, 3, 4, 5)((1, 3)(2, 4)) = (1, 2, 5, 3, 4).$$

Then

$$(1, 2, 3, 4, 5)^{-1}(1, 2, 5, 3, 4) = (2, 4, 5).$$

So N contains a 3-cycle, by relabeling we assume to be $(1, 2, 3)$. We claim that N contains all 3-cycles. We can replace $(1, 2, 3)$ by $(1, 2, \alpha)$ for any $\alpha \in \{4, 5\}$. Now

$$(2, 3, \alpha)^{-1}(1, 2, 3)^2(2, 3, \alpha) = (1, 2, \alpha).$$

So whenever $(\alpha, \beta, \gamma) \in N$, then so too does (α, β, δ) for $\delta \notin \{\alpha, \beta, \gamma\}$. By switching elements we obtain any 3-cycle.

Then N contains all 3-cycles, and 3-cycles generate A_5 , so $N = A_5$, so A_5 has no proper normal subgroup. Thus A_5 is simple. $\square$

Lecture 13

Wed 05 Oct 2022 10:37

Remark (Commutator). $[x, y] = xyx^{-1}y^{-1}$ gives a measure of how nonabelian a group is.

Theorem 30. (6.1.9) A_n is simple for $n \geq 5$.

Proof. The last theorem proves A_5 is simple. We prove by induction, and assume that A_m is simple for $5 \leq m < n$, and seek to prove that A_n is simple.

Suppose $\{e\} \neq N \triangleleft A_n$, and try to show that N contains an element σ which leaves an element of $\{1, 2, \dots, n\}$ fixed. Suppose that no such σ exists. Then there exists distinct elements $a, b, c, d \in \{1, 2, \dots, n\}$ such that

$$\sigma(a) = b, \sigma(c) = d.$$

Since $n \geq 6$, there is an element $\tau \in A_n$ with

$$\tau = (a, b)(c, d, e, f).$$

for $e, f \notin \{a, b, c, d\}$ with $e \neq f$. By normality of N in A_n , one has $\tau^{-1}\sigma\tau \in N$. Then

$$\begin{aligned} \tau^{-1}\sigma\tau(b) &= \tau^{-1}\sigma(a) = \tau^{-1}(b) = a \\ \tau^{-1}\sigma\tau(d) &= \tau^{-1}\sigma(e) = \tau^{-1}(e) = d. \end{aligned}$$

Then

$$\begin{aligned} \tau^{-1}\sigma\tau\sigma(a) &= \tau^{-1}\sigma\tau(b) = a \\ \tau^{-1}\sigma\tau\sigma(c) &= \tau^{-1}\sigma\tau(d) = d \end{aligned}$$

So $\tau^{-1}\sigma\tau\sigma$ fixes a and moves c , so it is not the identity, and there is an element of N fixing a . We have a contradiction.

We can suppose, by relabeling, that $\sigma \in N \setminus \{\text{id}\}$ and $\sigma(1) = 1$. The elements of A_n which fix 1 form a subgroup of A_n , say $H_1 \leq A_n$, with $H_1 \cong A_{n-1}$. Then $N \cap H_1 \neq \{\text{id}\}$ (since $\sigma \in N \cap H_1$), and $N \cap H_1 \triangleleft H_1$. But H_1 is simple (because A_{n-1} is simple), so $N \cap H_1 = H_1$. Then $H_1 \leq N$.

Let τ be any permutation with $\tau(2) = 1$. Then $\tau^{-1}\sigma\tau(2) = \tau^{-1}\sigma(1) = \tau^{-1}(1) = 2$. So $H_2 := \tau^{-1}H_1\tau$ is a subgroup of A_n that fixes 2. By normality of N in A_n we have $\tau^{-1}H_1\tau \leq N$, and $\tau^{-1}H_1\tau \cong A_{n-1}$. Also,

$$H_1 \cap \tau H_1 \tau \cong A_{n-2}$$

, since it fixes both 1 and 2.

Note, since $H_1 \cap H_2$ fixes both 1 and 2, so $H_1 \cap H_2 \cong A_{n-2}$. Consider the group generated by H_1 and H_2 :

$$\text{gp}(H_1, H_2) = \{g_1, \dots, g_k : g_i \in H_1 \vee g_i \in H_2\}.$$

Observe that $\text{gp}(H_1, H_2) \leq N$ since $H_1 \leq N$ and $H_2 \leq N$. Also, the set $H_1 H_2$ is contained in $\text{gp}(H_1, H_2)$ and hence in N .

We compute $|H_1 H_2|$. We look at how many representations an element $n \in H_1 H_2$ has in the shape $n = h_1 h_2$, with $h_1 \in H_1$ and $h_2 \in H_2$.

If $n = g_1 g_2$ and $n = h_1 h_2$, $g_1, h_1 \in H_1$, $g_2, h_2 \in H_2$, then $g_1 g_2 = h_1 h_2$. So $H_1 \ni h_1^{-1} g_1 = h_2 g_2^{-1} \in H_2$, and hence, for some $t \in H_1 \cap H_2$, we have $h_1^{-1} g_1 = t = h_2 g_2^{-1}$.

Then the number of ways that $n \in H_1 H_2$ is represented as a product $h_1 h_2$ with $h_1 \in H_1$ and $h_2 \in H_2$ is at most $|H_1 \cap H_2|$. Then

$$|H_1 H_2| \geq |H_1| |H_2| / |H_1 \cap H_2|.$$

Hence

$$\begin{aligned} |N| &\geq |H_1 H_2| \geq |H_1| |H_2| / |H_1 \cap H_2| \\ &= \frac{|A_{n-1}| |A_{n-1}|}{|A_{n-2}|} \\ &= \frac{\frac{1}{2}(n-1)! \frac{1}{2}(n-1)!}{\frac{1}{2}(n-2)!} \\ &= \frac{1}{2}(n-1)(n-1)!. \end{aligned}$$

But $N \triangleleft A_n$, so $|N| \leq \frac{1}{2}n!$. Thus $1 \leq |A_n/N| \leq \frac{\frac{1}{2}n!}{\frac{1}{2}(n-1)(n-1)!} = \frac{n}{n-1} < 2$ when $n > 2$. But now $n \geq 6$, so $|A_n/N| = 1$ and $N = A_n$. Thus A_n is simple for $n \geq 5$. $\square$

Lecture 14: Cauchy's Theorem

Fri 07 Oct 2022 11:10

4 Finite Groups

4.1 Cauchy's Theorem

Definition 26. Let S be a set and $f \in A(S)$. Then the *orbit* of an element $s \in S$ under the action f is $\text{Orb}_f(s) := \{f^i(s) : i \in \mathbb{Z}\}$.

We define an equivalence relation such that

$$s \sim t \Leftrightarrow t = f^i(s) \text{ for some } i \in \mathbb{Z}.$$

Equivalence classes $[s]$ are then orbits $\text{Orb}_f(s)$.

Lemma 16. (2.8.1) Suppose that $f \in A(S)$ has order p , where p is prime, then for all $s \in S$, we have $|\text{Orb}_f(s)| = 1$ or p .

Proof. Two cases:

1. If $f(s) = s$, then $\text{Orb}_f(s) = \{s\}$, so $|\text{Orb}_f(s)| = 1$.
2. If $f(s) \neq s$, then

$$\text{Orb}_f(s) = \{s, f(s), f^2(s), \dots, f^{p-1}(s)\}.$$

We claim that these elements are all distinct. If not, say $f^i(s) = f^j(s)$, for some $0 \leq i < j \leq p-1$, one has $f^{j-i}(s) = s$. Since also $f^p(s) = s$. But $(p, j-i) = 1$, so there exist $a, b \in \mathbb{Z}$ with $a(j-i) + bp = 1$, and then

$$f(s) = f^{a(j-i)+bp}(s) = f^{bp}f^{(j-i)a}(s) = f^{bp}(s) = s.$$

Therefore $|\text{Orb}_f(s)| = p$ after all.

□

Theorem 31 (Cauchy's Theorem). (2.8.2) Suppose p is prime and $p \mid |G|$, then G has an element of order p .

Thus G has a subgroup of order p , say $\langle a \rangle \leq G$ with $o(a) = p$.

Proof. Two cases.

1. $p = 2$. If $2 \mid |G|$, if $a^2 = e$ has no solution with $a \neq e$, then for no $a \in G \setminus \{e\}$ does one have $a = a^{-1}$. But for each $b \in G$ there is a distinct $b^{-1} \in G$ with $bb^{-1} = e$. Hence we can partition G into m subsets of shape $\{b, b^{-1}\}$, say together with $\{e\}$. Then $|G| = 2m + 1$, which is a contradiction since $2 \mid |G|$. Thus there exists $a \in G$ with $o(a) = 2$.
2. p prime, $p > 2$, $p \mid |G|$. Let $S = \{(a_1, a_2, \dots, a_p) \in G^p : a_1 \dots a_p = e\}$. Note that if $(a_1, a_2, \dots, a_p) \in S$, then given $a_1, \dots, a_{p-1}$, we have $a_p = (a_1, \dots, a_{p-1})^{-1} \in G$ is uniquely defined, so $|S| = |G|^{p-1}$.

We define

$$\begin{aligned} f : S &\longrightarrow S \\ (a_1, \dots, a_p) &\longmapsto f((a_1, \dots, a_p)) = (a_p, a_1, \dots, a_{p-1}). \end{aligned}$$

Plainly $f \in A(S)$. Note that $f \neq \text{id}$, and $f^p = \text{id}$, so f has order p . Also, one has $f(s) = s$ if and only if $s = (a, \dots, a)$ for some $a \in G$.

Observe that if $f(s) = s$ for precisely m elements $s \in S$, then

$$|S| - m = \text{card}\{s \in S : f(s) \neq s\}.$$

where any element in the set satisfies $|\text{Orb}_f(s)| = p$, so $|S| - m \equiv 0 \pmod{p}$ (recall orbits are disjoint, so p elements such that $f(s) \neq s$ comes together as one orbit).

Also, $|S| = |G|^{p-1} \equiv 0 \pmod{p}$ since $p \mid |G|$. Thus $m \equiv 0 \pmod{p}$. Since $f((e, \dots, e)) = (e, \dots, e)$ and $(e, \dots, e) \in S$, $m \geq 1$. Hence $m \geq p > 1$. Hence there exists $a \in G$ with property $(a, \dots, a) \in S$ (by definition of S), whence $a^p = e$. But $o(a) \mid p$, and $a \neq e$, so $o(a) = p$.

□

Lemma 17. (2.8.3) Let G be a group with $|G| = pq$, where p and q are primes with $p > q$. Whenever $a \in G$ has order p , then $\langle a \rangle \triangleleft G$.

Proof. We claim $A := \langle a \rangle$ is the **only** subgroup of G having order p . Suppose, by the way of deriving a contradiction, that $B \leq G$ and $|B| \neq p$ with $B \neq A$.

Observe that $AB = \{ab : a \in A, b \in B\} \subset G$. Therefore $|AB| \leq pq < p^2$. But if a_1, a_2 are distinct elements of A and b_1, b_2 are distinct elements from B , then $a_1b_1 = a_2b_2$ if and only if $a_2^{-1}a_1 = b_2b_1^{-1}$, so $a_2^{-1}a_1 \in A \cap B$.

But if $B \neq A$, since $A \cap B \leq G$, and A (and B) have prime order, so we must have $A \cap B = \{e\}$. Therefore $a_2^{-1}a_1 = e$, so $a_1 = a_2$ and similarly $b_1 = b_2$. Thus $|AB| = p^2$. But $|AB| < p^2$, a contradiction. Thus A is the only subgroup of G of order p .

Thus if $g \in G$, we have $g^{-1}Ag \leq G$ of order p , so $g^{-1}Ag = A$. Then $A \triangleleft G$. □

Lecture 15

Fri 14 Oct 2022 10:33

Corollary 7. Under assumption of last lemma, for each $g \in G$, one has $g^{-1}ag = a^i$ for some $i = i(g)$ with $1 \leq i < p$.

Proof. We know $g^{-1}ag \in \langle a \rangle$ since $\langle a \rangle \triangleleft G$. So $g^{-1}ag = a^j$ for some $j \in \mathbb{Z}$. We can assume $0 \leq j < p$ since $o(a) = p$, and $j \neq 0$ (if $j = 0$ then $ag = g \Rightarrow a = e$, a contradiction). □

Theorem 32. (2.8.5) Suppose G is a group of order pq with p and q primes with $p > q$. Provided that $q \nmid (p-1)$, one has that G is cyclic.

Proof. By Cauchy's Theorem, since $p \mid |G|$ and $q \mid |G|$, so there exists $a, b \in G$ with $o(a) = p$ and $o(b) = q$. The corollary shows that $b^{-1}ab = a^i$ for some i with $1 \leq i < p$. Then $b^{-2}ab^2 = b^{-1}a^ib = (b^{-1}ab)^i = (a^i)^i = a^{i^2}$. By induction, $b^{-r}ab^r = a^{i^r}$. In particular,

$$a = b^{-q}ab^q = a^{i^q}.$$

Thus $i^q \equiv 1 \pmod{p}$, since a has order p . But $i^{p-1} \equiv 1 \pmod{p}$ by Fermat's Little Theorem. Then since $i^q \equiv 1 \pmod{p}$ and $(q, q-1) = 1$, we must have integers u and v with $u(p-1) + vq = 1$, so $i = (i^{p-1})^u (i^q)^v \equiv 1 \pmod{p}$.

Thus $b^{-1}ab = a^1$. Thus G is abelian. So now we can assume G is abelian, $o(a) = p$ and $o(b) = q$ with $(p, q) = 1$. Note that $(ab)^m = e$ then $a^m = b^{-m}$. Now $a^{mq} = (b^1)^{-m} = e$, so $p|mq$, whence $p|m$. Similarly $b^{mp} = (a^{-p})^m = e$, so $q|m$. But then the order of ab is at least pq , and $(ab)^{pq} = (a^p)^q (b^q)^p = e$. So $o(ab) = pq$, and $\langle ab \rangle = G$. Thus G cyclic. $\square$

4.2 Direct Products (internal and external)

Definition 27. Suppose $G_1, G_2, \dots, G_n$ is any collection of groups. Then the (*external*) *direct product* $G_1 \times G_2 \times \dots \times G_n$ is the group

$$G = \{(g_1, \dots, g_n) : g_i \in G_i, 1 \leq i \leq n\}$$

and the group operation defined element-wise, i.e.

$$(g_1, \dots, g_n) * (h_1, \dots, h_n) = (g_1 h_1, \dots, g_n h_n).$$

It's easy to check that $(G, *)$ forms a group with identity $(e_1, \dots, e_n)$.

Remark. Define

$$\overline{G_i} = \{(e_1, \dots, e_{i-1}, g_i, e_{i+1}, \dots, e_n) : g_i \in G_i\}.$$

Then $G_i \cong \overline{G_i} \triangleleft G_1 \times \dots \times G_n =: G$.

Proof. Easily $\overline{G_i} \leq G_i$, and in fact we can define a projection mapping

$$\begin{aligned} \pi_i : \overline{G_i} &\longrightarrow G_i \\ (e_1, \dots, e_{i-1}, g_i, e_{i+1}, \dots, e_n) &\longmapsto \pi_i((e_1, \dots, e_{i-1}, g_i, e_{i+1}, \dots, e_n)) = g_i. \end{aligned}$$

Which is a surjective and injective homomorphism.

Also, if $\gamma \in G$, say $\gamma = (h_1, \dots, h_n)$, then

$$\begin{aligned} &\gamma^{-1}(e_1, \dots, e_{i-1}, g_i, e_{i+1}, \dots, e_n)\gamma \\ &= (e_1, \dots, e_{i-1}, h_i^{-1} g_i h_i, e_{i+1}, \dots, e_n) \\ &\in \overline{G_i} \end{aligned}$$

$\square$

Remark. If $\gamma = (g_1, \dots, g_n) \in G$, then

$$\gamma = (g_1, e_2, \dots, e_n)(e_1, g_2, e_3, \dots, e_n) \dots (e_1, e_2, \dots, e_{n-1}, g_n).$$

Therefore $\gamma \in \overline{G_1} \overline{G_2} \dots \overline{G_n}$ and the representation is unique.

Definition 28. We say G is the (*internal*) *direct product* of its **normal** subgroups $N_1, \dots, N_n$ if for every $g \in G$ there is a **unique** representation in the form $g = g_1 g_2 \dots g_n$ with $g_i \in N_i$ for $1 \leq i \leq n$.

Lemma 18. (2.9.1) If $G = G_1 \times G_2 \times \dots \times G_n$ is the external direct product, then G is the internal direct product $G = \overline{G_1 G_2 \dots G_n}$.

Proof. See 2 remarks. $\square$

Lecture 16

Mon 17 Oct 2022 10:38

Lemma 19. (2.9.3) If G is an internal direct product of its normal subgroups $N_1, \dots, N_n$, then $N_i \cap N_j = \{e\}$ for $i \neq j$.

Proof. If $a \in N_i \cap N_j$ and $a \neq e$ then

$$a = e_1 e_2 \dots a_i e_n \dots e = e_1 e_2 \dots a_j \dots e_n.$$

which contradicts the uniqueness of representation of a in the direct product. $\square$

Lemma 20. (2.9.2) Let G be a group with $M \triangleleft G$ and $N \triangleleft G$ and $M \cap N = \{e\}$. Then for all $m \in M$ and $n \in N$ one has $mn = nm$.

Proof. Observe that if $a = mnm^{-1}n^{-1}$, then $a = m(nm^{-1}n^{-1}) \in M$. Also, $a = (mnm^{-1})n \in N$. That means $a \in M \cap N$, so $a = e$. Thus $mn = nm$. $\square$

Theorem 33. (2.9.4) Let G be a group with $N_1, \dots, N_n$. Then the map

$$\begin{aligned} \psi : N_1 \times N_2 \times \dots \times N_n &\longrightarrow G \\ (g_1, g_2, \dots, g_n) &\longmapsto \psi((g_1, g_2, \dots, g_n)) = g_1 g_2 \dots g_n. \end{aligned}$$

is an isomorphism if and only if G is the internal direct product of $N_1, \dots, N_n$.

Proof. (Proof of 2.9.4)

Suppose ψ defined in hypothesis is an isomorphism. Then each $g \in G$ has a representation $g = g_1 g_2 \dots g_n$ for some $g_i \in N_i$ by surjectivity. Also by injectivity, whenever $g = g_1 \dots g_n$ with $(g_1, g_2, \dots, g_n) \in N_1 \times \dots \times N_n$, each g_i is unique, so representation is unique. Then G is the internal direct product of $N_1, \dots, N_n$.

Suppose G is the internal direct product of $N_1, \dots, N_n$. Then each $g \in G$ has a unique representation $g = g_1 g_2 \dots g_n$. So ψ is surjective. If $\psi(g_1, \dots, g_n) = \psi(h_1, \dots, h_n)$, then uniqueness of representation shows $g_1 \dots g_n = h_1 \dots h_n$ implies $g_i = h_i$ for each i . So the n -tuples are the same. So ψ is injective.

Remains to show ψ is a homomorphism. But whenever $(g_1, \dots, g_n)$ and $(h_1, \dots, h_n)$ lie in $N_1 \times \dots \times N_n$, then

$$\begin{aligned}
 & \psi((g_1, \dots, g_n)(h_1, \dots, h_n)) \\
 &= \psi(g_1 h_1, \dots, g_n h_n) \\
 &= g_1 h_1 g_2 h_2 \dots g_n h_n \\
 &= (g_1 g_2 \dots g_n)(h_1 h_2 \dots h_n) \quad \text{Lemma 2.9.2} \\
 &= \psi(g_1, \dots, g_n) \psi(h_1, \dots, h_n)
 \end{aligned}$$

Thus ψ is a homomorphism, it's injective and surjective, hence it is an isomorphism. $\square$

4.3 Finite Abelian Groups

Theorem 34 (Classification for Finite Abelian Groups). (2.10 A) Suppose that G is a finite abelian group. Then G is isomorphic to a direct product of cyclic groups of the shape

$$\mathbb{Z}_{p_1^{\alpha_1}} \times \mathbb{Z}_{p_2^{\alpha_2}} \times \dots \times \mathbb{Z}_{p_r^{\alpha_r}}.$$

where p_i are primes (not necessarily distinct), and $\alpha_i \in \mathbb{N}$, and $n = |G| = p_1^{\alpha_1} \dots p_r^{\alpha_r}$.

Corollary 8. (2.10 B) If G is a finite abelian group, then there are integers $m_1, \dots, m_k$ with $2 \leq m_1 \leq m_2 \leq \dots \leq m_k$, and $m_i | m_{i+1}$ for $1 \leq i < k$, such that

$$G \cong \mathbb{Z}_{m_1} \times \mathbb{Z}_{m_2} \times \dots \times \mathbb{Z}_{m_k}.$$

Example. Classify all abelian groups of order 1400.

Observe that $1400 = 2^3 5^2 7$. Then by classification theorem, if $|G| = 1400$, then G is isomorphic to one of:

- $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_5 \times \mathbb{Z}_5 \times \mathbb{Z}_7 \cong \mathbb{Z}_{70} \times \mathbb{Z}_{10} \times \mathbb{Z}_2$
- $\mathbb{Z}_2 \times \mathbb{Z}_4 \times \mathbb{Z}_5 \times \mathbb{Z}_5 \times \mathbb{Z}_7 \cong \mathbb{Z}_{140} \times \mathbb{Z}_{10}$
- $\mathbb{Z}_8 \times \mathbb{Z}_5 \times \mathbb{Z}_5 \times \mathbb{Z}_7 \cong \mathbb{Z}_{280} \times \mathbb{Z}_5$
- $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_{25} \times \mathbb{Z}_7 \cong \mathbb{Z}_{350} \times \mathbb{Z}_2 \times \mathbb{Z}_2$
- $\mathbb{Z}_2 \times \mathbb{Z}_4 \times \mathbb{Z}_{25} \times \mathbb{Z}_7 \cong \mathbb{Z}_{700} \times \mathbb{Z}_2$
- $\mathbb{Z}_8 \times \mathbb{Z}_{25} \times \mathbb{Z}_7 \cong \mathbb{Z}_{1400}$

There are 6 isomorphism classes for all finite abelian group of order 1400. $\diamond$

Lecture 17

Wed 19 Oct 2022 10:33

Lemma 21. Let G be a finite abelian group of order $n = p_1^{\alpha_1} \dots p_k^{\alpha_k}$, where the p_i are distinct primes and $\alpha_i \in \mathbb{N}$. Then G is an internal direct product of subgroups $G_1, \dots, G_k$ where G_i is the subgroup of G having all its elements g of order $p_i^{r_i}$ for some $r_i = r_i(g) \geq 0$.

Proof. Let $G_i = \{g \in G : g^{p_i^{r_i}} = e \text{ for some } r_i \in \mathbb{N}\}$. Then (homework problem) $G_i \leq G$, and since G is abelian, one has $G_i \triangleleft G$.

Next we show that when $g \in G$, then $g = g_1 g_2 \dots g_k$ with $g_i \in G_i$. We have (by Lagrange) that $o(g) | G$, so $o(g) = p_1^{\beta_1} p_2^{\beta_2} \dots p_k^{\beta_k}$, say with $0 \leq \beta_i \leq \alpha_i$. Put $a_i = o(g) / p_i^{\beta_i}$, so $(a_1, a_2, \dots, a_k) = 1$. Then, by Euclidean algorithm, there exist $b_i \in \mathbb{Z}$ with $a_1 b_1 + \dots + a_k b_k = 1$. Thus

$$g = g^{a_1 b_1} g^{a_2 b_2} \dots g^{a_k b_k}.$$

But

$$(g^{a_i b_i})^{p_i^{\beta_i}} = (g^{o(g)})^{b_i} = e$$

So $g^{a_i b_i} \in G_i$. Thus $g = g_1 \dots g_k$ with $g_i = g^{a_i b_i} \in G_i$.

To show $G = G_1 \dots G_k$ as a direct product, we must show uniqueness of the decomposition, which amounts to showing $G_i \cap G_j = \{e\}$ for $i \neq j$. But if $g \in G_i \cap G_j$, then

$$g^{p_i^{\alpha_i}} = e = g^{p_j^{\alpha_j}}.$$

So $g^{(p_i^{\alpha_i}, p_j^{\alpha_j})} = g^1 = e$.

So $G_i \cap G_j = \{e\}$ for $i \neq j$ and $G = G_1 \dots G_k$ as a direct product. $\square$

We now classify abelian p -groups, groups having all elements a power of p .

Lemma 22. Let G be a finite abelian p -group and suppose $g \in G$ has maximal order. Then for some $H \leq G$, one has $G = \langle g \rangle H$ as an internal direct product.

Proof. By Cauchy's Theorem, if G is a p -group when $|G| = p^n$, some $n \in \mathbb{Z}_{\geq 0}$. Proceed by induction on n . When $n = 1$ then G is cyclic of order p , so $G = \langle g \rangle$ and we are done taking $H = \{e\}$.

Suppose $n > 1$, and the desired conclusion holds for all groups G' of order p^k where $1 \leq k < n$. Let $g \in G$ have maximal order, say $o(g) = p^m$ with $1 \leq m < n$ (without loss of generality since if $m = n$ we again have G cyclic).

Put $A = \langle g \rangle$, and let $h \in G/A$ be the element of least possible order in G/A . Note that G/A is nonempty, for else $G = \langle g \rangle \times \{e\}$.

Claim. $o(h) = p$.

To check this, note that the order of h is a power of p , so $o(h^p) = o(h)/p$. By minimality of h , $h^p \in A = \langle g \rangle$, so $h^p = g^r$ for some $r \in \mathbb{Z}$. But $o(g) = p^m$, and

$$(g^r)^{p^{m-1}} = (h^p)^{p^{m-1}} = h^{p^m} = e$$

since $o(g) = p^m$ is maximal. Whence $p^m | rp^{m-1}$. Thus $p | r$, say $r = ps$ for some $s \in \mathbb{Z}$. Put $a = g^{-s}h$, and observe that if $a \in \langle g \rangle$ then $h \in \langle g \rangle$, a contradiction. So $a \notin \langle g \rangle$ and $a^p = g^{-sp}h^p = g^{-r}h^p = e$. Then $o(a) = p$, so by the minimality of $o(h)$, we have $o(h) = p$. This proves the claim.

Now study $H = \langle h \rangle$ and G/H (noting $H \triangleleft G$). We claim that the order of Hg in G/H is $o(g) = p^m$ (Exercise). If $H = (Hg)^{p^{m-1}}$, so $g^{p^{m-1}} \in H$, whence $g^{p^{m-1}} = h^u$ for some $u \in \mathbb{Z}$, so $h = g^{up^{m-1}}$ for $v \in \mathbb{Z}$ satisfying $uv = 1 \pmod{p}$. This is a contradiction since $h \notin \langle g \rangle$.

We have Hg has order $p^m = o(g)$ in G/H . Observe that $|G/H| < |G|$, so by inductive hypothesis, $G/H = \langle Hg \rangle \times T$, for some subgroup $T \leq G/H$. By the correspondence theorem, one has $T = K/H$ for some subgroup K of G such that $H \triangleleft K \leq G$. Thus $G/H = \langle Hg \rangle \times K/H$.

Now we aim to prove $G = \langle g \rangle \times K$. To see this, observe that when $a \in G$, one has $Ha = aH \in \langle Hg \rangle \times K/H$, so $g^r kh$, for some $r \in \mathbb{Z}$ and $k \in K$, whence

$$\begin{aligned} a &= g^r kh' \text{ for some } h' \in H \\ &= g^r k' \text{ for some } k' \in K, \text{ since } H \triangleleft K. \end{aligned}$$

Thus $G = \langle g \rangle K$. If $\langle g \rangle \cap K \neq \{e\}$, say $b \in \langle g \rangle \cap K$ with $b \neq e$, then $Hb \in \langle Hg \rangle \cap K/H = \{H\}$ (trivial intersection, since G/H is an internal direct product). Then $b \in H = \langle h \rangle$, and $b \in \langle g \rangle = A$, so $b \in A \cap \langle h \rangle$, and this gives $b = e$. Thus $G = \langle g \rangle K$ with $\langle g \rangle \cap K = \{e\}$, so $G = \langle g \rangle \times K$ as an internal direct product. This completes induction. $\square$

Lecture 18: Proof of classification theorem

Fri 21 Oct 2022 10:35

Theorem 35 (Classification for Finite Abelian Groups). (2.10 A) Suppose that G is a finite abelian group. Then G is isomorphic to a direct product of cyclic groups of the shape

$$\mathbb{Z}_{p_1^{\alpha_1}} \times \mathbb{Z}_{p_2^{\alpha_2}} \times \dots \times \mathbb{Z}_{p_r^{\alpha_r}}.$$

where p_i are primes (not necessarily distinct), and $\alpha_i \in \mathbb{N}$, and $n = |G| = p_1^{\alpha_1} \dots p_r^{\alpha_r}$.

Proof. Apply Lemma 2.10.C to show that if G is a finite abelian group of order $n = p_1^{\alpha_1} \dots p_k^{\alpha_k}$, then G is an internal direct product $G = G_1 \dots G_k$ of p_i -groups G_i , for distinct primes $p_1, \dots, p_k$. Fix a choice for i . Then by Lemma 2.10.D, show that G_i has an element $g_i \in G \setminus \{e\}$ of maximal order and a subgroup H_i such that $G_i = \langle g_i \rangle \times H_i$ and H_i is also a p_i -group. And by induction H_i has a similar decomposition, so

$$G_i = \langle g_{i1} \rangle \langle g_{i2} \rangle \dots \langle g_{ir_i} \rangle.$$

where $o(g_{ij}) = p_i^{n_j}$, say, and $n_1 \geq n_2 \geq \dots \geq n_{r_i} \geq 1$.

Thus

$$G_i = \mathbb{Z}_{p_i^{n_1}} \times \mathbb{Z}_{p_i^{n_2}} \times \dots \times \mathbb{Z}_{p_i^{n_{r_i}}}.$$

Hence G is an internal direct product of cyclic p -groups. $\square$

4.4 Conjugacy and Sylow Theorems

Recall: a is conjugate to b means $b = g^{-1}ag$ for some $g \in G$.

Conjugacy defines an equivalence relation. We define the *equivalence class*:

$$\text{cl}(a) := \{b \in G : b \text{ conjugate to } a\}.$$

If G is abelian, then $g^{-1}ag = a$ for all $g \in G$, so $\text{cl}(a) = \{a\}$. This is true also if $a \in Z(G)$. If $a \notin Z(G)$, the center can be bigger.

Also recall the *centralizer* $C(a) = \{g \in G : ga = ag\}$. They are related by

Theorem 36. Let G be a finite group and $a \in G$. Then

$$|\text{cl}(a)| = \frac{|G|}{|C(a)|} = i_G(C(a)).$$

Proof. If $g, h \in G$, and the conjugate of a by g and h are the same, then $gag^{-1} = hah^{-1}$, whence $h^{-1}gag^{-1}h = a$, hence $h^{-1}g \in C(a)$. Thus $g \in hC(a)$. Thus, g and h lie in $C(a)$ if and only if $g \in hC(a)$. Thus the number of conjugates of a in G is equal to $\{hC(a) \in G/C(a) : h \in G\} = |G|/|C(a)| = i_G(C(a))$. $\square$

Theorem 37 (Class equation). If G is a finite group, then

$$|G| = \sum_{a=1} i_G(C(a)) = \sum_a \frac{|G|}{|C(a)|}$$

, where the summation is over distinct conjugacy class representatives.

Proof. The equivalence classes partition G . $\square$

Consequences

- Any group G of order p^2 with p prime is abelian, so $G \cong \mathbb{Z}_{p^2}$ or $G \cong \mathbb{Z}_p \times \mathbb{Z}_p$.

Lecture 19

Wed 26 Oct 2022 10:35

Theorem 38. Let G be a nontrivial finite group whose order is a power of a prime p . Then $Z(G)$ has more than one element.

Theorem 39 (Sylow's First Theorem). Let G be a group of order $p^n m$ with p prime and $p \nmid m$, $n \geq 1$. Then G has a subgroup of order p^n .

Proof. Induction. Suppose true for all groups H with $1 \leq |H| < |G|$. Suppose $|G| = p^n m$, p prime, $p \nmid m$. Then $Z(G)$ has an element of a group of order p , put $A = \langle a \rangle$. We investigate $K = G/A$.

Since $|K| = |G|/|A| = p^n m/p < |G|$, so by inductive hypothesis that K has a subgroup M of order p^{n-1} . By the correspondence theorem, there exists $A \triangleleft P \leq G$ and $P/A = M$. Hence $|P|/|A| = |M| = p^{n-1}$, so $|P| = p^n$.

Then P is a subgroup G of order p^n . This completes the induction. $\square$

Lecture 20

Wed 26 Oct 2022 10:35

Theorem 40 (Second Sylow Theorem). Let P_1 and P_2 be Sylow p -subgroups of a finite group. Then P_1 and P_2 are conjugate. i.e., there exists $g \in G$ such that $P_1 = gP_2g^{-1}$.

Proof. Let P be a Sylow p -subgroup of G , where $|G| = p^n m$ with p prime and $p \nmid m$, $n \geq 1$. We suppose that Q is another Sylow p -subgroup of G with $Q \neq x^{-1}Px$ for any $x \in G$ (and seek to derive a contradiction.)

Define $S = \{y^{-1}Qy : y \in G\}$ and a relation $\sim$ on S by taking $Q_1 \sim Q_2$ where $Q_2 = a^{-1}Q_1a$ for some $a \in P$. We can check that this gives an equivalence relation.

How many elements in each equivalence class? We have

$$a^{-1}Q_1a = b^{-1}Q_1b \Leftrightarrow ab^{-1} \in N(Q_1) \cap P \Leftrightarrow a \in (N(Q_1) \cap P)b.$$

Then for a fixed choice of $b \in P$, there are $|N(Q_1) \cap P|$ choices for $a \in P$ with $a^{-1}Q_1a = b^{-1}Q_1b$.

So the number of elements in an equivalence class is

$$|P|/|N(Q_1) \cap P|.$$

We claim that this is divisible by p . If true, then $|S|$ is also divisible by p , which gives the conclusion, since each equivalence class contained in S has a number of elements divisible.

But since $|P| = p^n$, we have $|P|/|N(Q_1) \cap P|$ not divisible by p iff $|P|/|N(Q_1) \cap P| = 1$, iff $N(Q_1) = P$, i.e. $P \subset N(Q_1)$. But then $P \leq N(Q_1)$ and (by hw q2.11.9) we have $Q_1 \triangleleft N(Q_1)$, so $PQ_1 \leq N(Q_1)$ and by second homomorphism theorem, $PQ_1/Q_1 = P/(P \cap Q_1)$. Then $|PQ_1| = \frac{|Q_1||P|}{|P \cap Q_1|}$. But every element of $P \cap Q_1$ has order a power of p , and likewise for P, Q_1 , so each group has order of a power of p .

In particular, PQ_1 has order a power of p . Then PQ_1 is a p -group that contains P , which is a Sylow p -subgroup that has maximal p -power order in G . But $PQ_1 \subset G$, then $PQ_1 = P$. Thus $P = Q_1$, since $|Q_1| = |P|$, and hence $a^{-1}Qa = P$ for some $a \in P$, whence $P = Q$. This is a contradiction.

Thus $|S| \equiv 0 \pmod{p}$. Then also, by (HW Q 2.11.13), the number of distinct subgroups $x^{-1}Qx$ of G is equal to

$$|S| = i_G(N(Q)) = |G|/|N(Q)|.$$

Since $Q \triangleleft N(Q)$, we have $|Q| \mid |N(Q)|$, whence $p^n \mid |N(Q)|$. But $|G| = p^n m$ with $p \nmid m$, so

$$|S| \not\equiv 0 \pmod{p}.$$

Thus this contradiction shows that Q has to be equal to $x^{-1}Px$ for some $x \in G$. So any two Sylow p -subgroups are conjugate. $\square$

Lecture 21: Third Sylow Theorem

Fri 28 Oct 2022 10:33

Theorem 41 (Third Sylow Theorem). The number of Sylow p -subgroups of a finite group G of order $p^n m$, where p prime, $n \geq 1$ and $p \nmid m$, is equal to $kp + 1$ for some $k \in \mathbb{Z}$ with $(kp + 1) \mid |G|$.

Proof. Using the proof of Second Sylow Theorem, if there is more than one Sylow p -subgroup, say P and $Q \neq P$, then each equivalence class of the set S from that proof has a number of elements divisible by p . Then $|S| = kp$ for some $k \in \mathbb{Z}_{\geq 0}$.

But the number of Sylow p -subgroups conjugate to P by the action of P is 1. Thus the number of Sylow p -subgroups of G is $1 + |S| = 1 + kp$.

We still need to show $(1 + kp) \mid |G|$. Suppose P is any Sylow p -subgroup of G . If Q is any other Sylow p -subgroup (possibly P itself) then $Q = g^{-1}Pg$ for some $g \in G$ (by 2nd Sylow Theorem).

With what multiplicity is Q represented in this way? If $h^{-1}Ph = Q = g^{-1}Pg$ where $h \in G$, then $hg^{-1} \in N(P)$, and thus $h \in N(P)g$. So there are $|N(P)|$ elements g with $Q = g^{-1}Pg$, for each $Q \in S \cup \{P\}$. Then $|G| = |N(P)| \cdot (1 + kp)$. Hence $(kp + 1) \mid |G|$. $\square$

4.5 Small Groups

We try to classify all isomorphism classes of groups of a given order.

Example. Classify all groups of order $2p$ where $p \neq 2$ is prime.

G has a Sylow p -subgroup of order p . Since $(kp+1) \nmid 2$ if $k \geq 1$, by Sylow's Third Theorem, there is precisely one Sylow p -subgroup of G , fixed under conjugation, hence G has a normal subgroup P of order p . Say $P = \langle a \rangle \triangleleft G$ with $a^p = e$. If $b \in G \setminus P$, then b has order 2. Thus $b^{-1}ab = a^r$ for some $r \in \mathbb{Z}$.

Then $a = b^{-2}ab^2 = b^{-1}a^r b = a^{r^2}$. So $r^2 \equiv 1 \pmod{p}$. Then $r = \pm 1$ without loss of generality. If $r = 1$ then $ab = ba$ thus G is abelian. If $r = -1$ then $b^{-1}ab = a^{-1}$, thus G is dihedral. $\diamond$

Lecture 22: Ring Theory

Fri 04 Nov 2022 10:31

Definition 29 (Ring). Let R be a nonempty set. Then R is a *ring* if there are two binary operations $+$ and $\cdot$ satisfying:

R0 $a, b \in R$ implies $a + b \in R$.

R1 $+$ is commutative.

R2 $+$ is associative.

R3 There exists $0 \in R$ as additive identity.

R4 For all $a \in R$ there exists $-a \in R$ such that $a + (-a) = (-a) + a = 0$

M0 $a, b \in R$ implies $a \cdot b \in R$.

M1 $\cdot$ is associative.

D1 (distributivity) $a \cdot (b + c) = a \cdot b + a \cdot c$

D2 $(a + b) \cdot c = a \cdot c + b \cdot c$

Example. $\mathbb{Z}$ and $\mathbb{Z}_n$ are rings with usual addition and multiplication (modulo n for $\mathbb{Z}_n$)

$M_2(n\mathbb{Z})$ (2 by 2 matrices with entries from $n\mathbb{Z}$) form a ring with usual matrix addition and multiplication. $\diamond$

Remark. R is a ring when $(R, +)$ is an abelian group and $(R, \cdot)$ is associative, distributive.

Rings with extra properties:

1. (Rings with unity) If R is a ring with also

M2 There exists $1 \in R$ with $1 \neq 0$ such that $1 \cdot a = a \cdot 1$ for all $a \in R$

then R is a ring with unity.

Nonexample. $2\mathbb{Z}$

Example. $M_2(\mathbb{Z})$, identity matrix is the multiplicative 1. $\diamond$

Notation. Rings without 1 are sometimes called “rngs”.

2. (Commutative Ring) If $(R, +, \cdot)$ also satisfies

M3 For all $a, b \in R$ we have $a \cdot b = b \cdot a$

then R is a *commutative ring*.

Nonexample. $M_2(\mathbb{Z})$

Example. $\mathbb{Z}, 2\mathbb{Z}$ $\diamond$

3. (Domains) If $(R, +, \cdot)$ satisfies

M4 whenever $a \cdot b = 0$ then $a = 0$ or $b = 0$

then R is a *domain*.

Nonexample. $M_2(\mathbb{Z})$. ($e_1 \cdot e_2 = 0$)

4. (Integral Domain) If $(R, +, \cdot)$ is a commutative domain, then R is an integral domain.
5. (Division Ring) A ring $(R, +, \cdot)$ with unity is a *division ring* if for all $a \in R \setminus \{0\}$, there exists $b \in R$ such that $a \cdot b = b \cdot a = 1$.
6. (Fields) A ring $(R, +, \cdot)$ is a *field* if it is a commutative division ring.

Example. $\mathbb{Q}, \mathbb{R}, \mathbb{C}, \mathbb{Z}_p(+, \cdot \pmod{p})$ form a field. $\diamond$

Example. Quaternions (Hamilton, 19th century). Consider

$$\mathbb{H} = \{a + bi + cj + dk : a, b, c, d \in \mathbb{R}\}$$

subject to: $+$ coordinate-wise, and $i^2 = j^2 = k^2 = -1$, $ij = k$, $jk = i$, $ki = j$.

$\mathbb{H}$ is noncommutative but a division ring. Observe that

$$(a + bi + cj + dk)(a - bi - cj - dk) = a^2 + b^2 + c^2 + d^2.$$

Hence if $a + bi + cj + dk \neq 0$, then there is an element

$$(a + bi + cj + dk)^{-1} = \frac{a - bi - cj - dk}{a^2 + b^2 + c^2 + d^2}.$$

$\diamond$

Definition 30 (Subring). A *subring* S of a ring R is a subset $S \subset R$ with S satisfying ring axioms.

To verify subring, we have to check:

- $S \neq \emptyset$
- For all $a, b \in S$, $a \pm b \in S$ and $a \cdot b \in S$.

Example. $2\mathbb{Z}$ is a subring of $\mathbb{Z}$. $\diamond$

Lecture 23

Mon 07 Nov 2022 10:30

4.6 Ideals, Homomorphisms, and Quotient Ring

Remark. Subgroups are exactly images of homomorphisms of groups; normal subgroups are exactly kernels of homomorphism of groups. The same corresponding structure are there for ring theory, namely subrings and ideals.

Definition 31 (Homomorphism). Let R and R' be two rings. A map $\varphi : R \rightarrow R'$ is a *homomorphism* if

- for all $a \in R$, $b \in R$: $\varphi(a + b) = \varphi(a) + \varphi(b)$
- for all $a \in R$, $b \in R$: $\varphi(ab) = \varphi(a)\varphi(b)$

Definition 32 (kernel). The kernel of a homomorphism $\varphi : R \rightarrow R'$ of rings is

$$\ker \varphi = \{r \in R \mid \varphi(r) = 0\}.$$

Remark. The kernel does not know about the other operation $\cdot$. The definition is exactly same as for groups.

Observation. If $\varphi : R \rightarrow R'$ is a homomorphism of rings, then $\varphi(R)$ is a subring of R'

Proof. Let $\varphi(a)$ and $\varphi(b)$ be elements from $\varphi(R)$. Then $\varphi(a) - \varphi(b) = \varphi(a - b) \in \varphi(R)$ and $\varphi(a)\varphi(b) = \varphi(ab) \in \varphi(R)$. $\square$

Example. Let p be prime number, define

$$\begin{aligned} \varphi : \mathbb{Z} &\longrightarrow \mathbb{Z}_p \\ n &\longmapsto \varphi(n) = [n]. \end{aligned}$$

Then φ is a homomorphism of rings. $\varphi(\mathbb{Z}) = \mathbb{Z}_p$, $\ker \varphi = p\mathbb{Z} = [0]$.

Note in this case, $k(p\mathbb{Z}) = p\mathbb{Z}$ whenever $k \in \mathbb{Z}$. This property motivates the definition of $\diamond$

Definition 33 (Ideal). Let R be a ring. An *ideal* of R is a subset $I \subset R$ so that

- I is a subgroup of additive abelian group R . In particular, I is nonempty.
- For every $r \in R$, $a \in I$, we have $ra \in I$ and $ar \in I$.

Example. $p\mathbb{Z}$ is an ideal of $\mathbb{Z}$.

Note: Every ideal is a subring, but not conversely. $\diamond$

Notation. If I is an ideal of R , then we write $I \triangleleft R$, like for normal subgroups.

Lemma 23. Let $\varphi : R \rightarrow R'$ be a homomorphism of rings, then $\ker \varphi \triangleleft R$.

Proof. Obviously $\ker \varphi$ is an additive subgroup of the additive abelian group R . Now let $r \in R$ and $a \in \ker \varphi$. Then

$$\varphi(ra) = \varphi(r)\varphi(a) = \varphi(r) \cdot 0 = 0.$$

And same for ar . Hence $ra \in \ker \varphi$ and $ar \in \ker \varphi$. $\square$

Definition 34 (One-sided Ideal). Let R be a ring. A *left ideal* (right ideal) is a subset $I \subset R$ such that

- I is a subgroup of the additive group R
- For all $r \in R$ and $a \in I$, $ra \in I$ ($ar \in I$).

Example. Let F be a field, $n \geq 2$, $R = M_{n \times n}(F) = \{n \times n \text{ matrices with entries in } F\}$. Then R is a ring.

Let $1 \leq k \leq n$, and $I = \{\text{matrices that have non-zero entries only in the } k\text{-th row}\}$. The I is a right ideal, not a left ideal. $\diamond$

Let R be a ring and $I \triangleleft R$. Since I is a normal subgroup of the additive group R , R/I is an additive subgroup of additive group R . We want to make it a ring.

Theorem 42 (Quotient Ring). The additive group R/I becomes a ring with respect to the operation

$$(a + I)(b + I) := ab + I.$$

And the natural map

$$\begin{aligned} \varphi : R &\longrightarrow R/I \\ r &\longmapsto \varphi(r) = r + I. \end{aligned}$$

is a homomorphism of rings with $\ker \varphi = I$. In particular, every ideal is a kernel.

Proof. We show this is an operation, i.e. φ is well-defined. Assume $a + I = a' + I$ and $b + I = b' + I$. We have $a = a' + x$ for some $x \in I$ and likewise $b = b' + y$ for some $y \in I$. Now $ab = (a' + x)(b' + y) = a'b' + a'y + xb' + xy$. Since $I \triangleleft R$, $a'y \in I$ and $xb' \in I$. Also $xy \in I$. Hence $ab \in a'b' + I$. Hence $ab + I = a'b' + I$. $\square$

Definition 35 (Isomorphism). An isomorphism of rings is a bijective homomorphism of rings. We say two rings R, R' are *isomorphic*, and write $R \cong R'$ if there exists an isomorphism $R \rightarrow R'$.

Definition 36 (Automorphism). An automorphism of ring is an isomorphism of rings where the domain and range are the same.

Lecture 24

Wed 09 Nov 2022 10:31

Example.

$$\begin{aligned} \varphi : \mathbb{C} &\longrightarrow M_{2 \times 2}(\mathbb{R}) \\ a + bi &\longmapsto \varphi(a + bi) = \begin{pmatrix} a & b \\ -b & a \end{pmatrix} \end{aligned}$$

This is a homomorphism of rings. We can verify that the map is injective.

So

$$\mathbb{C} \cong \varphi(\mathbb{C}) = \left\{ \begin{pmatrix} a & b \\ -b & a \end{pmatrix} : a, b \in \mathbb{R} \right\} \subset M_{2 \times 2}(\mathbb{R})$$

◇

Theorem 43 (First Homomorphism Theorem). Let $\varphi : R \rightarrow R'$ be a surjective homomorphism of rings with $K = \ker \varphi$. Then $R/K' \cong R'$. In fact,

$$\begin{aligned} \psi : R/K &\longrightarrow R' \\ a + K &\longmapsto \psi(a + K) = \varphi(a) \end{aligned}$$

is an isomorphism.

Proof. From the First Homomorphism Theorem of groups, we know that ψ is an isomorphism of additive groups. It remains to show ψ preserves multiplication. But

$$\psi((a + K)(b + K)) = \psi(ab + K) = \varphi(ab) = \varphi(a)\varphi(b) = \psi(a + K)\psi(b + K).$$

□

Example. $\mathbb{Z}_p \cong \mathbb{Z}/p\mathbb{Z}$.

◇

Theorem 44 (Correspondence Theorem). Let $\varphi : R \rightarrow R'$ be surjective homomorphism of rings and $K = \ker \varphi$. Let $I' \triangleleft R'$. Let $I := \varphi^{-1}(I') = \{a \in R : \varphi(a) \in I'\}$. Then $I \triangleleft R$ and $I' = \varphi(I)$. In fact this gives a 1-1 inclusion-preserving correspondence between ideals of R containing K and ideals of R' .

Proof. We only need to check I , in addition to being a normal subgroup, is an ideal. □

Theorem 45 (Second Homomorphism Theorem). Let R be a ring, A a subring and I an ideal of R . Then $A + I$ is a subring of R and I is an ideal of $A + I$, and $I \cap A$ is an ideal of A . Also,

$$(A + I)/I \cong A/(A \cap I).$$

Theorem 46 (Third Homomorphism Theorem). If $K \triangleleft R$ and $I \triangleleft R$ and $I \supset K$, then

$$R/K \cong (R/K)/(I/K).$$

Example.

$$K = p^2\mathbb{Z} \triangleleft \mathbb{Z}, I = p\mathbb{Z} \triangleleft \mathbb{Z}, K \subset I.$$

Hence

$$\mathbb{Z}/p\mathbb{Z} \cong (\mathbb{Z}/p^2\mathbb{Z}) / (p\mathbb{Z}/p^2\mathbb{Z}).$$

◇

Example. 1. Let F be a field. Then $\{0\}$ and F are the only ideals.

Proof. Let $I \neq \{0\}$ be an ideal. Then there exists $a \in I$, $a \neq 0$. So there exists $a^{-1} \in F$. Now $1 = a^{-1}a \in I$ since $a \in I$ and $a^{-1} \in F$. Let $r \in F$. Then $r = r1 \in I$ since $1 \in I$. So $I = F$. $\square$

Consequence. Let $\varphi : F \rightarrow R'$ be a homomorphism of rings, where F is a field. Then $\ker \varphi \triangleleft F$. So $\varphi = 0$ ($\ker \varphi = F$) or φ is injective ($\ker \varphi = \{0\}$). In the second case, $F \cong \varphi(F)$.

2. Let R be a commutative ring with 1. Fix $a \in R$. Then $Ra = \{ra \mid r \in R\}$. Notice $Ra \triangleleft R$ and $a = 1a \in Ra$. This is the unique smallest ideal containing a . This is called the [Principal Ideal](#) generated by a .
3. Let $\mathbb{R}[t] = \{\text{polynomials with real coefficients}\}$ and

$$\begin{aligned} \varphi : \mathbb{R}[t] &\longrightarrow \mathbb{C} \\ f &\longmapsto \varphi(f) = f(i). \end{aligned}$$

Then φ is a surjective homomorphism of rings and $\ker \varphi = \mathbb{R}[t](t^2 + 1)$. So by first homomorphism theorem,

$$\mathbb{C} \cong \mathbb{R}[t] / (\mathbb{R}[t](t^2 + 1)).$$

◇

Lecture 25

Wed 09 Nov 2022 11:30

Definition 37 (Maximal Ideal). An ideal M of R is a *maximal ideal* if $M \neq R$ and for any $N \triangleleft R$, $N \neq R$ we have $N \subset M$.

Corollary 9. A ring R is a field if and only if R is commutative with 1 and $\{0\}$ is a maximal ideal of R .

Theorem 47. Let R be a commutative ring with 1, and let $M \triangleleft R$. Then R/M is a field if and only if M is a maximal ideal of R .

Proof. In either case, $M \neq R$, equivalently, $R/M \neq \{0\}$. So R/M is a commutative with $1 \neq 0$. By the corollary, R/M is a field and M/M is a maximal ideal of R/M . By correspondence theorem, this happens iff there is no ideal of R between M and R , iff M is a maximal ideal of R . $\square$

Example. 1. $n \in \mathbb{Z}, n \geq 2 : n\mathbb{Z} \triangleleft \mathbb{Z}$. Note $n\mathbb{Z} \neq \mathbb{Z}$. Whenever $n = ab$ for $a, b \in \mathbb{Z}$, we have $n\mathbb{Z} \subset a\mathbb{Z} \subset \mathbb{Z}$, so $n\mathbb{Z}$. If n is a prime, then there exists

such a decomposition with $1 < a < n$, so $n\mathbb{Z}$ is not a maximal ideal. So $\mathbb{Z}/n\mathbb{Z}$ is not a field.

If $n = p$ is a prime number, then $p \neq \pm 1$, so $p\mathbb{Z} \neq \mathbb{Z}$, and in fact $p\mathbb{Z}$ is a maximal ideal, so $\mathbb{Z}/p\mathbb{Z}$ is a field.

So $\mathbb{Z}_n$ is a field iff n is prime.

2. $\mathbb{Z}_9$ is not a field. However, there exists a field of order 9.

Fact. Any field must have prime-power order. Conversely, for any prime power there exists up to isomorphism a unique field of that order.

(Field of 9 elements) Let $R = \mathbb{Z}_3[t]$. Let $J = R(t^2 + 1)$. Then R/J is a field with 9 elements.

Proof. $R/J = \{f + J : f \in R\} = \{a + bt + J : a, b \in \mathbb{Z}_3\}$. There are 9 such elements. How does multiplication work?

$$\begin{aligned} ((a + bt) + J)((c + dt) + J) &= ((a + bt)(c + dt)) + J \\ &= ac + bdt^2 + (ad + bc)t + J \\ &= (ac - bd) + (ad + bc)t + J \quad t^2 \equiv -1 \pmod{J}. \end{aligned}$$

Let $0 \neq (a + bt) + J \in R/J$. Then $a \neq 0$ or $b \neq 0$ in $\mathbb{Z}_3$. We have $a^2 + b^2 \neq 0$ in $\mathbb{Z}_3$. Therefore $\frac{1}{a^2 + b^2}$ exists in $\mathbb{Z}_3$. Now

$$\begin{aligned} (a + bt + J) \left(\frac{a - bt}{a^2 + b^2} + J \right) &= \frac{a^2 - b^2 t^2}{a^2 + b^2} + J \\ &= \frac{a^2 + b^2}{a^2 + b^2} + J \\ &= J \end{aligned}$$

Hence R/J is a field with 9 elements, and J is a maximal ideal. □

◇

Lecture 26

Mon 14 Nov 2022 10:38

Fact. Every finite field has p^n elements for prime p and natural number n ; and for every p, n we have a finite field with p^n elements. Furthermore, every field with p^n elements are isomorphic.

4.7 Polynomial Rings

Given R a ring, we can define $R[x] = \{a_0 + a_1x + \dots + a_nx^n : a_i \in R, n \geq 0\}$. We concentrate on situation when R is a field.

1. $F[x]$ forms a commutative ring with unit.

Proof. Consider $p(x) = a_0 + a_1x + \dots + a_nx^n \in F[x]$, and $q(x) = b_0 + b_1x + \dots + b_mx^m \in F[x]$, $n \geq m \geq 0$.

- (equality) $p(x) = q(x)$ iff $a_0 = b_0, \dots, a_m = b_m, a_j = 0$ for $j > m$.
- (addition)

$$p(x) + q(x) = (a_0 + b_0) + (a_1 + b_1)x + \dots + (a_m + b_m)x^m + a_{m+1}x^{m+1} + \dots + a_n x^n.$$

- (multiplication) Define $p(x)q(x) = c_0 + c_1x + \dots + c_r x^r \in F[x]$ where $c_i = \sum_{l=0}^i a^l b^{i-l}$, adopt convention that $a_i = 0$ for $i > n$, $b_j = 0$ for $j > m$.
- (Additive Identity) $0 \in F[x]$
- (Additive Inverse) $-(a_0 + a_1x + \dots + a_n x^n) = (-a_0) + (-a_1)x + \dots + (-a_n)x^n$
- Associativity and Commutativity of addition inherited from F .
- (Multiplicative Identity) $1 \in F[x]$ since $1 \in F$
- (Commutativity of Multiplication) inherited from F .

□

We measure the size of polynomials with “degree”, which is analogous to $\log |n|$ for $n \in \mathbb{Z}$.

Definition 38 (Degree of Polynomial). Suppose $f(x) = a_0 + a_1x + \dots + a_n x^n$ with $a_n \neq 0$, then the *degree* of $f(x)$ is $\deg f = n$. By convention, $\deg(0) = -\infty$.

Example. One has $3x^2 + 2 \in \mathbb{Z}_5[x]$, and $\deg(3x^2 + 2) = 2$, and $\deg(2) = 0$. ◇

Proposition 2. The following statements hold for $p(x), q(x) \in F[x]$:

- $\deg(p(x) + q(x)) \leq \max\{\deg p, \deg q\}$
- $\deg(pq) = \deg p + \deg q$

Proof. exercise. □

Lemma 24. When F is a field, then $F[x]$ is an integral domain.

Proof. Need to check $F[x]$, which is commutative ring with unity, has no zero divisors. But if $p(x), q(x) \in F[x]$ and $p(x)q(x) = 0$, then

$$-\infty = \deg(p(x)q(x)) = \deg p + \deg q.$$

Which is possible when either $\deg p = -\infty$ or $\deg q = -\infty$, or both. So one of them is 0. □

Theorem 48 (Division Algorithm). Suppose that $f(x), g(x) \in F[x]$ and that $g(x) \neq 0$. Then

$$f(x) = q(x)g(x) + r(x)$$

where the *quotient* $q(x)$ and *remainder* $r(x) \in F[x]$, and $r(x) = 0$ or $\deg r < \deg g$.

Proof. If $\deg f < \deg g$, take $q = 0$ and $r = f$ and we are done. So we assume $\deg f \geq \deg g$, and say

$$\begin{aligned} f(x) &= a_0 + a_1x + \dots + a_mx^m \\ g(x) &= b_0 + b_1x + \dots + b_nx^n \end{aligned} \quad m \geq n \geq 0, a_m \neq 0, b_n \neq 0$$

Proceed induction and assume conclusion holds for $\deg(f) < m$ for some $m \geq 1$. Now assume $\deg f = m$ as above, and observe

$$\begin{aligned} f(x) - \frac{a_m}{b^n} x^{m-n} g(x) &= \left(a_{m-1} - \frac{a_m}{b^n} b_{n-1} \right) x^{m-1} + \dots + \left(a_{m-n} - \frac{a_m}{b^n} b_0 \right) x^{m-n} + a_{m-n-1} x^{m-n-1} + \dots + a_0 \\ &= h(x) \in F[x] \end{aligned}$$

When $\deg h \leq m-1 < \deg f$, so inductive hypothesis applies, so $h(x) = q_1(x)g(x) + r_1(x)$ where $q_1, r_1 \in F[x]$, $r_1(x) = 0$ or $\deg r_1 < \deg g$. But then $f(x) = \left(\frac{a_m}{b^n} x^{m-n} q_1(x) \right) g(x) + r_1(x) =: q(x)g(x) + r(x)$. Where $q(x), r(x) \in F[x]$, and q, r satisfies needed properties. This complete inductive step. $\square$

Remark. One can also prove by linear algebra.

Lecture 27

Wed 16 Nov 2022 10:48

Remark. The situation $r = 0$ is important, where g “divides” f , and $f(x) \in g(x)F[x] = (g)$. g divides f iff f lies in ideal generated by g .

Theorem 49. (Polynomials rings are PID) Suppose that $I \neq 0$ in an ideal of $F[x]$. Then $I = (g) = \{f(x)g(x) : f(x) \in F[x]\}$ for some $g \in F[x]$.

Proof. Try to identify a polynomial g to generate I , by considering

$$d = \min\{\deg h : h \in I, \deg h \geq 0\}.$$

and take g to be any polynomial of degree d . Now, if $h \in I$ we have $\deg h \geq \deg g$, and by the division algorithm we have $h = qg + r$, where $q, r \in F[x]$ and $\deg r < \deg g$ or $r = 0$. But $h \in I$ and $qg \in I$, so $r = h - qg \in I$. And so I contains the polynomial r which has degree smaller than $\deg g$ (contradiction) or $r = 0$. Thus $h = qg \in (g)$. $\square$

Definition 39 (Principal Ideal Domain (PID)). A *principal ideal domain* is an integral domain R satisfying the condition that whenever I is an ideal of R , then $I = (a) = \{xa : x \in R\}$ for some $a \in R$.

Remark. Every principal ideal domain (PID) is a unique factorization domain (UFD)

Remark. Generator of an ideal need not be unique. But this can be resolved by focusing on *monic* polynomials with lead coefficient 1. To see this, note that if $I \triangleleft F[x]$ and $(p(x)) = I = (m(x))$, then since $p(x) \in (m(x))$, $p(x) = f(x)m(x)$, in particular $\deg p \leq \deg m$. Similarly $\deg m \leq \deg p$. Hence $\deg p = \deg m$, so $\deg f = 0$. Hence they are same up to scaling.

Definition 40 (Divisibility). 1. When $f, g \in F[x]$ and $g \neq 0$, we say g divides f (and write $g|f$) if $f = ag$ for some $a \in F[x]$.
2. The *greatest common divisor* of f and g is the **monic** divisor $d \in F[x]$ of f and g divisible by all other such common divisors. This is written as (f, g) or $\gcd(f, g)$.

Theorem 50. (Analog of consequence of Euclid's Algorithm) Let $f, g \in F[x]$ and suppose $g \neq 0$. Then $d = (f, g)$ exists and there exists $a, b \in F[x]$ such that $d = af + bg$.

Proof. Let $I = \{af + bg : a, b \in F[x]\}$. We

Claim. $I \triangleleft F[x]$.

To check this: whenever $a, b \in I$, say $a = u_1f + v_1g$ and $b = u_2f + v_2g$, one has

$$a + b = (u_1 + u_2)f + (v_1 + v_2)g \in I.$$

and for all $t \in F[x]$,

$$ta = (tu_1)f + (tv_1)g \in I.$$

Also $I \neq 0$ since $f \in I$. Hence $I \triangleleft F[x]$.

Then since $F[x]$ is a PID, we know that $I = (d)$ for some $d \in F[x]$. But then, since $f, g \in I$ we have f and g are both multiples of d . Then $d|f$ and $d|g$, so d is a common divisor of f and g .

But the definition of I shows that

$$d = uf + vg,$$

for some $u, v \in F[x]$. Thus, whenever $h|f$ and $h|g$, we have $h|d$. If we take (as we can) d monic, then we have shown that $d = (f, g)$. $\square$

From the proof we see that any generating monic polynomial is automatically GCD.

Say f and g are *relatively prime* when $(f(x), g(x)) = 1$, iff there exists $u, v \in F[x]$ s.t. $uf + vg = 1$.

Lemma 25. Suppose $(q(x), f(x)) = 1$ and $q|fg$. Then $q|g$.

Proof. There exists $a, b \in F[x]$ such that $aq + bf = 1$. Then

$$(aq)g + b(fg) = g.$$

Now the lhs is divisible by q since $q|aq$ and $q|fg$. Hence $q|g$. $\square$

Lemma 26. Suppose that $(q(x), f(x)) = 1$ and $q(x)|f(x)g(x)$. Then $q(x)|g(x)$.

Proof. We have that there exists $a, b \in F[x]$ such that $af + bq = 1$. Thus

$$g = a(fg) + b(qg)$$

which is divisible by q . $\square$

Definition 41. We say $p \in F[x]$ is *irreducible* if

1. $\deg p \geq 1$
2. whenever $f \in F[x]$, either $p|f$ or $(p, f) = 1$.

As a consequence, p irreducible and $p = fg$ with $\deg f \geq \deg g \geq 1$, then either $f = cp$ or $g = cp$, some $c \in F$.

Lecture 28: Factorization

Mon 21 Nov 2022 20:26

Proposition 3. (Consequence of irreducibility) p is irreducible if and only if there does not exist $f, g \in F[x]$ such that $\deg f \geq \deg g \geq 1$ and $p = fg$.

Proof. Suppose p is irreducible and $p = fg$ with $\deg f \geq \deg g \geq 1$. Then either $p|f$ or $(p, f) = 1$, in which case $p|g$. But then $\deg p \geq \deg p + \min\{\deg f, \deg g\} > \deg g$, a contradiction.

Suppose $p(x)$ cannot be written as $p = fg$ with $\deg f \geq \deg g \geq 1$, and yet $\deg p \geq 1$. Suppose that $h \in F[x]$. Then either $(p, h) = 1$ or else $(p, h) = cp$ for some $c \in F$ (in other cases $(p, h)|p$ and $\deg(p, h) \geq 1$, so we have contradiction). Then $(p, h) = 1$ or $p|h$, so p irreducible. $\square$

Lemma 27. Suppose that $p \in F[x]$ and p is irreducible. Then whenever $p|a_1a_2 \dots a_k$, with $a_i \in F[x]$, one has $p|a_i$ for some index i .

Proof. We proceed inductively, noting that the conclusion is trivial for $k = 1$. Suppose that the conclusion has been proved for $1 \leq k < K$ with $K \geq 2$, and that $p|a_1a_2 \dots a_K$. Then $p|(a_1 \dots a_{K-1})a_K$, so either $p|a_K$ or $(p, a_K) = 1$, in the later case we have $p|a_1 \dots a_{K-1}$, and the inductive hypothesis shows $p|a_i$ for some i . This completes the induction. $\square$

Theorem 51. (Analogue of [\[\[Fundamental Theorem of Arithmetic \]\]](#)) Suppose that $f(x) \in F[x]$ has degree at least 1. Then $f(x)$ is the product of irreducible polynomials in $F[x]$, and this product is unique in the following sense. One has

$$f(x) = cp_1(x)^{m_1}p_2(x)^{m_2} \dots p_n(x)^{m_n}.$$

where c is the leading coefficient of f , the polynomials $p_1(x), \dots, p_n(x)$ are distinct, monic and irreducible in $F[x]$, and the integers $m_1, \dots, m_n$ are positive, and up to reordering, the list of tuples $(p_1, m_1), \dots, (p_n, m_n)$ is unique.

Proof. We proceed by induction on $n = \deg(f)$. When $\deg f = 1$, we have $f(x) = ax + b$ for some $a, b \in F$, and thus $f(x) = a(x + b/a)$, with $x + b/a$ irreducible.

Now suppose that the conclusion holds whenever $\deg f < n$, with $n \geq 2$. If $f(x)$ is irreducible, we are done. Otherwise, we have $f(x) = a(x)b(x)$, some $a, b \in F[x]$ with $\deg b \geq \deg a \geq 1$ and $\deg b < \deg f$. By induction, both a and b factor into a product of irreducibles, and so therefore does f . This completes the inductive step showing that f factors as a product of irreducibles.

Now we consider the uniqueness claim. Suppose that f has two representations of the claimed type. Say

$$cp_1(x)^{m_1}p_2(x)^{m_2} \dots p_k(x)^{m_k} = f(x) = c'p'_1(x)^{m'_1}p'_2(x)^{m'_2} \dots p_{k'}(x)^{m'_{k'}}.$$

with p_i, p'_i monic and irreducible, $m_i, m'_i \in \mathbb{N}$ and $c, c' \in F$. Then we have $p_k(x)|c'p'_1(x)^{m'_1}p'_2(x)^{m'_2} \dots p_{k'}(x)^{m'_{k'}}$, whence $p_k(x)|p'_i(x)$ for some i . But p_i is monic and irreducible, so $p_k(x) = p'_i(x)$. Dividing left and right hand sides by $p_k(x)$, we find that $f(x)/p_k(x)$ has two factorizations as a product of irreducibles. But $\deg(f(x)/p_k(x)) < \deg f(x)$, so by induction we see that $f(x)/p_k(x)$ and hence $f(x)$ have unique factorization. $\square$

One can use these ideas to develop arithmetic over $F[x]$. For example,

Fact. If F is a finite field having p elements, p prime, then $\text{card}\{f(x) \in F[x] : \deg(f) \leq n, f \text{ monic and irreducible}\}$ is approximately p^n/n .

This is analogue to the theorem about prime numbers: $\varphi(x) \sim x/\log x$

Irreducible polynomials can be used to generate new fields:

Theorem 52. Suppose that $p(x) \in F[x]$. Then $(p(x))$ is a maximal ideal of $F[x]$ if and only if $p(x)$ is irreducible over $F[x]$.

Proof. Suppose $p(x)$ is irreducible over $F[x]$. We try to show that $(p(x))$ is a maximal ideal of $F[x]$ by considering what it means for some other ideal N to satisfy $(p(x)) \subset N$. But $N = (f(x))$ for some $f \in F[x]$, and since $p(x) \in (p(x)) \subset N$, we have $p(x) = a(x)f(x)$ for some $a \in F[x]$. By irreducibility of p , we see that either a or f is constant, whence $f(x) = a^{-1}p(x)$ for some $a \in F \setminus \{0\}$, or $f(x) = f_0 \in F$.

In the first case we have $f(x) \in (p(x))$, so $N \subset (p(x))$, which yields $N = (p(x))$. In the second case we have $1 = f_0^{-1}f_0 \in (f(x))$, whence $g(x) \in (f(x))$ for all $g \in F[x]$, yielding $(f(x)) = F[x]$. Hence $(p(x))$ is a maximal ideal, since whenever $(p(x)) \subset N$, we have $N = (p(x))$ or $N = F[x]$. $\square$

Remark. This conclusion has the consequence that

$$F[x]/(p(x)) \text{ is a field} \Leftrightarrow p(x) \text{ is irreducible.}$$

In particular, $\mathbb{R}[x]/(x^2 + 1) \cong \mathbb{C}$ is a field.

Note that if $p(x)$ is irreducible of degree d , then (by division algorithm)

$$F[x]/(p(x)) = \{a_0 + a_1x + \dots + a_{d-1}x^{d-1} + (p(x)) : a_i \in F\}.$$

Hence, the element $x + (p(x))$ “solves” $p(x)$, since

$$p(x + (p(x))) = p(x) + (p(x)) = (p(x)),$$

which is the zero element in $F[x]/(p(x))$.

Remark. Since there are infinitely many (monic) irreducible polynomials over $F[x]$, we can find infinitely many fields of the shape $F[x]/(p(x))$.

We have used the division algorithm (which can be used to generate the Euclidean Algorithm) to establish results on divisibility in $F[x]$. This is not the only route, but it is convenient algorithmically when available. This motivates:

Definition 42 (Euclidean Ring). An integral domain R is called a *Euclidean ring* when there exists a function $d : R \setminus \{0\} \rightarrow \mathbb{Z}_{\geq 0}$ satisfying:

1. For all $a, b \in R \setminus \{0\}$, one has $d(a) \leq d(ab)$
2. Whenever $a, b \in R \setminus \{0\}$, there exist $q, r \in R$ such that $b = qa + r$, where $r = 0$ or $d(r) < d(a)$.

Example. $\mathbb{Z}, F[x], \mathbb{Z}[\sqrt{-1}] = \{a + b\sqrt{-1} : a, b \in \mathbb{Z}\}$ with $d(a + b\sqrt{-1}) = a^2 + b^2$ $\diamond$

Nonexample. $\mathbb{Z}[\sqrt{-5}]$

4.8 Polynomials over Rationals

We develop the theory of (irreducible) polynomials further in $\mathbb{Z}[x]$ and $\mathbb{Q}[x]$. Our first goal is to show that irreducibility over $\mathbb{Z}[x]$ is sufficient to guarantee irreducibility over $\mathbb{Q}[x]$.

Definition 43 (primitive polynomial). $f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_0 \in \mathbb{Z}[x]$ is a polynomial in which $(a_n, a_{n-1}, \dots, a_0) = 1$.

Theorem 53 (Gauss' Lemma). Let $f(x) \in \mathbb{Z}[x]$ be a primitive polynomial, and suppose that $f(x) = A(x)B(x)$, where $A(x), B(x) \in \mathbb{Q}[x]$. Then there exist primitive polynomials $a(x), b(x) \in \mathbb{Z}[x]$ for which $f(x) = a(x)b(x)$, where $\deg a = \deg A$ and $\deg b = \deg B$.

Proof. Let $f(x) \in \mathbb{Z}[x]$ be primitive of degree at least 2, for otherwise there is nothing to prove. Suppose that $f(x) = A(x)B(x)$, where $A(x), B(x) \in \mathbb{Q}[x]$. By considering the least common multiple of the denominators of the coefficients of $A(x)$, we can write

$$A(x) = \frac{u_1}{v_1} a(x),$$

where $a(x) \in \mathbb{Z}[x]$ is primitive and $u_1, v_1 \in \mathbb{Z}$ satisfy $(u_1, v_1) = 1$. (Explanation: move common factor in a into the fraction, and simplify the fraction in order to obtain $(u_1, v_1) = 1$).

Similarly, we can write

$$B(x) = \frac{u_2}{v_2} b(x),$$

where $b(x) \in \mathbb{Z}[x]$ is primitive and $u_2, v_2 \in \mathbb{Z}$ satisfy $(u_2, v_2) = 1$. Thus, there exist integers $u, v \in \mathbb{Z}$ with $(u, v) = 1$ for which

$$f(x) = \frac{u}{v} a(x)b(x),$$

where, we emphasize, the polynomials a and b are primitive in $\mathbb{Z}[x]$.

We claim that $u = v = 1$ or $u = v = -1$. Otherwise, there is a prime p with $p|u$, or a prime v with $p|v$.

First consider the scenario with $p|v$, in which case

$$ua(x)b(x) = vf(x), \quad p|v.$$

so since $(u, p) = 1$ we see that all coefficients of $a(x)b(x)$ are divisible by p . Let the highest degree term of $a(x)$ with coefficient *not* divisible by p be $a_r x^r$, and the highest degree term of $b(x)$ with coefficient *not* divisible by p be $b_s x^s$. Then we see that

$$a(x)b(x) = a_r b_s x^{r+s} + c_{r+s-1} x^{r+s-1} + \dots + c_0 + ph(x)$$

for some $h(x) \in \mathbb{Z}[x]$ and some $c_i \in \mathbb{Z}$. But *all* coefficients of $a(x)b(x)$ are supposed to be divisible by p , so $p|a_r$ or $p|b_s$, a contradiction. This means either all coefficients of $a(x)$ are divisible by p , contradicting primitivity of $a(x)$, or all coefficients of $b(x)$ are divisible by p , contradicting primitivity. Thus $v = \pm 1$.

When $p|u$, we see that all coefficients of $f(x)$ are divisible by p , contradicting primitivity of f . Hence $u = \pm 1$,

Then we conclude that $f(x) = \pm a(x)b(x)$ with a, b primitive, and $\deg a = \deg A, \deg b = \deg B$. This suffices to prove the lemma. $\square$

Example. Already this is useful in finding irreducible polynomials over $\mathbb{Q}[x]$. Thus, for example, the polynomial $x^3 - x - 1$ is irreducible over $\mathbb{Q}[x]$, for if not, it would have a factor of degree 1 over $\mathbb{Q}[x]$, and Gauss' Lemma shows then that such a factor (primitive) exists over $\mathbb{Z}[x]$. This factor takes the form $ax + b$ with $a, b \in \mathbb{Z}$. But then

$$x^3 - x - 1 = (ax + b)g(x).$$

for some $g \in \mathbb{Z}[x]$, and we must have:

- (lead coefficient): $a|1 \Rightarrow a = \pm 1$
- (final coefficient): $b|(-1) \Rightarrow b = \pm 1$

But by trial and error, none of $\pm(x \pm 1)$ is a factor of $x^3 - x - 1$, so $x^3 - x - 1$ is irreducible over $\mathbb{Q}[x]$. $\diamond$

Notice that the final part of the argument applied in the proof of Gauss' Lemma could have been made more concise by reducing modulo p . We had, for example

$$ua(x)b(x) = vf(x) \quad \text{with } p|v \text{ and } p \nmid u.$$

So in a sense $a(x)b(x) \equiv 0 \pmod{p}$. But we are working with polynomials, so need to be more careful.

Lemma 28. Suppose that R is a ring and $I \triangleleft R$. Then

1. $I[x] \triangleleft R[x]$, and
2. $R[x]/I[x] \cong (R/I)[x]$

Proof. Write $\overline{R} = R/I$. Recall that there exists a homomorphism

$$\begin{aligned} \varphi : R &\longrightarrow \overline{R} \\ a &\longmapsto \varphi(a) = a + I. \end{aligned}$$

having kernel I . We now define

$$\begin{aligned} \Phi : R[x] &\longrightarrow (R/I)[x] \\ a_n x^n + \dots + a_0 &\longmapsto \Phi(a_n x^n + \dots + a_0) = \varphi(a_n) x^n + \dots + \varphi(a_0). \end{aligned}$$

It is easy to show that Φ is a surjective homomorphism of $R[x]$ into $(R/I)[x]$. $\square$

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